

Cooperative Control for Mitigation of Voltage Fluctuations in Power Distribution Systems

Report

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Application: Renewable Distributed Generation

System: Electrical Power Distribution Network

Key Focus

Voltage regulation using coordinated reactive-power control

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1 Abstract – Detailed Explanation

The research focuses on the problem of **voltage fluctuations in electrical power distribution systems caused by distributed generators (DGs), particularly inverter-based renewable energy sources.**

With the increasing penetration of renewable energy technologies such as solar photovoltaic (PV) systems, distributed generation is becoming an important part of modern distribution networks. However, renewable sources such as solar power are inherently intermittent (**naturally variable and not continuously constant**) because their power output changes with variations in solar irradiance. These rapid changes in generated active power can produce undesirable fluctuations in distribution-system voltage.

Here we specifically address this challenge and propose a cooperative control strategy for mitigating these voltage variations.

What are Distributed Generators (DGs)?

Distributed Generators (DGs) are **small- or medium-sized electricity generation sources that are connected close to the electrical loads or within the distribution network**, rather than generating electricity only at large centralized power plants.

In simple terms:

DG = Electricity generation located near where the electricity is being consumed.

Examples include:

- Solar PV systems
- Wind turbines
- Small gas turbines
- Fuel cells
- Other small-scale generation systems

Here we specifically mention solar PV, small gas turbines, fuel cells, and wind energy as examples of DGs.

Why are DGs important?

Here we are particularly concerned with **inverter-based DGs**, especially **solar PV systems**. A solar PV system produces **active power** P depending on the available solar irradiance. If the irradiance changes rapidly—for example, when a cloud passes over the solar panels—the generated power also changes rapidly.

This can cause:

Change in solar irradiance $\rightarrow$ **Change in PV active power** $\rightarrow$ **Voltage fluctuation**

Therefore, while DGs help provide renewable energy and meet increasing electricity demand, their intermittent generation can create **voltage-regulation problems** in the distribution network.

How does a DG help control voltage?

In this research, the important point is that the **DG inverter can control reactive power Q** .

The inverter can:

- **Inject reactive power** $\rightarrow$ can help increase/support voltage.
- **Absorb reactive power** $\rightarrow$ can help reduce voltage.

The proposed cooperative controller determines how much reactive power each DG should provide by adjusting its **reactive power utilization ratio α_q** .

Solar PV (DG) $\rightarrow$ Inverter $\rightarrow$ Reactive Power Control $\rightarrow$ Voltage Regulation

This is the key role of the DG in the paper.

What does “Cooperative Control Strategy” mean?

A **cooperative control strategy** means that **multiple devices or controllers work together and share information to achieve a common objective**, instead of each device operating independently.

In the context of your paper, the devices are mainly **DG inverters and capacitor banks**, and the common objective is **voltage regulation**.

Example

For example, suppose there are several DGs in different locations:

$$DG_1 \quad DG_2 \quad DG_3$$

Instead of each DG independently deciding how much reactive power to inject or absorb, they **exchange information** through a communication network:

$$DG_1 \longleftrightarrow DG_2 \longleftrightarrow DG_3$$

Based on this shared information, they coordinate their reactive-power outputs.

Cooperative control = Multiple devices communicate, coordinate, and work together to solve one common problem.

In Your Project

The cooperative-control process can be represented as:

Information Exchange → Coordination → Reactive Power Control → Voltage Regulation

Why is it called “cooperative”?

Because one DG does **not** try to solve the voltage problem alone.

For example:

- **DG 1** may inject reactive power.
- **DG 2** may inject a smaller amount.
- **DG 3** may absorb reactive power.
- The decisions depend on the **network voltage conditions** and information exchanged between the participating nodes.

Together, they try to achieve a better overall voltage profile while also avoiding unnecessary reactive-power utilization.

Definition for the Report

A **cooperative control strategy** is a control approach in which multiple distributed energy resources communicate and coordinate their control actions to achieve a common objective, such as mitigating voltage fluctuations, rather than operating independently.

Central Research Problem:

The intermittent active-power output of renewable distributed generators can produce rapid voltage fluctuations in distribution networks. The proposed research investigates cooperative reactive-power control as a means of mitigating these fluctuations.

1.1 Increasing Use of Distributed Generators

A **distributed generator (DG)** is a generation source connected within the distribution network rather than being located only at a large centralized power plant. Here we consider distributed generation from renewable sources, with particular emphasis on inverter-based generation.

Solar PV generation is highly dependent on **solar irradiance**. For example, when a cloud passes over a solar PV installation, the available solar irradiance can change rapidly.

This causes the active power produced by the PV system to change correspondingly. Because the electrical network must continuously maintain an appropriate balance between generation and consumption, these rapid changes can result in variations in the voltage magnitude of distribution-system buses.

Important Observation:

Although renewable DGs provide significant benefits in terms of renewable-energy utilization and distributed power generation, their intermittent nature introduces an additional voltage-regulation challenge.

1.2 Problem with Conventional Voltage Regulation

Traditionally, voltage regulation in distribution systems is performed using devices such as

- **On-load tap changers (OLTCs)**
- **Capacitor banks (CBs)**

These devices are effective for relatively slower voltage changes, but they are not well suited to rapidly changing voltage fluctuations associated with fast variations in DG output.

For instance, a sudden change in solar irradiance can produce a rapid change in PV power output.

Attempting to continuously compensate for every such fluctuation using mechanical voltage-regulating equipment would require frequent switching or tap-changing operations.

This can increase equipment wear and tear and consequently reduce the expected lifetime of the equipment.

Problem with Mechanical Regulation:

Frequent operation of OLTCs and capacitor banks in response to rapid voltage fluctuations can increase switching operations, equipment wear, and maintenance requirements.

Another possible solution is to **curtail the active power generated by the DG**.

Although this can help control voltage, it reduces the amount of renewable energy that can be delivered to the grid.

From the perspective of the generator, this represents a loss of available generation and potentially a reduction in revenue.

A battery energy storage system can also be used to compensate for fluctuations, but this introduces additional system complexity and cost.

Consequently, here we investigate another approach in which the available **reactive-power capability of DG inverters** is utilized for voltage regulation.

1.3 Proposed Cooperative Control Approach

The central contribution of the paper is a **cooperative control-based method** for controlling the reactive power of distributed generators.

Instead of relying on a single centralized controller, the proposed method allows different nodes participating in voltage regulation to cooperate with one another.

The DG inverters can either:

- **Inject reactive power** into the distribution network.
- **Consume reactive power** from the network.

Reactive-power injection or absorption changes the voltage conditions in the network and can therefore be used to mitigate voltage fluctuations.

Cooperative Control Concept

Different reactive-power resources located at different points in the distribution network cooperate with one another instead of operating independently.

The important feature of the proposed method is that the optimization does not depend exclusively on information available at one central controller.

Each participating node uses its **local information** together with information received from other nodes through the defined communication network.

The paper considers communication between reactive-power source nodes and also communication between reactive-power source nodes and other nodes that do not contain reactive-power sources.

This distributed/cooperative structure allows the reactive-power resources available at different locations in the distribution system to work together rather than operating independently.

1.4 Role of Reactive Power

Reactive power is particularly important in this research because it provides a means of controlling voltage without necessarily reducing the active power generated by the renewable DG.

The paper introduces a parameter called the **reactive power utilization ratio**, denoted by α_q .

This parameter represents the fraction of the available reactive-power capability of a DG or capacitor bank that is being utilized.

Conceptually,

$$\alpha_q = \frac{\text{Reactive power generated by the DG or CB}}{\text{Maximum available reactive power}} \quad (1)$$

Reactive Power Utilization Ratio

The parameter α_q represents the degree to which the available reactive-power capability of a distributed generator or capacitor bank is being utilized.

A positive value of α_q represents reactive-power injection, whereas a negative value represents reactive-power consumption.

The cooperative controller optimizes this utilization ratio to determine how much reactive-power support should be provided by the participating devices.

Thus, α_q becomes one of the principal control variables of the proposed cooperative-control algorithm.

1.5 Optimization Objective

The proposed controller does not simply attempt to minimize voltage deviation.

It considers **two important objectives simultaneously**.

The first objective is to minimize the deviation of the voltage magnitude from its desired/base value.

This ensures that the voltage profile remains as close as possible to the desired operating condition.

The second objective is to minimize the amount of reactive power generated or consumed by the DGs and capacitor banks.

This is important because excessive reactive-power operation increases the utilization of inverter power-electronic components and can contribute to accelerated aging.

Two Main Optimization Objectives

1. **Voltage-deviation minimization**
2. **Reactive-power utilization minimization**

The objective function therefore combines:

1. **Voltage-deviation minimization**, and
2. **Reactive-power utilization minimization**.

The paper assigns weighting factors to these two objectives.

By changing the weights, the controller can place greater importance on voltage regulation, reactive-power minimization, or a combination of both.

This makes the proposed method a **multi-objective optimization approach** in which voltage performance and equipment utilization are considered together.

1.6 Cooperative Information Exchange

The proposed approach uses a communication topology to exchange information among participating nodes.

The reactive-power-source nodes, referred to as **QG nodes**, can communicate with other QG nodes in both directions.

In contrast, non-QG nodes can send information to QG nodes.

Communication Structure

The communication network enables QG nodes to exchange information and coordinate their reactive-power decisions using information obtained from neighboring nodes.

This communication structure enables each QG node to make its control decision using information about the state of other relevant nodes in the network.

The paper represents this communication structure mathematically using communication-topology matrices.

The cooperative controller then uses an **iterative consensus-based optimization process** to update the reactive-power utilization ratios.

During each iteration, information from neighboring QG nodes and participating non-QG nodes contributes to the update of the control variable.

The objective is to reach a condition in which the participating reactive-power resources are utilized cooperatively and the overall voltage profile is improved.

1.7 Phase-Level Implementation

An important feature highlighted in the abstract and further explained in the paper is that the proposed cooperative-control method is implemented at the **phase level**.

This is significant because practical distribution systems can be **unbalanced three-phase systems**.

A control method that treats the entire three-phase system as perfectly balanced may not accurately represent the behavior of individual phases.

The paper therefore extends the cooperative-control approach to individual phases.

This provides greater control flexibility and allows the method to address voltage variations on a phase-by-phase basis.

According to the paper, this approach can also help reduce voltage unbalance between phases.

Phase-Level Advantage

Phase-level implementation provides the ability to address voltage variations separately on individual phases, which is particularly important for unbalanced three-phase distribution systems.

1.8 Main Advantage of the Proposed Method

The major advantage of the proposed method is that it attempts to regulate voltage **without relying primarily on rapid mechanical switching of conventional voltage-regulation equipment and without requiring active-power curtailment of the renewable generators.**

Instead, the available reactive-power capability of DG inverters and, where applicable, capacitor banks is coordinated through the cooperative algorithm.

The method therefore provides a way to:

- mitigate fast voltage fluctuations,
- utilize the reactive-power capability of DG inverters,
- coordinate multiple reactive-power sources,
- reduce unnecessary reactive-power utilization,
- consider the effect of reactive-power operation on inverter aging,
- operate at the phase level, and
- improve voltage profiles at participating QG and selected non-QG nodes.

1.9 Simulation-Based Validation

The effectiveness of the proposed method is evaluated through simulation studies.

The paper uses a **modified IEEE 13-bus distribution system** containing distributed generators and capacitor banks.

Three single-phase DGs are added at nodes

632,	670,	671
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while the capacitor-bank arrangement is also modified to support phase-level control.

The simulations consider solar PV generators whose active-power output varies according to changes in solar irradiance.

The results are evaluated over a 15-minute period, corresponding to

$$15 \text{ minutes} = 900 \text{ seconds}$$

Two main control scenarios are considered:

Scenario 1

Both DGs and capacitor banks participate in cooperative control.

Scenario 2

Only DGs participate in cooperative control.

Different objective-function weighting combinations are then used to examine the effects of emphasizing voltage regulation, minimizing reactive-power utilization, or simultaneously considering both objectives.

1.10 Overall Outcome

The simulation results demonstrate that the proposed cooperative-control method can produce **smoother voltage profiles**.

The paper reports improvement in voltage at both QG nodes and selected non-QG nodes when cooperative control is implemented.

The results therefore demonstrate the potential of coordinated reactive-power control for mitigating fast voltage fluctuations associated with intermittent renewable generation.

Overall Result

Coordinated reactive-power control can be used to mitigate voltage fluctuations while simultaneously considering the utilization of available reactive-power resources.

2 Distributed Generator (DG) and Inverter-Based DG

What is a Distributed Generator (DG)?

A **Distributed Generator (DG)** is a relatively small generation source connected to the **distribution network**, usually close to the consumers, rather than only at a large centralized power station.

DG = Local electricity generation located close to the electrical loads.

The paper defines DG broadly and gives examples such as:

- Solar PV

- Wind energy
- Small gas turbines
- Fuel cells

Thus, **DG is a broad category**. Different types of generation technologies can be classified as distributed generators.

What does “Inverter-Based Distributed Generator” mean?

An **inverter-based distributed generator** is a distributed generation source that is connected to the AC distribution grid through a **power-electronic inverter**.

The basic idea is:

$$\text{Generation Source} \longrightarrow \text{Inverter} \longrightarrow \text{AC Grid}$$

The inverter acts as an **interface between the generator and the electrical grid**.

Example: Solar PV as an Inverter-Based DG

Solar panels produce **DC power**. However, the distribution grid operates using **AC power**. Therefore, an inverter is required to interface the solar PV system with the AC distribution grid. The basic conversion process is:

$$\text{Sunlight} \longrightarrow \text{Solar PV} \longrightarrow \text{DC}$$

followed by:

$$\text{Solar PV} \longrightarrow \text{DC} \longrightarrow \text{Inverter} \longrightarrow \text{AC} \longrightarrow \text{Distribution Grid}$$

This is why a solar PV system is an excellent example of an **inverter-based distributed generator**.

Why is the Inverter Important in This Paper?

This is particularly important for the research because the paper is concerned with controlling the **reactive power of DG inverters**.

The inverter can control the reactive power:

$$Q$$

by either injecting or absorbing reactive power.

Reactive-Power Injection

When the inverter supplies reactive power to the grid,

$$Q > 0$$

the reactive power can provide voltage support and help increase or maintain the voltage.

Reactive-Power Absorption

When the inverter absorbs reactive power from the grid,

$$Q < 0$$

it can help reduce excessive voltage.

Therefore, the paper uses the inverter's reactive-power capability to help control voltage fluctuations caused by changing solar generation.

The inverter provides the controllable interface through which reactive power can be used for voltage regulation.

3 Types of Distributed Generators

There is **not one universally fixed number of DG types**. DGs can be classified in different ways.

For this paper, a useful classification is based on the **generation technology and grid interface**.

A. Solar PV DG

Solar PV systems are normally connected to the AC grid through an inverter and are therefore an important example of **inverter-based DGs**.

$$\text{Solar PV} \longrightarrow \text{Inverter} \longrightarrow \text{Grid}$$

The paper focuses particularly on this type of DG.

B. Wind DG

Wind generators can be connected to the grid in different ways.

Modern wind turbines commonly use power-electronic converters or inverters, making them **inverter-based DGs**.

A simplified representation is:



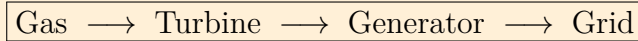
The paper identifies wind energy as another type of DG and notes that wind and solar generation can be intermittent.

C. Small Gas Turbine DG

A small gas turbine can be used as a distributed generation source.

Depending on its electrical configuration, it may be connected using a conventional rotating electrical generator or through power electronics.

Conceptually:



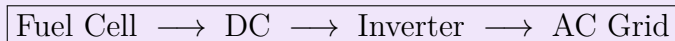
It is **not necessarily inverter-based**.

The paper specifically lists small gas turbines as examples of DGs.

D. Fuel Cell DG

Fuel cells produce electrical power through an electrochemical process and typically produce DC electricity.

Therefore, they generally require a power-electronic interface:



Thus, fuel cells are generally considered **inverter-based DGs**.

The paper also lists fuel cells among DG technologies.

4 Classification Based on Grid Interface

DGs can also be divided into two broad groups based on how they are connected to the electrical grid.

Type	Basic Idea	Examples
Inverter-based DG	Connected to the grid through power electronics	Solar PV, modern wind, fuel cell
Conventional rotating-machine DG	Uses a rotating electrical generator directly or through conventional equipment	Small gas turbine generator

Therefore,

DG

can contain both:

Inverter-Based DG

and

Non-Inverter-Based DG

The Most Important Distinction for Your Project

Distributed Generator

This tells you **what the device is doing**:

It generates electrical power locally in the distribution network.

Inverter-Based Distributed Generator

This tells you **how it is connected to the grid**:

The generator is interfaced with the AC distribution network using a power-electronic inverter/converter.

For example:

Solar PV

is a **DG**.

Because it uses an inverter to connect its DC output to the AC grid:

Solar PV + Inverter = Inverter-Based DG

5 DG in the Proposed Research

Here we are primarily dealing with **inverter-based intermittent DGs**, particularly **solar PV DGs**.

The important chain is:

Solar Irradiance $\rightarrow$ **PV Active Power Variation** $\rightarrow$ **Voltage Fluctuation**

The proposed solution is:

Measure System $\rightarrow$ **Cooperative Control** $\rightarrow$ **Calculate α_q** $\rightarrow$ **Control DG Inverter Reactive Power** $\rightarrow$ **Improve Voltage**

This is why the **inverter is central to the paper**: the controller is not mainly changing the solar panel's active-power production.

Instead, it uses the inverter's **reactive-power capability** to support voltage regulation.

Definition for the Report

A **distributed generator (DG)** is a small-scale generation source connected to the distribution network near the load, while an **inverter-based distributed generator** is a DG that uses a power-electronic inverter as the interface between the generation source and the AC distribution grid.

5.1 Summary of the Abstract

In summary, the research addresses the voltage fluctuations introduced by the increasing penetration of intermittent inverter-based renewable DGs, particularly solar PV systems.

Conventional mechanical voltage-regulation devices are not well suited to rapidly changing fluctuations, while alternatives such as active-power curtailment and battery storage introduce their own disadvantages.

To address this problem, the paper proposes a **cooperative distributed control strategy** in which DG inverters and capacitor banks coordinate their reactive-power utilization using local measurements and information exchanged through a communication network.

The control algorithm optimizes the reactive-power utilization ratio while minimizing both voltage deviation and unnecessary reactive-power operation.

The method is implemented at the phase level, making it applicable to unbalanced three-phase distribution systems.

Simulation results on a modified IEEE 13-bus system demonstrate that the proposed approach can improve and smooth the voltage profiles while coordinating the available reactive-power resources.

The overall objective is therefore to achieve effective voltage regulation while also considering the utilization and aging implications of reactive-power operation in power-electronic equipment.

6 Key Takeaways

Key Points

- Renewable DGs, particularly solar PV, can introduce rapid voltage fluctuations.
- Conventional OLTCs and capacitor banks are not ideally suited for very fast fluctuations.
- Active-power curtailment reduces the renewable energy delivered to the grid.
- Reactive-power capability of DG inverters provides an alternative means of voltage regulation.
- Cooperative control allows multiple reactive-power sources to coordinate their operation.
- The reactive power utilization ratio α_q is an important control variable.
- The optimization considers both voltage deviation and reactive-power utilization.
- Communication between QG and non-QG nodes enables distributed information exchange.
- Phase-level implementation makes the method suitable for unbalanced distribution systems.
- Simulation results using a modified IEEE 13-bus system demonstrate smoother voltage profiles.

7 Introduction

The increasing demand for electrical energy has led to the growing integration of **distributed generators (DGs)** into power distribution systems.

Distributed generation allows electrical power to be produced at locations within or close to the distribution network instead of relying exclusively on large centralized power plants.

According to the paper, DGs can include **solar photovoltaic (PV) systems, small gas turbines, fuel cells, and wind-energy systems.**

Among these, solar and wind generation are particularly important because their power output is intermittent and depends on environmental conditions.

7.1 Increasing Penetration of Distributed Generators

The main reasons for the increasing use of DGs are the need to **meet increasing electrical demand** and to make greater use of **renewable energy resources** such as solar and wind.

Unlike conventional centralized generation, renewable DGs can be installed at different locations throughout the distribution system.

This distributed arrangement can provide useful local generation; however, it also introduces new technical challenges for the operation and control of distribution networks.

The most important issue considered in this research is the effect of **intermittent DG generation on distribution-system voltage**.

Key Issue

The increasing penetration of renewable DGs provides valuable local generation, but their intermittent power output introduces additional challenges for voltage regulation in distribution systems.

7.2 Intermittency of Renewable Generation

Solar and wind power are described in the paper as **intermittent generation sources**.

This means that their electrical power output is not constant and can change according to environmental conditions.

For a solar PV system, the generated active power depends strongly on the available **solar irradiance**.

Under stable sunlight, the PV output may remain relatively steady. However, if a cloud passes over the PV installation, the solar irradiance can change rapidly.

Consequently, the active power produced by the DG can also change rapidly.

The resulting sequence can be represented as:

Change in Solar Irradiance → Change in DG Active Power → Voltage Fluctuation

This rapid variation in DG output is the fundamental problem addressed by the proposed control strategy.

The paper specifically focuses on the challenges associated with **inverter-based intermittent DGs**.

7.3 Conventional Voltage-Regulation Equipment

Traditionally, voltage fluctuations in distribution systems are controlled using devices such as:

- On-Load Tap Changers (OLTCs)
- Capacitor Banks (CBs)

An OLTC regulates voltage by changing the transformer tap position, while capacitor banks provide reactive-power support to influence the voltage level.

These devices are useful for conventional voltage regulation, but the paper points out that they are **not suitable for following very rapid voltage fluctuations** caused by fast variations in DG output.

For example, consider a solar PV system experiencing rapidly changing irradiance because of moving clouds.

If the voltage changes repeatedly, an OLTC or capacitor bank would need to perform many switching operations to continuously compensate for those changes.

This creates another problem.

Limitation of Conventional Regulation

OLTCs and capacitor banks are effective for conventional voltage regulation, but their mechanical switching mechanisms are not well suited to continuously following very fast voltage fluctuations.

7.4 Equipment Wear and Reduced Lifetime

If conventional mechanical voltage-regulating devices are continuously operated to follow rapid fluctuations, the **number of switching operations increases**.

Repeated operation causes:

More Switching → More Wear and Tear → Reduced Equipment Lifetime

Therefore, although OLTCs and capacitor banks can regulate voltage, using them to continuously chase fast DG-induced fluctuations can increase maintenance requirements and accelerate equipment degradation.

The paper specifically states that attempting to follow these rapid fluctuations can increase equipment wear and tear and reduce the lifetime of the voltage-regulating equipment.

7.5 Alternative Solution: Curtailing DG Power

Another possible method of reducing voltage fluctuations is to **curtail the active power output of the distributed generator**.

In this approach, instead of allowing the DG to produce all the power available from the renewable resource, its active-power generation is intentionally reduced when necessary.

Conceptually:

$$\text{Available PV Power} > \text{Allowed Grid Power}$$

Therefore, some available generation is intentionally not delivered to the grid.

Although this can help mitigate voltage problems, it has an important disadvantage:

Power Curtailment → Reduced Power Availability → Potential Revenue Loss

From the generator's perspective, curtailing renewable generation is undesirable because available renewable energy is being wasted or not fully utilized.

The paper therefore does not consider active-power curtailment as the preferred solution.

7.6 Alternative Solution: Battery Energy Storage

A battery energy storage system can also be used to mitigate voltage fluctuations.

The basic idea is that the battery can absorb or supply electrical power to compensate for changes in renewable generation.

For example:

Excess Generation

Excess Generation → Battery Charging

and during insufficient generation:

Insufficient Generation

Insufficient Generation → Battery Discharging

Although energy storage can provide effective support, the paper identifies an important disadvantage: the addition of a battery energy storage system **increases system complexity and cost**.

Therefore, while battery storage is technically capable of helping with voltage fluctuations, it may not always be the most economical or simplest solution.

7.7 Need for a New Control Strategy

The limitations of the existing approaches create the need for a control method capable of responding to **fast voltage fluctuations** without unnecessarily:

1. increasing mechanical switching operations,
2. curtailing renewable active-power generation, or
3. adding expensive and complex energy-storage systems.

This motivates the cooperative-control approach proposed in the paper.

The proposed method focuses on **inverter-based DGs** and uses their available **reactive-power capability** to regulate voltage.

Instead of changing the active power generated by the DG, the inverter can be controlled to either inject or consume reactive power as required by the network.

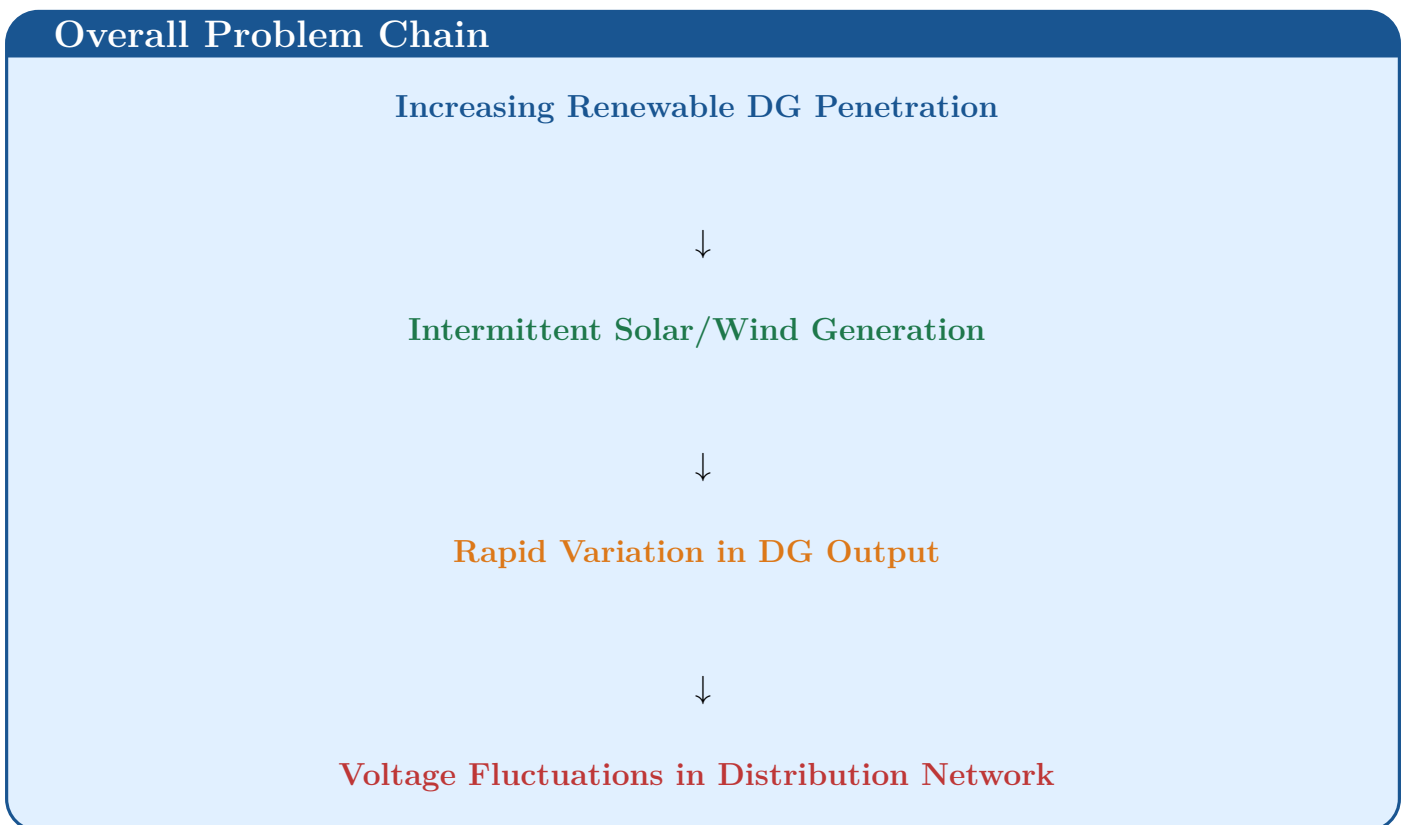
The basic concept is:

DG Inverter → **Reactive Power Control** → **Voltage Regulation**

The paper proposes a cooperative control strategy in which multiple reactive-power sources work together using local information and information received from other nodes.

7.8 Overall Problem Identified in the Introduction

The introduction can therefore be summarized as a chain of problems:



Traditional solutions have limitations:

OLTC / CB → **Frequent Switching and Wear**

Power Curtailment → **Reduced Renewable Generation**

Battery Storage → Higher Cost and Complexity

Therefore, the research investigates **coordinated reactive-power control of inverter-based DGs** as an alternative method for mitigating these fast voltage fluctuations.

7.9 Significance of the Proposed Research

The significance of this research is that modern distribution networks are increasingly incorporating renewable DGs, particularly solar PV systems.

As the penetration of these sources increases, maintaining acceptable voltage levels becomes more challenging because renewable generation can vary rapidly.

The proposed cooperative-control strategy attempts to make use of the **existing reactive-power capability of DG inverters** rather than depending solely on mechanical voltage-regulation devices or reducing renewable active-power production.

Central Motivation of the Research

To mitigate fast voltage fluctuations caused by intermittent inverter-based distributed generation while avoiding:

1. excessive mechanical switching,
2. unnecessary renewable-power curtailment, and
3. additional energy-storage complexity.

This problem statement forms the foundation for the cooperative control methodology developed in the subsequent sections of the research.

The paper later extends the cooperative approach to **phase-level control** and incorporates both **voltage deviation** and **reactive-power utilization** into its optimization objective.

8 Review of Previous Research

Before proposing the cooperative control method, the paper reviews several existing studies related to **volt-var control**, **volt-watt control**, **distributed energy resources**, **voltage regulation**, **reactive-power sharing**, and **cooperative energy management**.

These previous studies demonstrate that several control strategies have already been investigated for mitigating voltage problems caused by distributed generation. However, they also reveal the need for further development of cooperative control methods for rapidly varying renewable generation.

8.1 Volt-Var-Watt Proportional Control

Reference [1] W. De Carvalho, A. Attarha, and H. R. Pota, “Local Volt-Var-Watt Control for Voltage Regulation in Distribution Networks,” in Asia Pacific Power and Energy Engineering Conference, APPEEC,(2022) developed a **volt-var-watt proportional controller** based on a linear model of the distribution network.

The main purpose of this controller was to determine suitable control coefficients that would maintain overall system stability while also optimizing local operation.

An important objective of this approach was to **reduce power losses in the distribution network**.

The basic idea behind volt-var-watt control is to adjust the reactive and/or active power of distributed energy resources according to the voltage condition of the network.

Therefore, when the voltage changes, the controller can modify the power output of the distributed resource to help maintain an acceptable voltage level.

Reference [1] – Main Contribution

The study demonstrates the importance of designing appropriate control coefficients for achieving both **voltage regulation** and **network-loss reduction**.

8.2 Volt-Watt Control

Reference [2] Y. Alsharief, “Optimization of volt-watt control parameters to maximize power injection by DERs,” in 2021 IEEE PES Innovative Smart Grid Technologies Conference - Latin America, I investigated a **volt-watt control method** for reducing voltage fluctuations.

Unlike volt-var control, which primarily uses reactive power, volt-watt control modifies the **active-power injection** of distributed energy resources.

The method proposed in [2] reduces voltage fluctuations by optimizing the active power injected by distributed energy resources.

The basic concept can be represented as:

Voltage Condition $\longrightarrow$ **Volt-Watt Controller** $\longrightarrow$ **Active-Power Adjustment**

Although active-power adjustment can be effective for voltage regulation, reducing active-power injection can mean that available renewable energy is not fully utilized.

This is an important consideration for renewable-energy-based DG systems.

Limitation: Active-power adjustment can reduce the utilization of available renewable energy.

8.3 Sensitivity-Based Volt-Var Control for Distributed PV

Reference [3] Q. Shi, W. Feng, Q. Zhang, X. Wang, and F. Li, “Overvoltage Mitigation through Volt-VAR Control of Distributed PV Systems,” in Proceedings of the IEEE Power Engineering Conference, performed a **simulation-based sensitivity study** of volt-var control for distributed photovoltaic systems.

The objective was to investigate how volt-var control could be used to mitigate voltage fluctuations produced by distributed PV generation.

The simulation model included several important components of a distribution system, including:

- ZIP loads,
- shunt capacitors,
- on-load tap changers (OLTCs), and
- distributed PV systems.

The inclusion of these components is significant because voltage regulation in a practical distribution network is affected by the interaction between **loads, capacitor banks, OLTCs, and distributed renewable generation**.

This study therefore examined volt-var control within a more complete distribution-system environment rather than considering PV generation independently.

Reference [3] – Key Point

The study investigated the ability of sensitivity-based volt-var control to mitigate voltage fluctuations in distribution systems containing distributed PV generation and conventional voltage-regulation equipment.

8.4 Centralized Volt-Var-Watt Control

Reference [4] G. Svenda, I. Krstic, S. Kanjuh, M. Jajcanin, and D. Vuletic, “Volt Var Watt Optimization in Distribution Network with High Penetration of Renewable Energy Sources and Electric Vehicles,” in IEEE PES Innovative Smart Grid Technologies Conference Europe, 2022 proposed a **centralized, automatic, real-time volt-var-watt control procedure**.

The method was designed for an **unbalanced distribution network** with high penetration of renewable energy sources and electric vehicles.

Its objectives included:

1. mitigating constraint violations, and
2. improving power quality.

The consideration of an unbalanced distribution system is particularly relevant because actual distribution networks do not always operate under perfectly balanced three-phase conditions.

The study therefore demonstrates the importance of considering network imbalance when developing voltage-control strategies.

Centralized Control

High Renewable Penetration → Real-Time Control → Constraint Mitigation → Improved Power Quality

8.5 Local Measurement-Based Droop Control

Reference [5] D. F. Lima, L. F. De Souza, and J. c. Vieira, “Impact of Photovoltaic Systems in Distribution Networks Using Droop Control,” in 2020 IEEE PES Transmission and Distribution Conference, investigated the impact of photovoltaic systems on a distribution network using **two local measurement-based droop-control methods**.

The study focused particularly on residential loads and attempted to reduce the **overvoltage problem** associated with photovoltaic generation.

The important feature of this approach is the use of **local measurements**.

Instead of requiring a large amount of information from a centralized controller, the controller can respond to the voltage measured locally.

This provides a relatively decentralized approach to voltage regulation.

Main Feature

Local voltage measurements are used to determine the control response, reducing the dependence on a centralized controller.

8.6 Distributed Generator Hosting Capacity

Reference [6] W. Hu, Y. Guo, J. Jing, and L. Chen, “Dynamic hosting capacity evaluation of DGs in active distribution network,” in 2020 12th IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), 2020, focused on **hosting capacity evaluation for distributed generators**.

Hosting capacity can be understood as the amount of distributed generation that a distribution network can accommodate while maintaining acceptable operating conditions.

The study proposed a **multi-level hosting-capacity evaluation indicator system** to determine the maximum hosting capacity of a distribution network and encourage efficient utilization of DGs.

The authors also developed an evaluation model based on **optimal power flow (OPF)** to identify weak factors in a typical regional grid.

Hosting Capacity

Hosting capacity represents the amount of distributed generation that can be connected to a distribution network while maintaining acceptable operating conditions.

This research is important because simply adding more DGs to a distribution system is not sufficient. The network must have adequate capacity to accommodate the additional generation without violating voltage and other operating constraints.

8.7 Stochastic Cooperative Energy Management

Reference [7]"V. Khaligh, A. Ghezelbash, J. Liu, and J. H. Ryu, "Cooperative energy management of a microgrid with bidirectional electricity-hydrogen energy conversion," in 2022 14th IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), 2022" proposed a **stochastic co-operative energy-management model** for a microgrid containing bidirectional electricity-hydrogen conversion.

Unlike a deterministic approach, a stochastic model considers uncertainties associated with different operating conditions.

The model considered uncertainty in:

- hydrogen-vehicle behavior,
- wind-power output,
- PV output,
- electrical demand, and
- electricity price.

This study demonstrates the importance of **cooperative energy management under uncertain renewable generation and demand conditions**.

Although the specific application differs from the voltage-control problem addressed in the present paper, the concept of cooperation under uncertainty is relevant to modern renewable-energy-based power systems.

Cooperation + Uncertainty

Renewable Generation + Demand + Energy Price $\rightarrow$ Cooperative Energy Management

8.8 Optimal Sharing of Active and Reactive Power

References [8] and [9] investigated the **optimal sharing of active and reactive power among distributed generators**.

This is particularly relevant to the present research because multiple DGs may be capable of providing reactive-power support.

If every DG operates independently, one DG may provide excessive reactive power while another available DG may remain underutilized.

Therefore, an appropriate sharing strategy is required to distribute the reactive-power requirement among available sources.

The studies in [8] and [9] contribute to the broader research area of **optimal power sharing and voltage regulation in systems containing multiple distributed generation sources**.

Main Idea

Multiple DGs $\rightarrow$ Optimal Power Sharing $\rightarrow$ Better Utilization of Resources

8.9 Overall Contribution of the Previous Studies

The studies reviewed in this section demonstrate that several different approaches have been investigated for addressing voltage problems associated with distributed generation.

Reference	Main Approach	Main Focus
[1]	Volt-var-watt proportional control	Voltage regulation, stability, and loss reduction
[2]	Volt-watt control	Active-power optimization for voltage fluctuation reduction
[3]	Sensitivity-based volt-var control	Voltage fluctuation mitigation in distributed PV
[4]	Centralized volt-var-watt control	Constraint mitigation and power quality
[5]	Local measurement-based droop control	PV-related overvoltage reduction
[6]	DG hosting-capacity evaluation	Maximum acceptable DG penetration
[7]	Stochastic cooperative energy management	Uncertainty in renewable generation, demand, and price
[8], [9]	Optimal active/reactive-power sharing	Sharing available DG power resources

The paper's literature review therefore shows that **voltage regulation using distributed energy**

resources is an established research area, with approaches ranging from local droop control and volt-var control to centralized optimization and cooperative energy management.

8.10 Motivation for the Present Research

Despite the development of these techniques, the paper identifies the need for further development of **cooperative control for fast voltage fluctuations caused by intermittent inverter-based DGs**.

The key motivation is that solar PV generation can change rapidly, for example because of changes in solar irradiance.

Conventional voltage-regulating equipment may not respond effectively to such rapid changes, while excessive operation of mechanical devices can increase wear and reduce their lifetime.

Research Gap

There is a need for a cooperative-control approach specifically capable of handling **fast voltage fluctuations caused by intermittent inverter-based DGs** while reducing unnecessary operation of conventional mechanical voltage-regulation equipment.

The paper builds upon earlier cooperative-control research in Reference [12].

That earlier work used cooperative control to determine the reactive-power output of DGs for voltage control, but it assumed a **balanced three-phase system** and therefore did not address voltage unbalance between phases.

Three Important Extensions Introduced by the Present Paper

1. Phase-Level Cooperative Control

Phase-level cooperative control is introduced so that voltage can be controlled separately for individual phases and voltage unbalance can also be addressed.

2. Expanded Objective Function

The objective function is expanded to minimize both **voltage deviation and reactive-power output**, thereby considering the potential aging effect of reactive-power operation on inverters.

3. Extended Communication

Communication is extended beyond DG and capacitor-bank nodes so that **non-DG and non-capacitor-bank nodes can also provide information** to the cooperative optimization process.

Conclusion of the Literature Review

Therefore, the literature review establishes the progression from conventional and local voltage-control techniques toward the **cooperative, phase-level reactive-power control strategy** developed in this research.

The previous studies provide the foundation for understanding volt-var control, volt-watt control, droop control, centralized optimization, hosting-capacity analysis, stochastic cooperative energy management, and optimal power sharing.

The present research builds upon these developments by focusing on **fast voltage fluctuations caused by intermittent inverter-based DGs** and by introducing a cooperative control framework that operates at the **phase level**.

9 Research Gap and Contributions of the Proposed Method

The previous studies discussed in the literature review have proposed several methods for voltage regulation in distribution systems. These include reactive-power control using energy storage systems, model predictive control, centralized voltage control, and cooperative control.

However, the authors identify several limitations in the existing approaches and develop an improved cooperative control strategy to address these limitations.

9.1 Reactive-Power Optimization Using Energy Storage

Reference [10]"L. Chen, H. Cui, G. Cui, B. Yang, M. Yang, and H. Liu, "Voltage optimization strategy of distribution network considering the reactive power of energy storage," in 2020 12th IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), 2020" proposed a voltage-regulation method based on **optimizing the reactive power of an energy storage system**.

The objective of this approach was to minimize voltage deviation while simultaneously considering the constraints associated with power flow and device operation.

The basic concept is:

Voltage Deviation → Optimization → Reactive Power from Energy Storage → Voltage Regulation

This approach demonstrates that reactive power from controllable devices can be effectively used to regulate distribution-system voltage.

However, an energy-storage-based solution requires additional storage equipment, which can increase system complexity and cost.

The present paper therefore investigates the use of reactive-power capabilities already available in **DG inverters and capacitor banks**.

Limitation:

Additional energy-storage equipment can increase system **complexity and cost**.

9.2 Model Predictive Control-Based Coordination

Reference [11] "A. Dutta, S. Ganguly, and C. Kumar, "Model predictive control based coordinated voltage control in active distribution networks utilizing OLTC and dstatcom," in 9th IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES 2020)," proposed a **model predictive control (MPC)-based coordinated voltage-control strategy** for a distribution network.

The approach coordinated different voltage-control resources, including:

- Distribution Static Synchronous Compensator (DSTATCOM),
- Active-power curtailment, and
- On-Load Tap Changer (OLTC).

The objective was to maintain the voltage magnitudes of the distribution network within acceptable limits.

Important Feature

Multiple voltage-control resources are coordinated rather than operated independently.

However, the present paper takes a different approach by developing a **cooperative optimization method specifically for coordinating reactive-power resources associated with DGs and capacitor banks**.

9.3 Earlier Cooperative Control Approach

Reference [12] "A. Maknouninejad and Z. Qu, "Realizing unified microgrid voltage profile and loss minimization: A cooperative distributed optimization and control approach" IEEE Transactions on Smart Grid*, vol. 10, no. 3, pp. 289–298, 2019. This is particularly important because it introduced a **cooperative control method for determining the reactive-power output of distributed generators for voltage control**. The fundamental idea behind cooperative control is that multiple DGs can work together instead of each DG independently deciding its reactive-power output.

Multiple DGs → Information Exchange → Coordinated Reactive-Power Control

However, the earlier method assumed a **balanced three-phase system**. This creates a limitation for practical distribution systems because distribution networks can experience voltage differences between phases. For example, in an unbalanced system:

$$V_A \neq V_B \neq V_C$$

Therefore, a control method that considers only the balanced three-phase condition may not be capable of independently addressing the voltage condition of each phase.

The authors identify this as one of the major research gaps and extend the cooperative-control concept to address it.

Research Gap

The earlier cooperative-control approach assumed a balanced three-phase system and therefore did not explicitly address phase-to-phase voltage unbalance.

10 Major Contributions of the Proposed Research

The paper identifies **three major improvements** over the previous cooperative-control approach.

10.1 Phase-Level Cooperative Control

The first major contribution is the development of cooperative control at the **phase level**.

Instead of treating the three phases as one balanced system, the proposed method can consider the voltage condition of individual phases.

This provides greater flexibility because the controller can attempt to achieve the desired voltage profile for each phase separately.

The concept can be represented as:

Phase A $\rightarrow$ Individual Control

Phase B $\rightarrow$ Individual Control

Phase C $\rightarrow$ Individual Control

This is particularly useful for an **unbalanced three-phase distribution system**.

The phase-level implementation can therefore provide two important benefits:

1. Better control of the voltage profile of each individual phase.

2. Reduction of voltage unbalance between the phases.

Phase-Level Advantage

Phase-level implementation provides control versatility and can help reduce voltage unbalance between phases.

10.2 New Objective Function

The second major contribution is the introduction of a **new objective function**.

The previous objective of cooperative voltage control was primarily concerned with achieving an appropriate voltage profile.

The proposed research expands this objective by considering not only voltage deviation but also **reactive-power output**.

The two objectives can be expressed conceptually as:

$$\text{Objective} = \text{Voltage Deviation} + \text{Reactive Power Utilization}$$

Objective 1: Minimize Voltage Deviation

The controller attempts to keep the voltage magnitude close to its desired or base value:

$$|V| \rightarrow V_b$$

where V_b represents the base voltage magnitude.

A smaller voltage deviation means a better voltage profile.

Objective 2: Minimize Reactive-Power Utilization

The controller also attempts to avoid unnecessary reactive-power generation or consumption.

This is important because excessive reactive-power operation results in increased electrical stress and utilization of the power-electronic components of the inverter.

The authors specifically consider the fact that additional reactive-power output can contribute to **accelerated inverter aging**.

Therefore, the controller must find a compromise:

$$\text{Good Voltage Regulation} + \text{Low Reactive-Power Utilization}$$

This makes the proposed control strategy more comprehensive than a controller that focuses exclusively on voltage deviation.

10.3 Weighting Factors

The proposed objective function allows different **weights** to be assigned to voltage deviation and reactive-power utilization.

For example, if greater importance is given to voltage regulation, the voltage-deviation weight can be increased.

Conversely, if minimizing reactive-power utilization is more important, the reactive-power weight can be increased.

Conceptually:

$$F = w_v(\text{Voltage Deviation}) + w_\alpha(\text{Reactive Power Utilization}) \quad (2)$$

where:

- F = total objective/cost function,
- w_v = weight associated with voltage deviation,
- w_α = weight associated with reactive-power utilization.

The paper later evaluates different combinations of these weights to investigate their effects on the control performance.

This weighting approach allows the proposed method to operate under different control priorities.

Effect of Weighting Factors

Higher w_v : Greater emphasis on voltage regulation.

Higher w_α : Greater emphasis on minimizing reactive-power utilization.

11 Cooperative Control of Capacitor Banks

The third major improvement involves the **capacitor banks (CBs)**.

Capacitor banks are capable of supplying reactive power and therefore can contribute to voltage regulation.

However, unlike inverter-based DGs, capacitor banks generally operate through **switching discrete capacitor units**.

Frequent switching of capacitor banks can result in:

Frequent Switching $\rightarrow$ Mechanical/Electrical Wear $\rightarrow$ Higher O&M Cost $\rightarrow$ Reduced Equipment Lifetime

Therefore, the proposed method does not simply use capacitor banks whenever the voltage changes. It considers restrictions on capacitor switching.

The paper states that the switching of capacitor banks is regulated to **reduce operation and maintenance costs and increase equipment lifetime**, while considering switching-interval restrictions.

Thus, the proposed method coordinates the reactive-power contribution of both:

DG Inverters

and

Capacitor Banks

while attempting to avoid unnecessary capacitor switching.

12 Communication with Non-DG and Non-CB Nodes

Another important contribution is the expansion of the **communication network**.

In a distribution system, not every node contains a reactive-power source.

Some nodes may contain only loads or other network components.

The proposed method therefore allows communication not only between DGs and capacitor banks but also between reactive-power-source nodes and **non-DG/non-CB nodes**.

The purpose is to allow information from these nodes to be incorporated into the optimization process.

The communication structure can be represented conceptually as:

QG Nodes $\longleftrightarrow$ QG Nodes

and

Non-QG Nodes $\rightarrow$ QG Nodes

where QG nodes are nodes containing reactive-power sources.

According to the paper, QG nodes can exchange information with other QG nodes, while non-QG nodes can provide information to QG nodes.

This means that the controller does not need to consider only the voltage at the locations where DGs or capacitor banks are installed.

Information from other relevant locations in the distribution network can also influence the optimization.

13 Overall Research Gap

The research gap identified by the authors can therefore be summarized as follows:

Existing Limitation	Proposed Improvement
Earlier cooperative control assumed a balanced three-phase system	Phase-level cooperative control
Primarily focused on voltage regulation	Voltage deviation + reactive-power utilization
Reactive-power operation can contribute to inverter aging	Reactive-power utilization is included in the objective
Capacitor switching can increase O&M costs and wear	Switching restrictions are considered
Communication was limited to reactive-power-source nodes	Information from non-DG/non-CB nodes is also considered

Therefore, the proposed research is not simply introducing another voltage-control technique.

It is **extending the existing cooperative-control concept to make it more suitable for practical, unbalanced distribution systems with intermittent renewable DGs.**

14 Overall Proposed Concept

The complete concept developed from these contributions can be represented as:

Overall Problem-to-Control Chain

Intermittent Solar DG



Rapid Active-Power Variation



Voltage Fluctuation



Phase-Level Cooperative Controller

The controller receives:

Local Information + Neighboring QG Information + Non-QG Information

and determines the appropriate reactive-power utilization:

$$\alpha_q$$

for the available DGs and capacitor banks.

The resulting reactive-power adjustment is then used to improve the voltage profile while avoiding unnecessary reactive-power utilization and excessive capacitor switching.

Fundamental Contribution of the Paper

A **phase-level cooperative reactive-power control strategy** that coordinates DG inverters and capacitor banks using exchanged network information, while simultaneously considering **voltage regulation, reactive-power utilization, capacitor switching restrictions, and the characteristics of unbalanced distribution systems.**

This forms the basis for the methodology presented in the next section of the paper.

15 Problem Formulation

15.1 Standard IEEE 13-Bus System

What is the Standard IEEE 13-Bus System?

The **IEEE 13-bus test feeder** is a standard benchmark distribution system developed by the **Institute of Electrical and Electronics Engineers (IEEE)** and widely used for testing and validating different distribution-system analysis and control techniques.

A test feeder can be understood as a **standardized representation of a practical electrical distribution network**.

Instead of developing a new distribution network model for every study, researchers can use the same IEEE test feeder to implement their proposed method and evaluate its performance under a consistent network configuration.

15.1.1 The 13 Principal Buses

The term **13-bus** refers to the 13 principal buses or nodes that form the standard feeder.

In the system considered in this study, the principal bus numbers are:

611, 632, 633, 634, 645, 646, 650, 652, 671, 675, 680, 684, 692

A **bus** represents an electrical connection point in the distribution network where components such as loads, lines, transformers, or other devices can be connected.

The IEEE 13-bus system is particularly useful for distribution-system studies because it represents several characteristics of an actual distribution network, including different feeder sections, loads, transformers, capacitor banks, and three-phase network conditions.

The system can therefore be used to study important distribution-network problems such as:

- **Voltage regulation**
- **Voltage variation**
- **Power flow**
- **Reactive-power control**
- **Unbalanced operation**

The IEEE 13-bus system provides a standardized test environment for evaluating distribution-system analysis and control techniques.

15.1.2 Role of the IEEE 13-Bus System in the Present Research

For the present research, the IEEE 13-bus feeder provides the **base electrical network** on which the proposed cooperative-control method is implemented.

The general concept can be represented as:

Standard IEEE 13-Bus Feeder → Apply Proposed Control → Perform Simulation → Evaluate Voltage Profile

It is important to distinguish between the **standard IEEE 13-bus system** and the system used in this research.

The network shown in Figure 1 is **not the unmodified standard feeder**.

Instead, the standard feeder has been modified by the authors to introduce additional distributed-generation and reactive-power resources required for the cooperative-control study.

15.2 Modified IEEE 13-Bus System

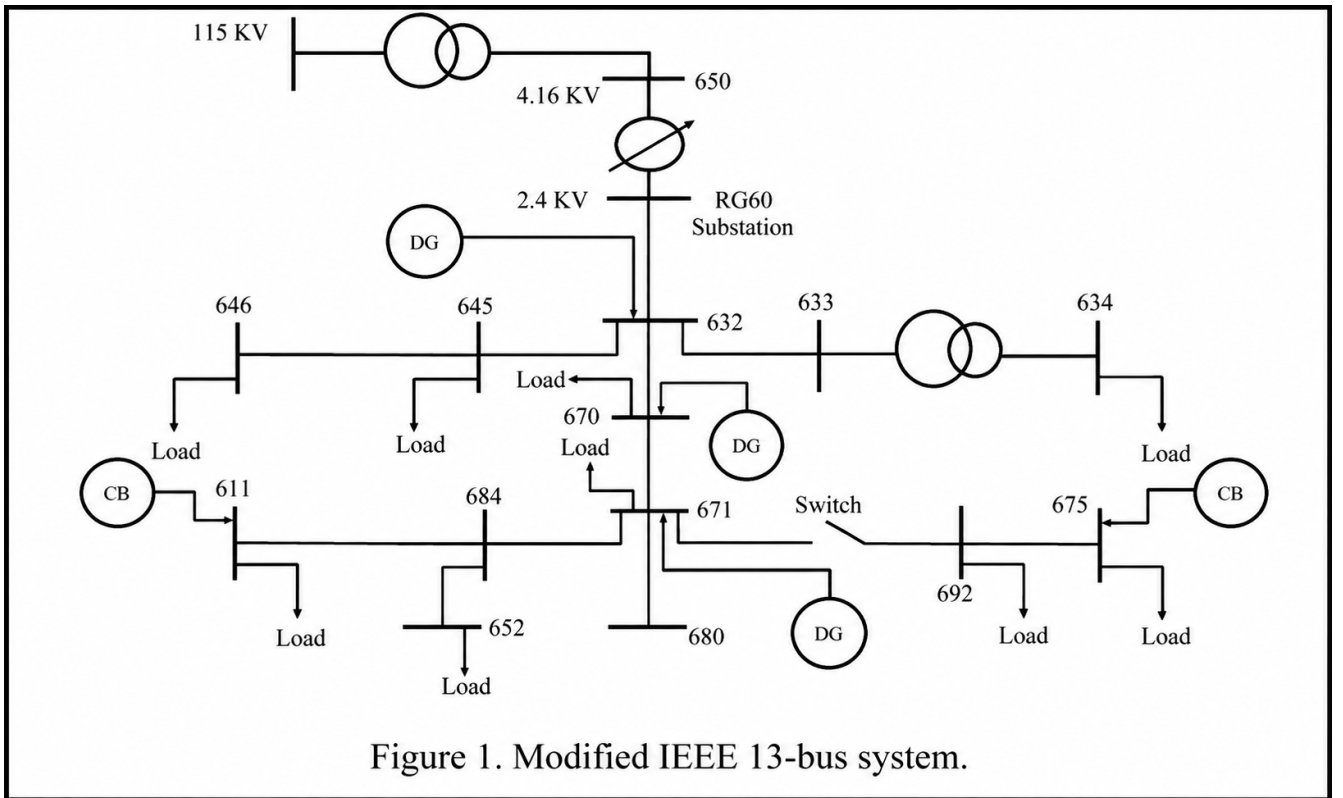


Figure 1: Modified IEEE 13-bus system.

For the implementation and evaluation of the proposed cooperative-control strategy, the authors use a **modified IEEE 13-bus distribution system**.

The modified system is shown in **Figure 1**.

The purpose of modifying the standard IEEE 13-bus feeder is to create a distribution network containing **distributed generators (DGs), capacitor banks (CBs), loads, and communication/control locations** that are suitable for studying voltage fluctuations and their mitigation.

According to the paper, the modified IEEE 13-bus system contains **41 nodes**.

The modification can be summarized as:

$$\text{Standard IEEE 13-Bus System} + \text{Single-Phase DGs} + \text{Modified CBs} = \text{Modified IEEE 13-Bus System}$$

15.2.1 Addition of Distributed Generators

The most important modification is the addition of distributed generators at three locations:

632, 670, 671

At each of these three nodes, three single-phase DGs are added.

Thus, the arrangement is:

632 → 3 single-phase DGs

670 → 3 single-phase DGs

671 → 3 single-phase DGs

The use of **single-phase DGs** is important because the proposed cooperative-control method is implemented at the **phase level**.

This allows the voltage and reactive-power behavior of the individual phases to be considered rather than treating the entire three-phase system as a single balanced system.

The DGs used in the study are assumed to be **solar PV generators**.

Their active-power output varies according to changes in solar irradiance.

Therefore, the modified system introduces the main disturbance being investigated:

Solar Irradiance Variation $\rightarrow$ PV Active-Power Variation $\rightarrow$ Voltage Fluctuation

15.2.2 Modification of the Capacitor Bank at Node 675

Another important modification is made at **node 675**.

In the standard IEEE 13-bus system, the capacitor bank connected at node 675 is a **three-phase capacitor bank**.

For the purpose of this research, it is converted into **three separate single-phase capacitor banks**.

Therefore, the modified arrangement can be represented as:

Node 675 $\rightarrow CB_A + CB_B + CB_C$

This modification is directly related to the **phase-level nature of the proposed controller**.

Instead of having one aggregated three-phase capacitor-bank representation, the reactive-power support can be considered individually for the three phases.

This provides additional flexibility when the system experiences unequal voltage conditions between phases.

15.2.3 Capacitor Bank at Node 611

The system also contains a **single-phase capacitor bank at node 611**.

The paper states that this capacitor bank already exists at node 611 in the system.

The capacitor banks are important because they provide **reactive-power support** to the distribution network.

Thus, the modified system has reactive-power resources in the form of:

DG Inverters

and

Capacitor Banks

These resources can be coordinated by the proposed cooperative-control strategy.

15.2.4 QG Nodes

The paper introduces the term **QG node** for nodes containing reactive-power sources.

Specifically:

Nodes on which DGs or CBs are connected are referred to as QG nodes.

Therefore, nodes containing the newly added DGs or the capacitor banks are treated as QG nodes.

The QG nodes are particularly important because they contain the resources that can directly influence the voltage through reactive-power control.

The main controllable resources are:

DG Inverter Reactive Power

and

Capacitor-Bank Reactive Power

The cooperative controller determines how these resources should be utilized.

15.2.5 Non-QG Nodes

All other nodes that do not contain DGs or capacitor banks are referred to as **non-QG nodes**.

These nodes do not directly provide the controllable reactive-power resources considered by the cooperative controller.

However, their voltage information is still important because changes in reactive power at one location can affect voltage at other locations throughout the feeder.

Therefore:

QG Reactive-Power Change $\rightarrow$ Network Voltage Change $\rightarrow$ Non-QG Voltage Change

The proposed method therefore also considers information from non-QG nodes during the cooperative optimization process.

The communication structure allows non-QG nodes to provide information to QG nodes.

15.2.6 Loads in the Modified System

The modified feeder contains several loads distributed across different nodes, as shown in Figure 1.

These loads represent the electrical demand supplied by the distribution network.

Their locations are important because voltage depends on the interaction between generation, load, line impedance, and reactive-power flow.

Therefore, a change in the reactive power supplied by a DG or capacitor bank at one location can influence the voltage at several other locations.

This makes the system suitable for studying **coordinated rather than independent voltage control**.

15.2.7 Overall Modification

The modifications made to the standard feeder can therefore be summarized as follows:

Component	Modification
Base network	Standard IEEE 13-bus feeder
DG locations	Nodes 632, 670 and 671
DG configuration	Three single-phase DGs at each location
DG technology	Solar PV
Capacitor bank at 675	Converted from three-phase to three single-phase CBs
Capacitor bank at 611	Existing single-phase CB
QG nodes	Nodes containing DGs or CBs
Non-QG nodes	Remaining nodes
Total nodes after modification	41

These modifications are explicitly described by the authors in the problem-formulation section.

15.2.8 Purpose of the Modified System

The modified IEEE 13-bus system is specifically constructed to reproduce the operating conditions required for the proposed cooperative-control study.

The basic problem is:

Solar PV Generation $\longrightarrow$ **Intermittent Active Power** $\longrightarrow$ **Voltage Fluctuations**

The modified network provides controllable reactive-power resources:

DG Inverters + Capacitor Banks

The cooperative controller can then coordinate these resources:

Network Information → Cooperative Control → Reactive-Power Adjustment → Voltage Regulation

Because the DGs and capacitor banks are represented at the single-phase level, the system is also suitable for studying **voltage differences between phases**.

15.2.9 Summary

In summary, the **standard IEEE 13-bus feeder** serves as the base distribution network for the research. The authors then modify this system by adding **three single-phase DGs at each of nodes 632, 670, and 671**. The DGs are modeled as solar PV generators whose active-power output varies with solar irradiance. The existing three-phase capacitor bank at node 675 is converted into three single-phase capacitor banks, while node 611 contains an existing single-phase capacitor bank. Nodes containing DGs or capacitor banks are classified as **QG nodes**, and the remaining nodes are classified as **non-QG nodes**. The resulting modified system contains **41 nodes**.

Summary of the Modified System

- **Base system:** Standard IEEE 13-bus feeder.
- **Total nodes after modification:** 41.
- **DG locations:** Nodes 632, 670, and 671.
- **DG configuration:** Three single-phase DGs at each location.
- **DG technology:** Solar PV.
- **Node 675:** Existing three-phase capacitor bank converted into three single-phase capacitor banks.
- **Node 611:** Existing single-phase capacitor bank.
- **QG nodes:** Nodes containing DGs or capacitor banks.
- **Non-QG nodes:** Remaining nodes.

This modified network provides the required test environment for the subsequent cooperative-control methodology, in which the reactive-power capabilities of DGs and capacitor banks are coordinated to mitigate voltage fluctuations in the distribution system.

15.3 Objective Function

The objective function is one of the most important components of the proposed cooperative-control strategy because it defines **what the controller is trying to optimize**.

In this research, the cooperative controller is designed to determine the appropriate reactive-power utilization of the available reactive-power sources so that the voltage profile of the distribution system is improved while unnecessary reactive-power utilization is avoided.

The overall objective is to **minimize the total cost function F** . The total cost consists of contributions from both **QG nodes** and **non-QG nodes**.

The total objective function is given by:

$$F = \sum_{i=1}^N f_i + \sum_{k=1}^M f_k \quad (1)$$

where:

- F = total objective/cost function,
- f_i = cost function associated with QG node i ,
- f_k = cost function associated with non-QG node k ,
- N = total number of QG nodes,
- M = total number of non-QG nodes.

Thus, the optimization considers the voltage conditions at **both types of nodes** rather than considering only the locations where reactive-power resources are installed.

15.3.1 Cost Function at QG Nodes

For a QG node i , the cost function is defined as:

$$f_i = w_{VQG} \left[\left| \frac{V_i}{V_{bi}} \right| - 1 \right]^2 + w_{\alpha} \alpha_{qi}^2 \bar{Q}_{Gi} \quad (2)$$

This equation contains **two separate terms**, and each term represents a different control objective.

Therefore,

$$f_i = \underbrace{w_{VQG} \left[\left| \frac{V_i}{V_{bi}} \right| - 1 \right]^2}_{\text{Voltage-deviation term}} + \underbrace{w_{\alpha} \alpha_{qi}^2 \bar{Q}_{Gi}}_{\text{Reactive-power utilization term}}$$

The first term attempts to keep the voltage close to its desired/base value, while the second term attempts to reduce unnecessary reactive-power utilization.

15.3.2 Voltage-Deviation Term at a QG Node

The first term in (2) is:

$$w_{VQG} \left[\left| \frac{V_i}{V_{bi}} \right| - 1 \right]^2$$

Here:

- V_i is the **complex voltage at QG node i** .
- $|V_i|$ is the magnitude of this complex voltage.
- V_{bi} is the **base voltage magnitude** associated with the DG or capacitor bank at QG node i .
- w_{VQG} is the weighting factor associated with voltage deviation at QG nodes.

The normalized voltage magnitude is:

$$\frac{|V_i|}{V_{bi}}$$

If the actual voltage magnitude is exactly equal to the base voltage,

$$|V_i| = V_{bi}$$

then:

$$\frac{|V_i|}{V_{bi}} = 1$$

and consequently:

$$\left[\frac{|V_i|}{V_{bi}} - 1 \right]^2 = 0$$

Therefore, the voltage-deviation cost becomes zero.

This means that the controller considers the base voltage magnitude as the desired reference and attempts to keep the actual voltage as close to this value as possible.

15.3.3 Why Is the Voltage Deviation Squared?

The deviation is squared:

$$\left[\frac{|V_i|}{V_{bi}} - 1 \right]^2$$

This has two important effects.

First, both overvoltage and undervoltage produce a **positive cost**.

For example, if:

$$\frac{|V_i|}{V_{bi}} = 1.05$$

then:

$$(1.05 - 1)^2 = 0.0025$$

Similarly, if:

$$\frac{|V_i|}{V_{bi}} = 0.95$$

then:

$$(0.95 - 1)^2 = 0.0025$$

Thus, both deviations are penalized.

Second, larger voltage deviations result in disproportionately larger costs. Therefore, the optimization gives greater importance to avoiding large deviations from the desired voltage.

15.3.4 Reactive-Power Utilization Term

The second term in the QG-node cost function is:

$$w_\alpha \alpha_{qi}^2 \overline{Q}_{Gi}$$

This term represents the cost associated with the **reactive-power utilization of the DG or capacitor bank at QG node i** .

The variables involved are:

- α_{qi} = reactive-power utilization ratio at QG node i ,
- $\overline{Q}_{Gi}$ = maximum available reactive power of the DG or capacitor bank at QG node i ,
- w_α = weighting factor associated with the level of reactive-power utilization.

The reactive-power utilization ratio indicates how much of the available reactive-power capability is being used.

According to the paper, a **positive value of α_{qi}** represents reactive-power injection, while a **negative value** represents reactive-power consumption.

Therefore:

$$\alpha_{qi} > 0 \Rightarrow \text{Reactive-power injection}$$

and:

$$\alpha_{qi} < 0 \Rightarrow \text{Reactive-power consumption}$$

15.3.5 Why Is α_{qi}^2 Used?

An important point in equation (2) is that the paper uses:

$$\alpha_{qi}^2$$

rather than simply α_{qi} .

This is necessary because the controller wants to minimize the **amount of reactive-power utilization**, regardless of whether the DG is injecting or consuming reactive power.

For example:

$$\alpha_{qi} = +0.5$$

represents reactive-power injection, whereas:

$$\alpha_{qi} = -0.5$$

represents reactive-power consumption.

If α_{qi} itself were used in the objective, the positive and negative values could have opposite effects on the cost.

By using:

$$\alpha_{qi}^2$$

both cases produce the same utilization penalty:

$$(+0.5)^2 = 0.25$$

and:

$$(-0.5)^2 = 0.25$$

Therefore, the optimization penalizes the **magnitude of reactive-power utilization**, irrespective of its direction.

The paper uses this term because excessive reactive-power operation can contribute to **accelerated inverter aging**. Therefore, the controller should provide enough reactive power to regulate voltage but should avoid unnecessary reactive-power output or consumption.

15.3.6 Role of the Weighting Factors

The objective function contains weighting factors that allow the relative importance of the different objectives to be adjusted.

For QG nodes, the two relevant weighting factors are:

$$w_{VQG}$$

and:

$$w_\alpha$$

where:

- w_{VQG} determines the importance assigned to voltage deviation at QG nodes.
- w_α determines the importance assigned to reactive-power utilization.

Therefore, the controller can prioritize either voltage regulation or reduced reactive-power utilization depending on the selected weighting values.

Conceptually:

Objective = Voltage Regulation Priority + Reactive-Power Utilization Priority

The paper later investigates different weighting combinations to study their influence on the resulting voltage profile and reactive-power utilization.

15.3.7 Cost Function at Non-QG Nodes

For a non-QG node k , the cost function is given by:

$$f_k = w_{VNQG} \left[\left| \frac{V_k}{V_{bk}} \right| - 1 \right]^2 \quad (3)$$

where:

- V_k = complex voltage at non-QG node k ,
- $|V_k|$ = magnitude of the voltage at non-QG node k ,
- V_{bk} = base voltage magnitude at non-QG node k ,
- w_{VNQG} = weighting factor associated with voltage deviation at non-QG nodes.

Notice an important difference between equations (2) and (3).

Equation (2) contains:

Voltage deviation + Reactive-power utilization

whereas equation (3) contains only:

Voltage deviation

The reason is that the non-QG nodes do not contain the DG or capacitor-bank reactive-power resources being directly controlled by the optimization.

Therefore, there is no corresponding α_{qk} reactive-power utilization term for a non-QG node.

15.3.8 Voltage Objective at Non-QG Nodes

The objective at a non-QG node is therefore simply to keep its voltage magnitude close to the corresponding base voltage.

If:

$$|V_k| = V_{bk}$$

then:

$$\frac{|V_k|}{V_{bk}} = 1$$

and:

$$f_k = 0$$

Thus, the controller is encouraged to maintain acceptable voltage not only at QG nodes but also at **non-QG nodes throughout the distribution network**.

This is important because reactive-power control at a QG node can influence the voltage of other nodes through the electrical network.

For example:

Reactive Power at QG Node $\longrightarrow$ Power Flow Changes $\longrightarrow$ Voltage Changes at Non-QG Nodes

Consequently, the optimization must consider the voltage effects beyond the locations where reactive-power sources are installed.

15.3.9 Meaning of the Base Voltage

The base voltage terms are:

$$V_{bi}$$

for a QG node and:

$$V_{bk}$$

for a non-QG node.

These values provide the voltage reference against which the actual voltage is evaluated.

The normalized voltage deviation is therefore calculated as:

$$\left| \frac{|V|}{V_b} - 1 \right|$$

This formulation makes the voltage deviation dimensionless and allows the optimization to evaluate voltage deviation relative to the corresponding base voltage.

15.3.10 Complete Interpretation of the Objective Function

Combining equations (1), (2), and (3), the optimization problem can be understood as having **three major considerations**.

1. Voltage regulation at QG nodes

$$w_{VQG} \left[\left| \frac{V_i}{V_{bi}} \right| - 1 \right]^2$$

The controller tries to keep the voltage at locations containing reactive-power sources close to the desired/base voltage.

2. Reactive-power utilization at QG nodes

$$w_\alpha \alpha_{qi}^2 \overline{Q}_{Gi}$$

The controller attempts to avoid unnecessary reactive-power utilization because excessive reactive-power operation can contribute to inverter aging.

3. Voltage regulation at non-QG nodes

$$w_{VNQG} \left[\left| \frac{V_k}{V_{bk}} \right| - 1 \right]^2$$

The controller also considers the voltage at locations that do not contain controllable DG or capacitor-bank resources.

Therefore, the overall optimization can be conceptually expressed as:

$$\begin{aligned}
\min F = & \underbrace{\text{Voltage deviation at QG nodes}}_{\text{Local voltage regulation}} \\
& + \underbrace{\text{Reactive-power utilization at QG nodes}}_{\text{Reduce unnecessary Q utilization}} \\
& + \underbrace{\text{Voltage deviation at non-QG nodes}}_{\text{Network-wide voltage regulation}}
\end{aligned}$$

15.3.11 Physical Meaning of the Optimization

The most important point is that the controller is **not simply trying to inject as much reactive power as possible**.

Instead, it must find an appropriate balance.

Suppose the voltage at a particular node is too low.

The controller can increase reactive-power injection from an available DG inverter or capacitor bank.

This can improve the voltage:

Low voltage $\rightarrow$ **Increase reactive-power support** $\rightarrow$ **Voltage improvement**

However, continuously increasing reactive power is not desirable because it increases the reactive-power utilization term:

$$\alpha_{qi}^2 \bar{Q}_{Gi}$$

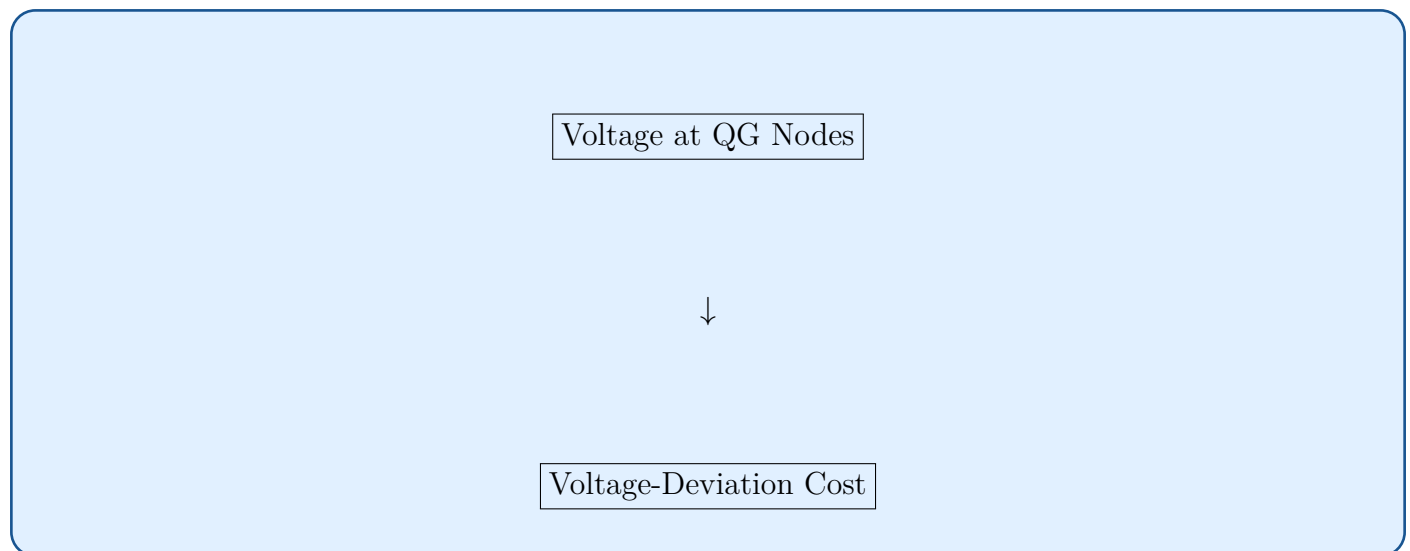
Therefore, the optimization searches for an appropriate compromise:

Sufficient Reactive Power **but** **Avoid Excessive Reactive Power**

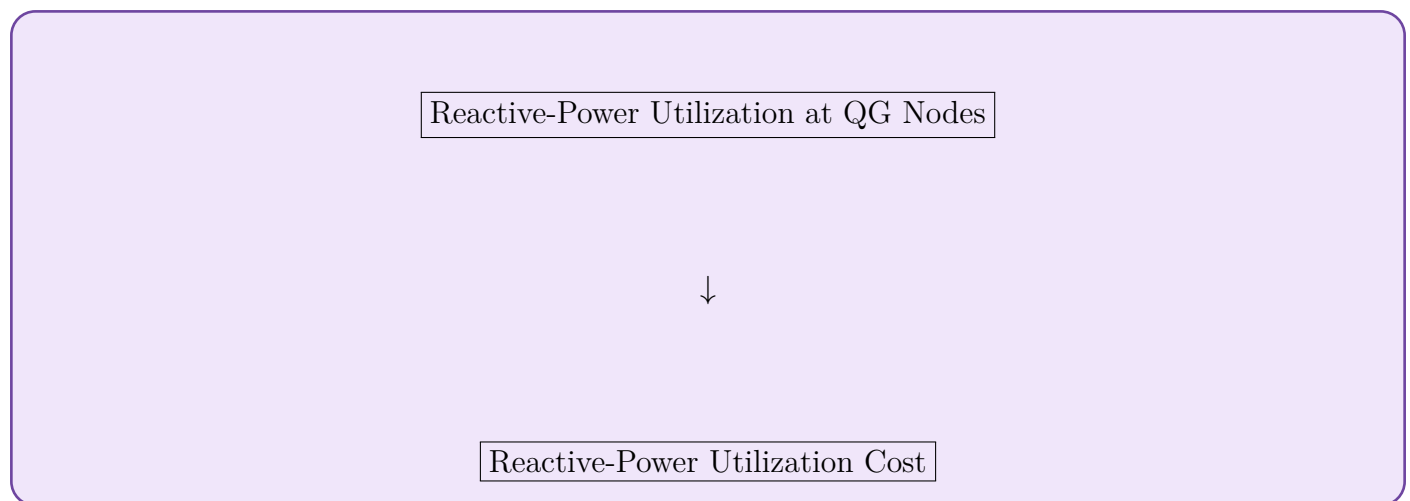
At the same time, the controller must ensure that the voltage conditions at other QG and non-QG nodes are also considered.

15.3.12 Overall Objective-Function Flow

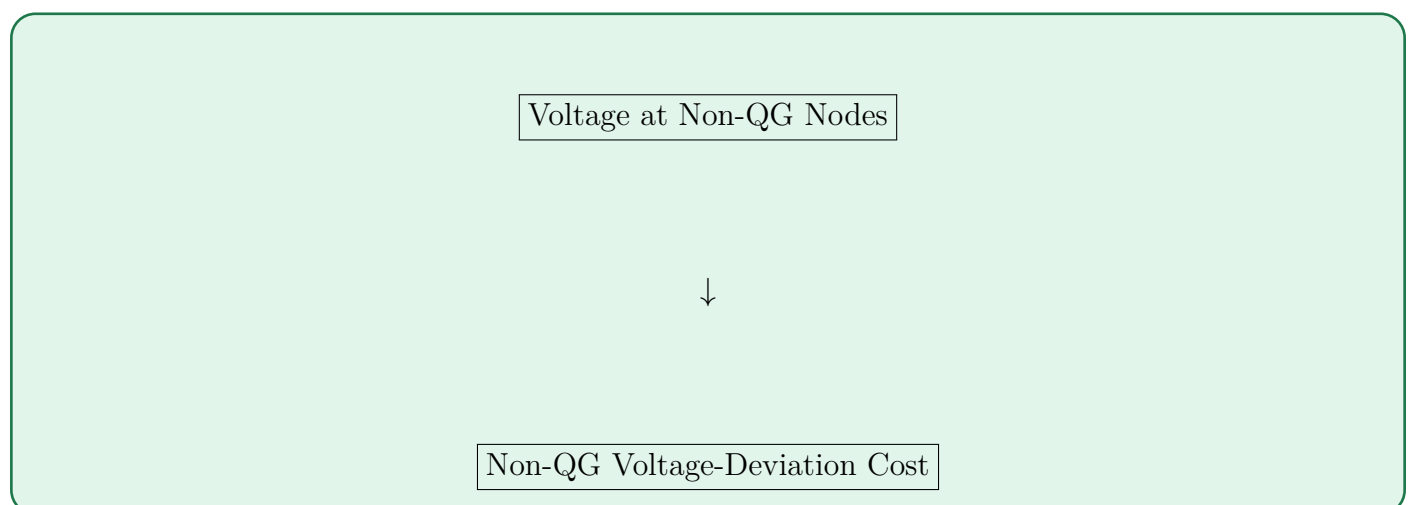
The complete concept can be represented as:



and simultaneously:



while:



All three contributions are combined:

$$F = \sum_{i=1}^N f_i + \sum_{k=1}^M f_k$$

and the cooperative controller attempts to find the control variables that **minimize** F .

summary of the Objective Function

The objective function of the proposed cooperative-control method is formulated to achieve an appropriate balance between voltage regulation and reactive-power utilization.

The total cost function consists of contributions from both QG and non-QG nodes.

At each QG node, the cost function contains a voltage-deviation term and a reactive-power-utilization term.

The voltage-deviation term penalizes deviations of the voltage magnitude from its base value, while the reactive-power term penalizes excessive utilization of the available reactive-power capability of the DG or capacitor bank.

The square of the reactive-power utilization ratio is used so that both reactive-power injection and consumption are penalized according to their magnitude.

At non-QG nodes, only the voltage-deviation term is considered.

Consequently, the optimization seeks to maintain voltages close to their desired values throughout the network while avoiding unnecessary reactive-power utilization at QG nodes.

This formulation allows the proposed cooperative controller to improve the voltage profile while also considering the impact of reactive-power operation on the utilization and aging of inverter-based resources.

15.4 Reactive Power Utilization Ratio

The **reactive power utilization ratio** is an important control variable in the proposed cooperative-control method. It represents the proportion of the available reactive-power capability of each **distributed generator (DG)** or **capacitor bank (CB)** that is actually being utilized for voltage regulation.

The paper denotes this parameter by:

$$\alpha_{qi}$$

where i represents the corresponding QG node.

The reactive-power utilization ratio is defined in (4) as:

$$\alpha_{qi} = \frac{Q_{Gi}}{\overline{Q}_{Gi}} \quad (4)$$

where:

- Q_{Gi} = reactive power **generated or supplied** by the i^{th} DG or CB,
- $\overline{Q}_{Gi}$ = **maximum available reactive power** of the i^{th} DG or CB,
- α_{qi} = reactive power utilization ratio.

15.4.1 Physical Meaning of α_{qi}

The ratio essentially answers the question:

What fraction of the maximum available reactive-power capability is currently being used?

For example, suppose a DG has a maximum available reactive-power capability of:

$$\overline{Q}_{Gi} = 100 \text{ kvar}$$

and the controller determines that it should provide:

$$Q_{Gi} = 50 \text{ kvar}$$

Then:

$$\alpha_{qi} = \frac{50}{100} = 0.5$$

Therefore, the DG is utilizing:

$$50\%$$

of its available reactive-power capability.

15.4.2 Interpretation of Different Values of α_{qi}

The value of α_{qi} indicates both the **magnitude and direction** of reactive-power operation.

Case 1: $\alpha_{qi} = 0$ If:

$$\alpha_{qi} = 0$$

then:

$$Q_{Gi} = 0$$

This means that the DG or CB is not supplying reactive power.

Case 2: $0 < \alpha_{qi} < 1$ A positive value indicates **reactive-power injection**.

For example:

$$\alpha_{qi} = 0.3$$

means that the resource is supplying 30% of its maximum available reactive-power capability.

$$Q_{Gi} = 0.3\overline{Q}_{Gi}$$

Similarly,

$$\alpha_{qi} = 0.8$$

means approximately 80% utilization in the injection direction.

Case 3: $\alpha_{qi} < 0$ A negative value represents **reactive-power consumption**.

For example:

$$\alpha_{qi} = -0.4$$

indicates that the resource is consuming reactive power corresponding to 40% of its available reactive-power capability.

Thus:

$$\alpha_{qi} > 0 \Rightarrow \text{Reactive-power injection}$$

and:

$$\alpha_{qi} < 0 \Rightarrow \text{Reactive-power consumption}$$

This sign convention is important because the controller may need either reactive-power injection or absorption depending on the voltage condition of the network.

15.4.3 Why Is the Reactive Power Utilization Ratio Used?

The purpose of introducing α_{qi} is to provide the cooperative controller with a convenient variable for determining **how much reactive power should be utilized by each DG or capacitor bank**.

The controller does not simply switch a DG between “reactive power ON” and “reactive power OFF.” Instead, the available reactive-power capability can be represented through the utilization ratio.

Therefore:

$$\alpha_{qi} \rightarrow \text{Determines the level of reactive-power utilization}$$

The cooperative-control algorithm optimizes α_{qi} so that the reactive-power resources installed at the QG nodes are used effectively for voltage regulation.

15.4.4 Relationship Between α_{qi} and Reactive Power

Equation (4) can also be rearranged to obtain the actual reactive power:

$$Q_{Gi} = \alpha_{qi} \overline{Q}_{Gi}$$

This equation is particularly useful for understanding the controller.

The optimization determines:

$$\alpha_{qi}$$

while the available capability is:

$$\overline{Q}_{Gi}$$

The resulting reactive-power output is then:

$$Q_{Gi} = \alpha_{qi} \overline{Q}_{Gi}$$

For example, if:

$$\overline{Q}_{Gi} = 80 \text{ kvar}$$

and:

$$\alpha_{qi} = 0.25$$

then:

$$Q_{Gi} = 0.25(80)$$

Therefore:

$$Q_{Gi} = 20 \text{ kvar}$$

Thus, the DG provides 20 kvar of reactive power.

15.4.5 Role in Cooperative Control

The paper states that the cooperative control aims to **fairly utilize the reactive power of DGs and CBs installed at QG nodes** in order to improve the voltage profile of the system.

This is an important point.

Suppose several DGs are available:

$$DG_1, \quad DG_2, \quad DG_3$$

and all of them have reactive-power capability.

If only DG_1 is used heavily while DG_2 and DG_3 are barely used, the available reactive-power resources may not be utilized efficiently.

The cooperative approach instead seeks an appropriate allocation of reactive-power utilization among the available resources.

Conceptually:

Multiple QG Nodes $\rightarrow$ Cooperative Optimization of $\alpha_q \rightarrow$ Reactive-Power Sharing

The resulting reactive-power allocation is intended to improve the overall voltage profile rather than focusing only on one particular node.

15.4.6 Connection with the Objective Function

The importance of α_{qi} becomes clearer when it is connected to the objective function discussed previously.

The QG-node cost function contains the term:

$$w_\alpha \alpha_{qi}^2 \overline{Q}_{Gi}$$

Therefore, α_{qi} is not only a measure of reactive-power utilization; it is also one of the **variables optimized by the cooperative controller**.

The optimization has to find a suitable value of α_{qi} that provides enough reactive-power support to regulate voltage without unnecessarily increasing reactive-power utilization.

Thus, there is a trade-off:

More Reactive Power $\rightarrow$ Potentially Better Voltage Regulation

but:

More Reactive Power $\rightarrow$ Higher Reactive-Power Utilization

The controller therefore seeks an appropriate balance between the two.

15.4.7 Determination of Maximum Available Reactive Power

The quantity

$$\overline{Q}_{Gi}$$

represents the **maximum available reactive power** of the DG or CB.

For a DG, this value is related to the generator's available operating capability.

The paper specifically states that the maximum available reactive power of a DG is calculated based on:

1. **The DG rating**, and
2. **The active power determined by solar irradiance**.

This is particularly important for the solar PV DGs used in the study.

The available reactive-power capability is therefore not necessarily constant throughout the simulation because the active power of the PV system changes with solar irradiance.

Conceptually:

$$\text{Solar Irradiance} \rightarrow P_{DG} \rightarrow \bar{Q}_{Gi} \rightarrow \alpha_{qi} \rightarrow Q_{Gi}$$

Therefore, changes in solar irradiance can affect not only the active-power generation of the PV system but also its available reactive-power capability.

15.4.8 Why Solar Irradiance Matters

The DGs in this study are solar PV generators. Solar PV output varies with solar irradiance.

When solar irradiance changes:

$$P_{DG} = P_{DG}(t)$$

also changes.

Since the available reactive-power capability of the DG is calculated based on its rating and active-power output, the available reactive-power capability can also change with operating conditions.

Therefore, the controller must continuously determine an appropriate reactive-power utilization level based on the current operating condition.

This makes the problem dynamic:

$$\text{Changing Solar Irradiance} \rightarrow \text{Changing } P_{DG} \rightarrow \text{Changing Reactive-Power Capability} \rightarrow \text{Changing Optimal } \alpha_q$$

15.4.9 Fair Utilization of DGs and Capacitor Banks

Another important purpose of the cooperative control is **fair utilization**.

The system may contain several reactive-power resources:

$$DG_{632}, \quad DG_{670}, \quad DG_{671}$$

along with capacitor banks.

Rather than independently controlling each resource, the cooperative method considers the resources collectively.

The objective is to determine suitable values of:

$$\alpha_{q1}, \alpha_{q2}, \dots, \alpha_{qN}$$

such that the overall system voltage profile is improved.

This cooperative allocation is particularly important because the effect of reactive power depends on the **location of the resource within the distribution network**. Reactive power supplied at one node may have a different influence on system voltage compared with the same amount supplied at another node.

15.4.10 Overall Mathematical Interpretation

The complete relationship can be summarized as follows.

Step 1: Determine Maximum Capability

$$\overline{Q}_{Gi} = \text{Maximum available reactive power}$$

Step 2: Optimize Utilization Ratio

$$\alpha_{qi} = \frac{Q_{Gi}}{\overline{Q}_{Gi}}$$

Step 3: Determine Actual Reactive Power

$$Q_{Gi} = \alpha_{qi} \overline{Q}_{Gi}$$

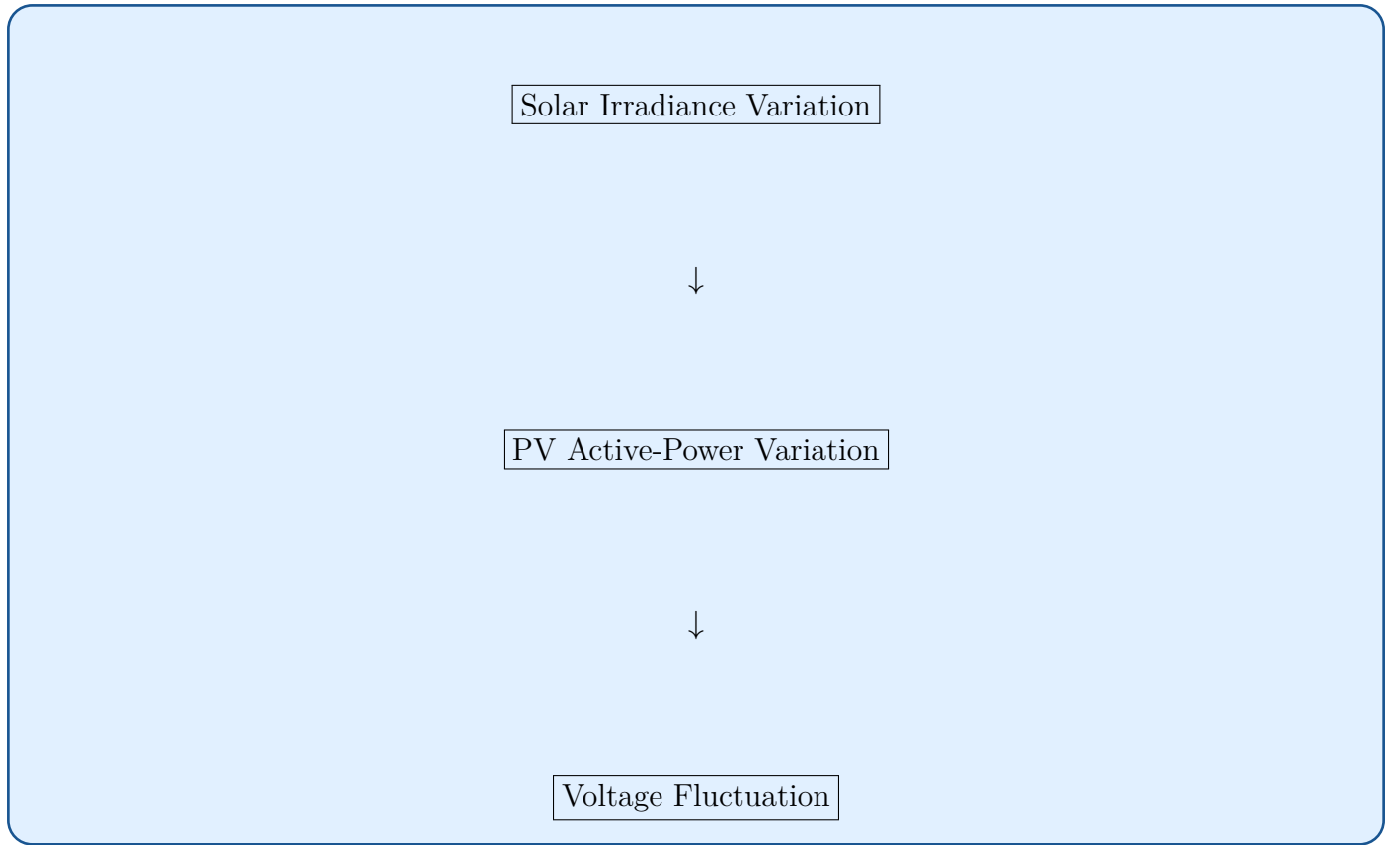
Step 4: Use Reactive Power for Voltage Regulation

$$Q_{Gi} \rightarrow \text{Voltage adjustment}$$

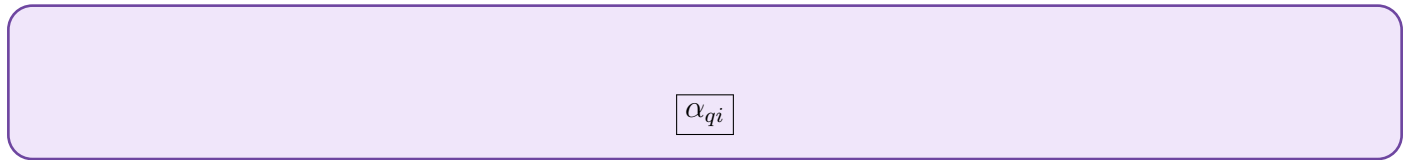
The cooperative controller searches for the appropriate α_{qi} values that minimize the total objective function while satisfying the system's operating requirements.

15.4.11 Relation to the Overall Research Problem

The entire purpose of introducing the reactive-power utilization ratio can be connected to the original voltage-fluctuation problem:

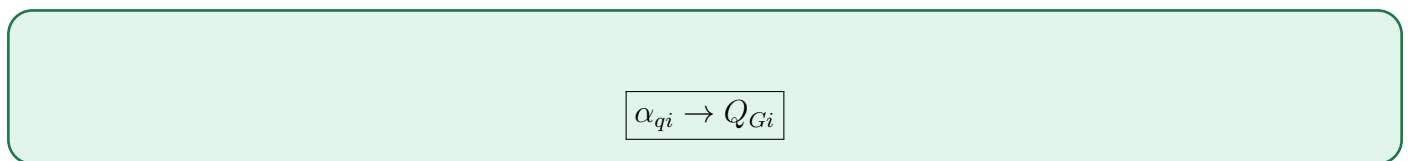


The proposed controller responds by determining:

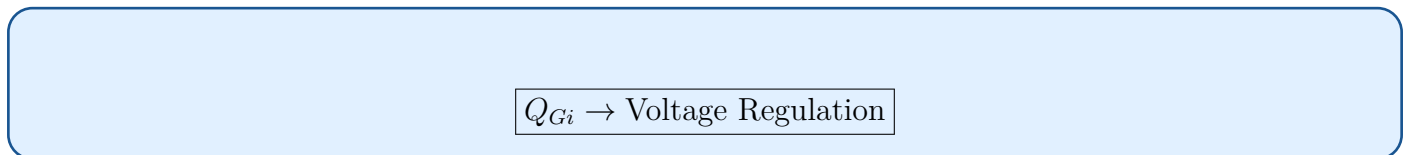


for the available DGs and capacitor banks.

This determines the required reactive-power response:



which is then used to improve the voltage profile:



At the same time, the objective function penalizes excessive reactive-power utilization, preventing the controller from simply maximizing reactive-power output.

summary of the Reactive-Power Utilization Ratio

The reactive power utilization ratio, denoted by α_{qi} , represents the proportion of the maximum available reactive-power capability of the i^{th} DG or capacitor bank that is utilized for voltage regulation.

It is defined as the ratio between the generated reactive power Q_{Gi} and the maximum available reactive power $\overline{Q}_{Gi}$.

A positive value of α_{qi} represents reactive-power injection, whereas a negative value represents reactive-power consumption.

The cooperative controller optimizes α_{qi} to fairly utilize the reactive-power capabilities of the DGs and capacitor banks installed at QG nodes and thereby improve the overall voltage profile. For DGs, the maximum available reactive power is determined from the DG rating and the active power produced according to the available solar irradiance.

Consequently, the reactive-power capability can vary with the operating condition of the solar PV system.

The utilization ratio therefore provides the key control variable through which the proposed cooperative strategy coordinates reactive-power resources while simultaneously limiting unnecessary reactive-power utilization.

15.5 Communication Topology

The **communication topology** defines how information is exchanged among the different nodes participating in the proposed cooperative-control algorithm. Since the optimal reactive-power utilization ratio α_{qi} is not obtained directly in a single step, the proposed method uses an **iterative optimization process**. During this process, each QG node uses its own local information together with information received from other QG and non-QG nodes.

The communication topology therefore determines **which nodes are allowed to exchange information with each other and in which direction the information flows**.

15.5.1 Purpose of the Communication Topology

The main purpose of the communication network is to allow the different QG nodes to **cooperate** while determining the optimal values of their reactive-power utilization ratios.

Recall that:

$$\alpha_{qi} = \frac{Q_{Gi}}{\overline{Q}_{Gi}}$$

is the reactive-power utilization ratio of a reactive-power source at QG node i .

The optimal value of α_{qi} depends not only on the voltage at that particular node but also on the voltage

conditions and reactive-power resources elsewhere in the distribution network.

Therefore, a QG node cannot necessarily determine the best value of α_{qi} using only its own local measurement.

Instead, the controller uses:

Local information + Information from neighboring QG nodes + Information from non-QG nodes

to perform the cooperative optimization.

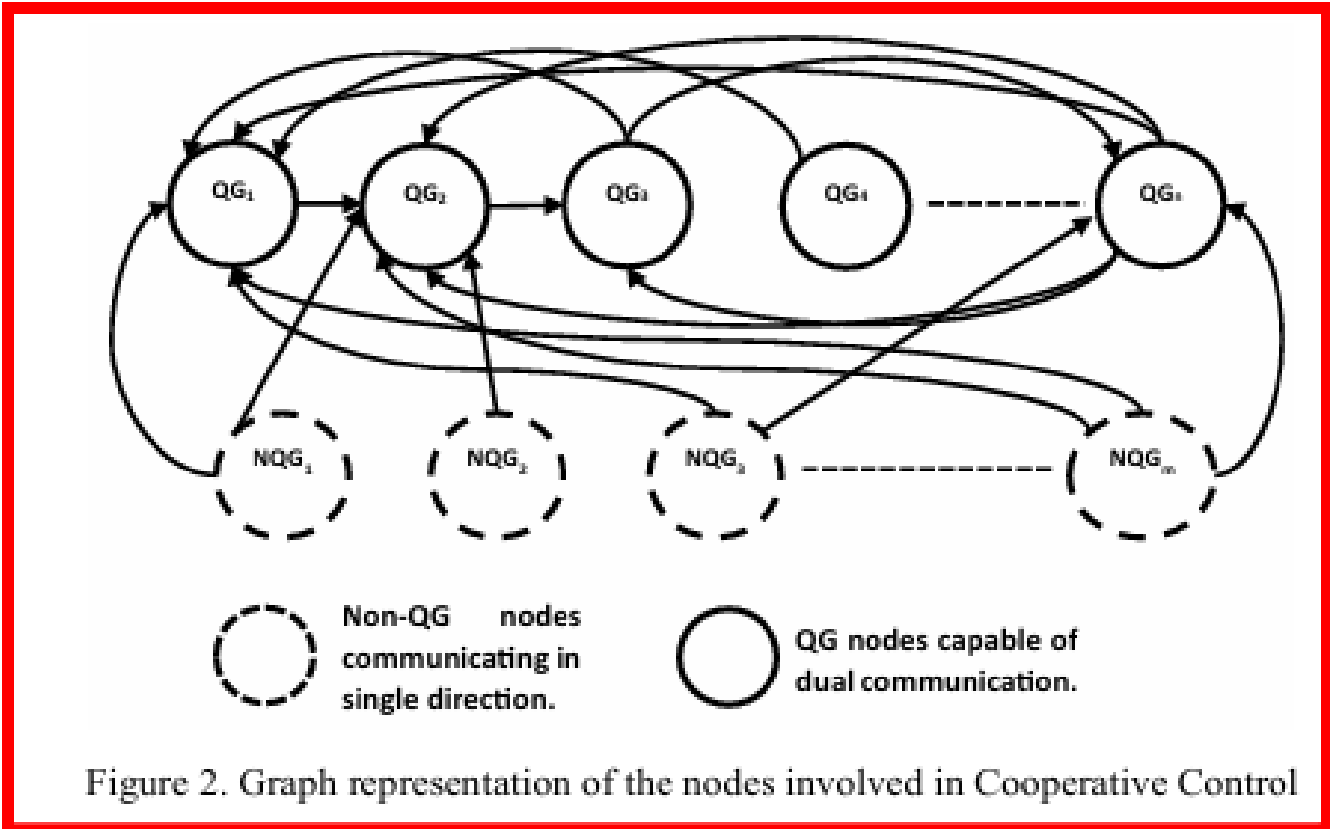


Figure 2: Graph representation of the nodes involved in Cooperative Control

15.5.2 QG Nodes and Non-QG Nodes

The communication network consists of two main types of nodes.

QG Nodes A **QG node** is a node at which a DG or capacitor bank is connected.

These nodes have reactive-power resources that can participate directly in the voltage-control process.

In the generalized communication graph shown in Figure 2, the QG nodes are represented by **solid circles**:

$QG_1, QG_2, QG_3, \dots, QG_n$

where n represents the total number of QG nodes participating in the cooperative control.

Non-QG Nodes The remaining nodes, which do not contain the reactive-power sources being controlled, are referred to as **non-QG nodes**.

In Figure 2, these nodes are represented by **dashed circles**:

$$NQG_1, NQG_2, NQG_3, \dots, NQG_m$$

where m represents the total number of non-QG nodes participating in the communication process.

Although these nodes do not directly control reactive-power resources, their voltage information can be important for determining the overall voltage condition of the distribution system.

15.5.3 Communication Between QG Nodes

The most important feature of the communication topology is the communication between QG nodes.

The QG nodes are capable of **dual communication**.

This means that a QG node can:

Send information

and:

Receive information

from another QG node when a communication link exists between them.

For example:

$$QG_1 \leftrightarrow QG_2$$

means that information can flow in both directions:

$$QG_1 \rightarrow QG_2$$

and:

$$QG_2 \rightarrow QG_1$$

Similarly, if a communication link exists between QG_2 and QG_3 , then:

$$QG_2 \leftrightarrow QG_3$$

This two-way communication enables the QG nodes to cooperate during the iterative optimization process.

15.5.4 Meaning of the Solid Arrows in Figure 2

In the figure, the **solid circles** represent QG nodes and the arrows between them represent communication links.

The presence of arrows in both directions indicates **dual communication**.

Therefore, the QG portion of the graph can be understood as:

$$QG_1 \leftrightarrow QG_2 \leftrightarrow QG_3 \leftrightarrow \cdots \leftrightarrow QG_n$$

The actual topology does not necessarily mean that every QG node communicates directly with every other QG node. Communication occurs according to the links defined in the topology.

This is important because the cooperative-control algorithm does not require every node to communicate with every other node.

15.5.5 Communication Between Non-QG and QG Nodes

The second type of communication is between **non-QG nodes** and **QG nodes**.

The non-QG nodes are represented by dashed circles in Figure 2.

Unlike QG nodes, the non-QG nodes can communicate in **only one direction**.

Specifically:

$$\text{Non-QG node} \rightarrow \text{QG node}$$

A non-QG node can therefore send its information to a QG node, but it does not receive information back from the QG node through this communication relationship.

For example:

$$NQG_1 \rightarrow QG_1$$

means that NQG_1 provides information to QG_1 .

Similarly:

$$NQG_2 \rightarrow QG_2$$

means that NQG_2 sends information to QG_2 .

15.5.6 Why Is Communication From Non-QG Nodes Needed?

Although non-QG nodes do not have controllable DGs or capacitor banks, the voltage at these nodes is still affected by the operation of the reactive-power sources located elsewhere in the network.

For example:

Reactive power change at QG node

can cause:



Change in power flow

which can result in:



Voltage change at a non-QG node

Therefore, ignoring the non-QG nodes could result in a controller that improves the voltage at QG locations while producing less desirable voltage conditions elsewhere in the feeder.

The proposed communication topology allows the voltage information from non-QG nodes to be incorporated into the optimization.

Thus:

Non-QG voltage information $\rightarrow$ QG node $\rightarrow$ Cooperative optimization

This allows the controller to consider a broader portion of the distribution system.

15.5.7 Local Information at Each QG Node

Each QG node first has access to its own **local information**.

This can include information associated with its local voltage and reactive-power resource.

For QG node i , the controller is concerned with quantities such as:

$$\boxed{V_i}$$

and:

$$\boxed{\alpha_{qi}}$$

as well as the available reactive-power capability:

$$\boxed{\overline{Q}_{Gi}}$$

Therefore, each QG node has information about its own operating condition.

However, local information alone is not sufficient because the distribution network is electrically interconnected.

15.5.8 Information From Neighboring QG Nodes

A QG node also receives information from its neighboring QG nodes.

For example, suppose QG_2 has communication links with QG_1 and QG_3 .

Then:

$$\boxed{QG_2}$$

can use its own information together with information received from:

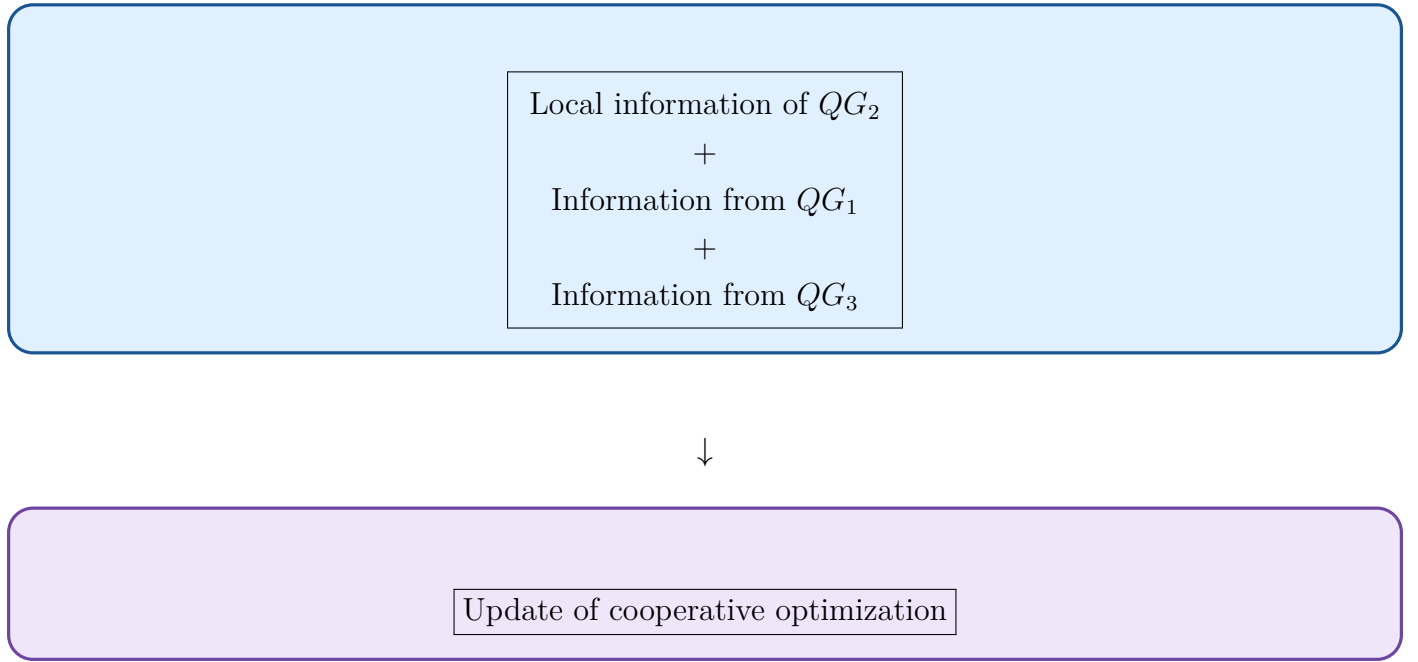
$$\boxed{QG_1}$$

and:

$$\boxed{QG_3}$$

The information is then used during the iterative cooperative-control process.

Conceptually:



This allows the control decision at one QG node to take the conditions of other QG nodes into account.

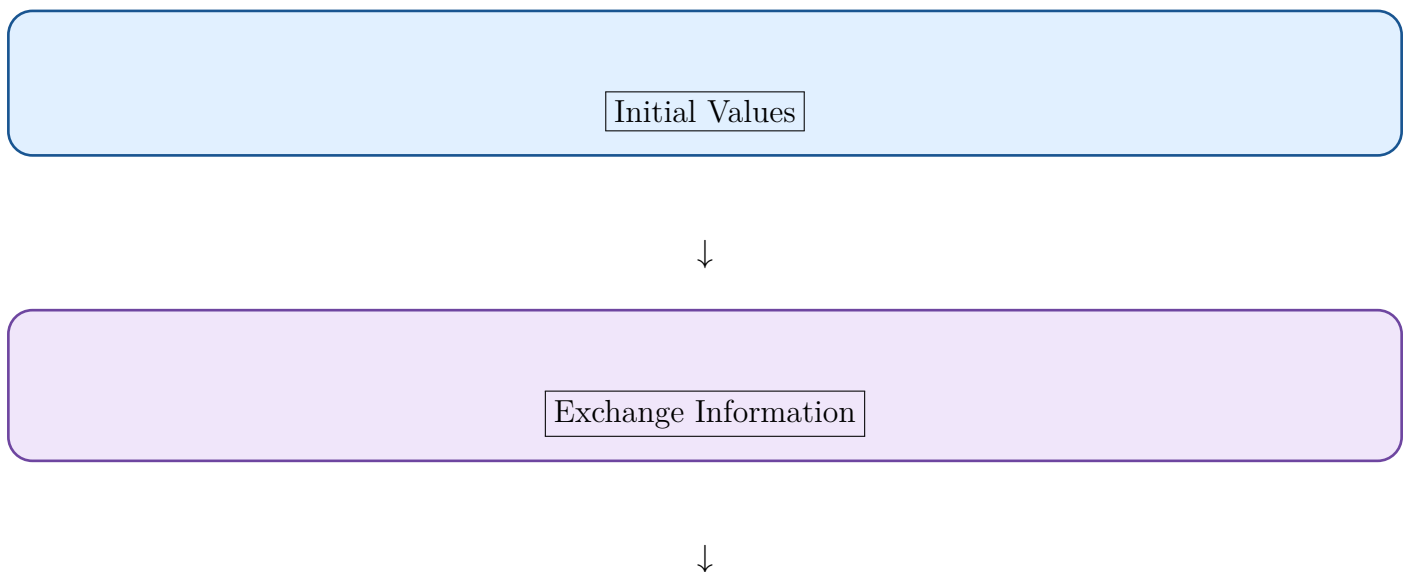
15.5.9 Iterative Optimization Process

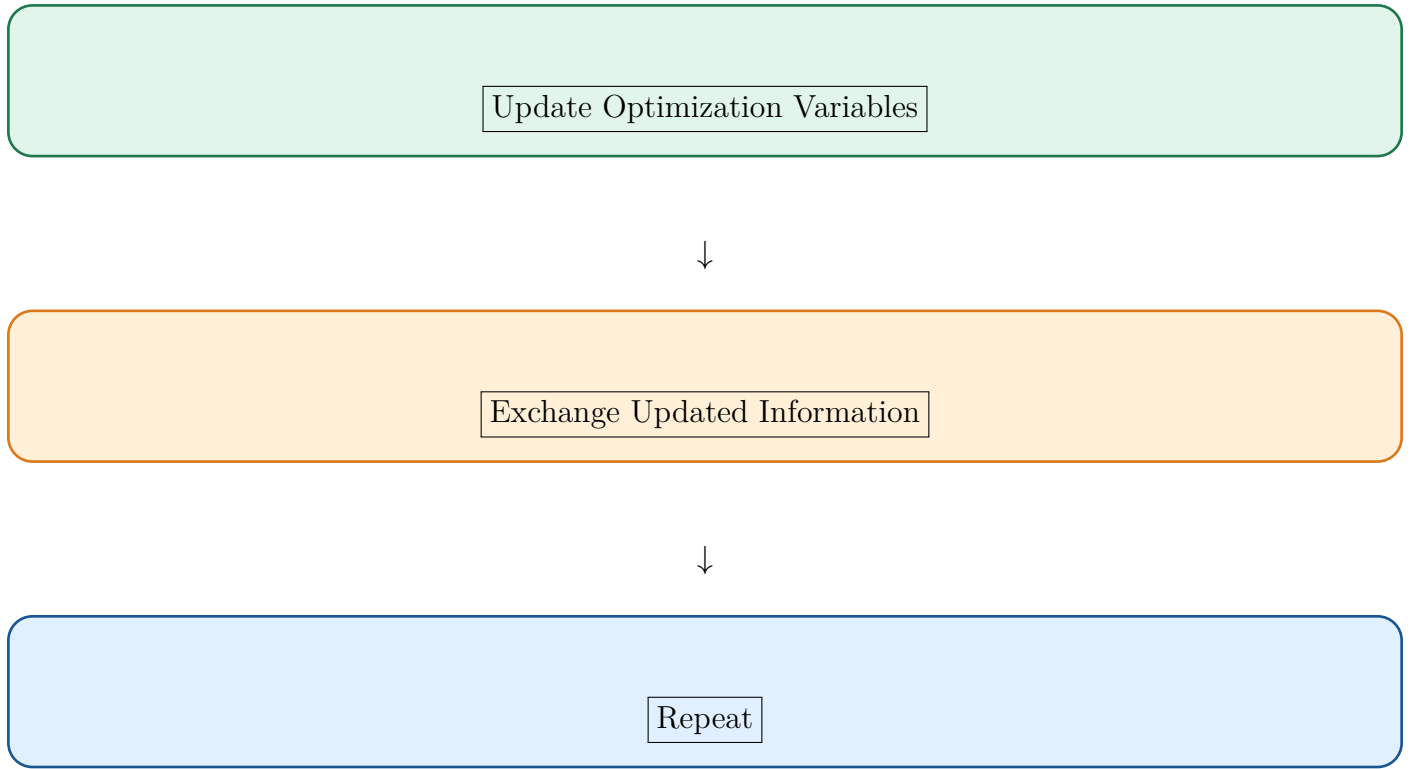
The paper states that the optimal α_{qi} is obtained through an **iterative process**.

This means that the final value is not determined immediately.

Instead, the QG nodes repeatedly exchange information and update the variables involved in the optimization.

The general process can be represented as:





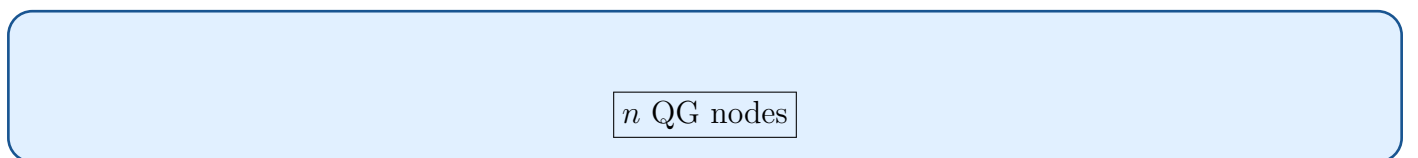
until the cooperative optimization reaches the required solution.

The important point is that the nodes **work together through information exchange** rather than independently optimizing their own reactive-power utilization.

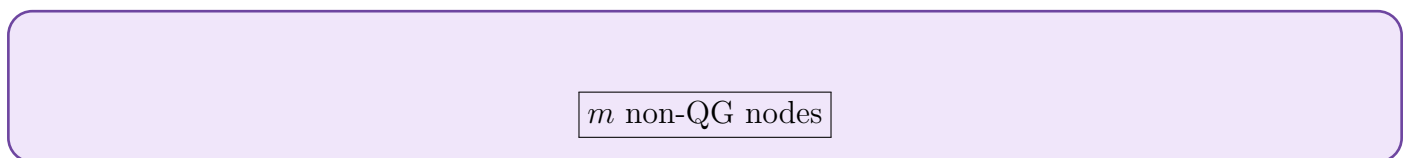
15.5.10 Generalized Graph Representation

Figure 2 represents the communication topology using a **graph representation**.

The graph consists of:



and:



participating in the cooperative control.

The QG nodes are shown as solid circles:

$$QG_1, QG_2, \dots, QG_n$$

while the non-QG nodes are shown as dashed circles:

$$NQG_1, NQG_2, \dots, NQG_m$$

The graph therefore provides a mathematical representation of the communication structure used by the cooperative-control algorithm.

15.5.11 Dual Communication vs. Single-Direction Communication

The distinction shown in Figure 2 can be summarized as follows:

Node relationship	Communication direction	Meaning
QG $\leftrightarrow$ QG	Two-way	QG nodes can send and receive information
Non-QG $\rightarrow$ QG	One-way	Non-QG node sends information to QG node
QG $\rightarrow$ Non-QG	Not shown as the permitted relationship	Non-QG node does not receive information through this link

Thus, the communication structure is **asymmetric** between QG and non-QG nodes.

The QG nodes form the main cooperative-control network, while non-QG nodes primarily act as **information providers**.

15.5.12 Communication Topology Matrices

The paper states that the communication topology matrices in equations (5) and (6) are derived from the generalized communication network shown in Figure 2.

These matrices provide a **mathematical representation of the communication relationships**.

In other words, Figure 2 provides the graphical representation:

Graph

while equations (5) and (6) provide the corresponding mathematical representation:

Communication matrices

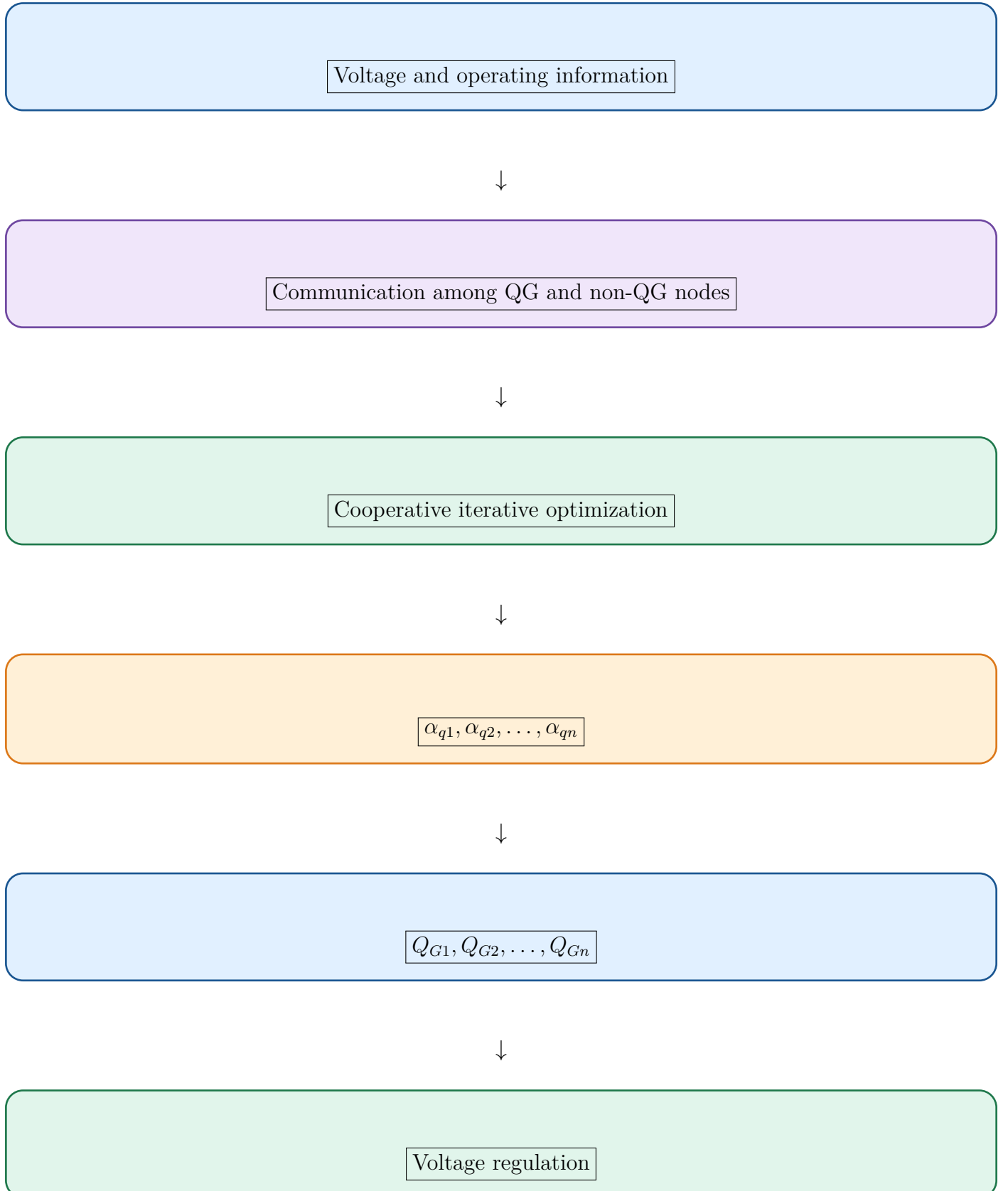
The matrices allow the communication relationships to be incorporated directly into the cooperative-control calculations.

They indicate which nodes are connected and how information is exchanged between them.

15.5.13 Relationship Between Communication and Reactive-Power Control

The communication topology is directly connected to the optimization of the reactive-power utilization ratio.

The overall process can be represented as:



Therefore, the communication network is what allows the reactive-power resources distributed across the feeder to operate **cooperatively**.

15.5.14 Importance of the Proposed Communication Structure

The communication topology is particularly important because the distribution system contains multiple reactive-power resources located at different electrical locations.

If each DG independently controlled its reactive power, the resulting control actions might not be optimal for the entire distribution system.

For example:

$$DG_1 \rightarrow Q_1$$

$$DG_2 \rightarrow Q_2$$

$$DG_3 \rightarrow Q_3$$

could be determined independently.

However, the proposed approach instead considers them cooperatively:

$$(Q_1, Q_2, Q_3, \dots, Q_n) \rightarrow \text{Joint Cooperative Optimization}$$

This allows the reactive-power contribution of one source to be considered together with the contributions of the other sources.

15.5.15 Interpretation of Figure 2

The figure can therefore be interpreted as follows.

At the top, the solid circles:

$$QG_1, QG_2, QG_3, \dots, QG_n$$

represent the **QG nodes**. These nodes are capable of two-way communication, as indicated by the communication arrows.

At the bottom, the dashed circles:

$$NQG_1, NQG_2, NQG_3, \dots, NQG_m$$

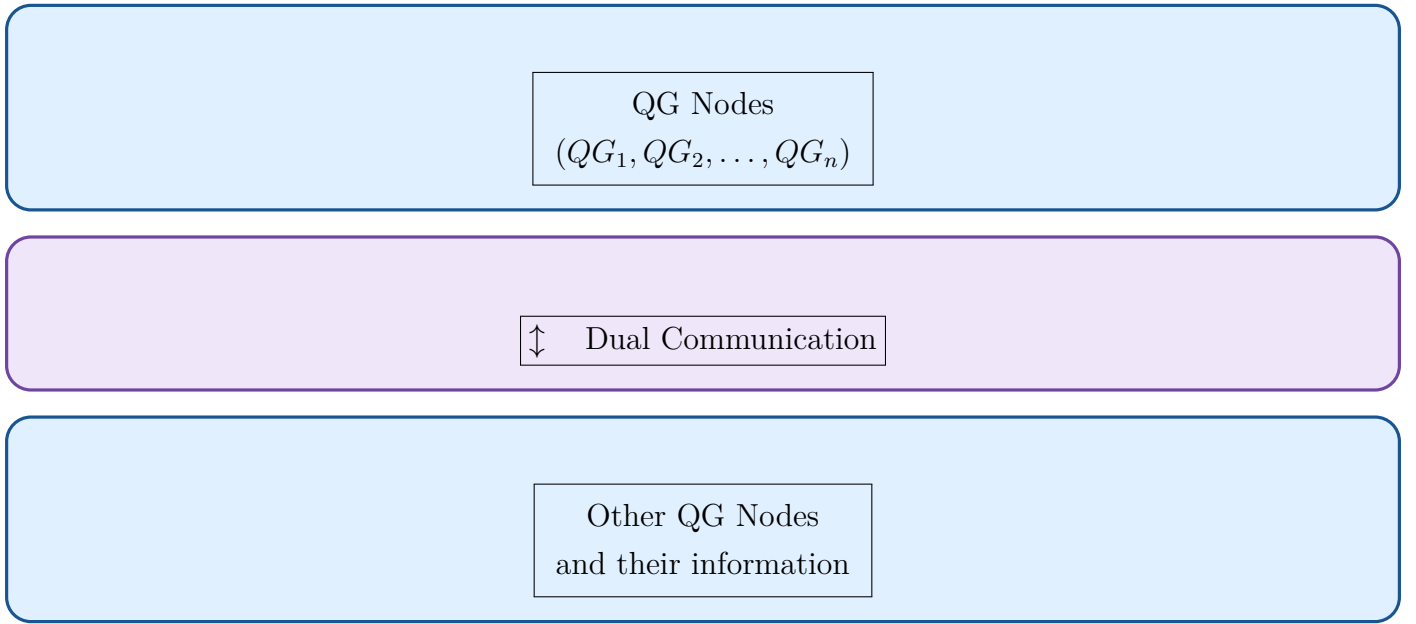
represent the **non-QG nodes**. These nodes provide information to the QG nodes through one-way communication links.

The multiple connections between QG nodes indicate that the cooperative-control algorithm allows information to propagate across the QG network.

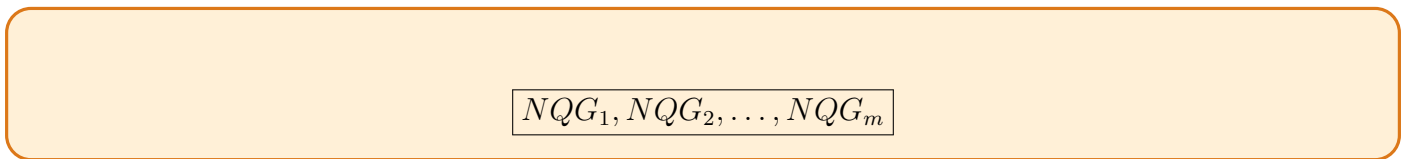
Therefore, the figure demonstrates that the proposed control system is **distributed and cooperative rather than completely centralized**.

15.5.16 Overall Communication Structure

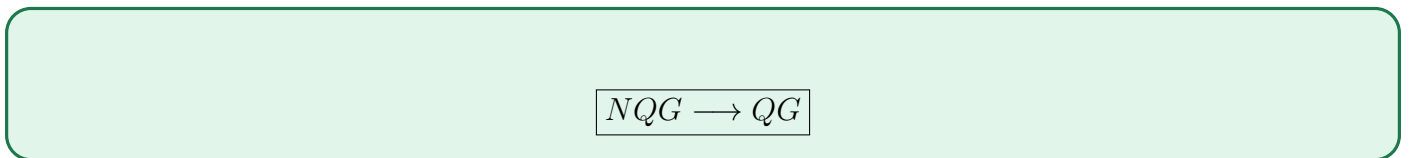
The complete communication concept can be summarized as:



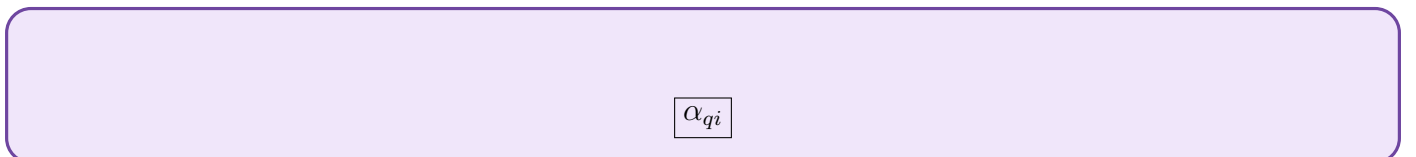
while:



provide information toward the QG network:



All of this information is used in the iterative optimization of:



which ultimately determines the reactive-power utilization of the DGs and capacitor banks.

summary of the Communication Topology

The communication topology defines the information-exchange structure used to obtain the optimal reactive-power utilization ratio α_{qi} through an iterative cooperative-control process. Each QG node uses its local information together with information received from neighboring QG nodes and non-QG nodes. The QG nodes are represented by solid circles in Figure 2 and are capable of dual communication, meaning that they can both send and receive information from other QG nodes when a communication link exists. The non-QG nodes are represented by dashed circles and can communicate only in a single direction, providing their information to the QG nodes. The generalized communication graph consists of n QG nodes and m non-QG nodes participating in the cooperative control. This communication structure enables information from different locations of the distribution network to be incorporated into the optimization rather than relying only on local measurements. The communication topology is mathematically represented through topology matrices, which are derived from the graph shown in Figure 2 and are subsequently used in the cooperative-control algorithm. Consequently, the proposed communication structure enables the DGs and capacitor banks at different QG nodes to coordinate their reactive-power utilization and collectively improve the voltage profile of the distribution system.

16 Methodology

The methodology describes the procedure used to implement the proposed **cooperative control strategy** for mitigating fast voltage fluctuations in a distribution system.

The fundamental problem considered is that the active-power output of distributed generators can vary rapidly, particularly for renewable sources such as solar PV. These active-power variations can produce voltage fluctuations in the distribution network.

The proposed methodology therefore establishes a **communication-based cooperative-control framework** in which information from different nodes is exchanged and used to coordinate the reactive-power control actions.

The first step of the methodology is to define the **communication topology matrices**.

16.1 Communication Between QG Nodes, QG and Non-QG Nodes

The first step is to define the communication relationships between the nodes participating in cooperative control.

There are two important types of nodes:

QG Nodes

QG nodes are the nodes associated with the available **reactive-power sources**, such as distributed generators (DGs) and capacitor banks (CBs), that participate directly in the cooperative control.

There are a total of n reactive-power sources participating in the cooperative control.

Therefore, the QG-node communication network can contain

$$QG_1, QG_2, QG_3, \dots, QG_n$$

The communication between these QG nodes is represented by the matrix d .

Non-QG Nodes

Non-QG nodes are nodes that do not directly contain the reactive-power sources being controlled. However, their voltage information can still be important because changes in reactive power at QG nodes can influence voltages at other locations in the distribution network.

The communication between QG nodes and non-QG nodes is represented by the matrix

$$d_{NQG}$$

There are m non-QG nodes involved in the cooperative control.

Therefore, the two matrices describe two different communication relationships:

$$d \rightarrow \text{Communication between QG nodes}$$

$$d_{NQG} \rightarrow \text{Communication between QG and non-QG nodes}$$

16.1.1 Communication Matrix Between QG Nodes

The communication between QG nodes is represented by the matrix d :

$$d = \begin{bmatrix} d_{11} & 0 & d_{13} & d_{14} & 0 & \cdots & d_{1n} \\ d_{21} & d_{22} & 0 & 0 & 0 & \cdots & d_{2n} \\ 0 & d_{32} & d_{33} & 0 & 0 & \cdots & d_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & d_{n3} & 0 & 0 & \cdots & d_{nn} \end{bmatrix} \quad (5)$$

This is an $n \times n$ matrix because there are n QG nodes and the matrix describes communication

relationships among those n QG nodes.

Therefore,

$$d \in \mathbb{R}^{n \times n}$$

16.1.2 Meaning of an Individual Element d_{il}

The element d_{il} represents the communication relationship between two QG nodes.

The direction of communication is given by

$$d_{il} \Rightarrow \text{Information flows from node } l \text{ to node } i$$

Therefore, the **first index** represents the receiving node, while the **second index** represents the sending node.

For example,

$$d_{21}$$

means that information is flowing from QG_1 to QG_2 :

$$QG_1 \rightarrow QG_2$$

Similarly,

$$d_{12}$$

represents information flowing from QG_2 to QG_1 :

$$QG_2 \rightarrow QG_1$$

This distinction is important because the communication relationship is directional.

16.1.3 Positive Value of d_{il}

A positive value of d_{il} indicates that a communication channel exists between the corresponding nodes.

Thus,

$$d_{il} > 0$$

↓

A weighted communication channel exists

The value of d_{il} acts as a weight associated with that communication relationship.

For example, if

$$d_{21} > 0,$$

then information flows from QG_1 to QG_2 :

$$QG_1 \rightarrow QG_2$$

and the information from QG_1 can be used by QG_2 in the cooperative-control process.

16.1.4 Zero Value of d_{il}

A zero element means that there is **no communication channel** between the corresponding nodes.

Therefore,

$$d_{il} = 0$$

↓

No communication from node l to node i

For example, if

$$d_{23} = 0,$$

then there is no communication channel represented from

$$QG_3 \rightarrow QG_2.$$

Therefore, the zero entries in the matrix identify the pairs of nodes that do not directly exchange information.

16.1.5 Diagonal Elements

The matrix contains diagonal elements such as

$$d_{11}, \quad d_{22}, \quad d_{33}, \dots, d_{nn}$$

These elements correspond to the respective QG nodes themselves.

For example,

$$d_{11}$$

corresponds to QG node 1, while

$$d_{22}$$

corresponds to QG node 2.

The diagonal elements are therefore included as part of the general communication-matrix structure.

16.1.6 Dimension of the QG Communication Matrix

The matrix d is an $n \times n$ matrix because there are n reactive-power sources participating in the cooperative control.

Thus,

$$d \in \mathbb{R}^{n \times n}$$

For example, if there are four QG nodes,

$$QG_1, \quad QG_2, \quad QG_3, \quad QG_4,$$

then the communication matrix has the form

$$d = \begin{bmatrix} d_{11} & d_{12} & d_{13} & d_{14} \\ d_{21} & d_{22} & d_{23} & d_{24} \\ d_{31} & d_{32} & d_{33} & d_{34} \\ d_{41} & d_{42} & d_{43} & d_{44} \end{bmatrix}.$$

Therefore, every row corresponds to a **receiving QG node**, while every column corresponds to a **transmitting QG node**.

16.1.7 Communication Matrix Between QG and Non-QG Nodes

The second communication matrix shown in the methodology is d_{NQG} :

$$d_{NQG} = \begin{bmatrix} d_{11} & 0 & d_{13} & \cdots & d_{1m} \\ d_{21} & d_{22} & 0 & \cdots & d_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & d_{n3} & \cdots & d_{nm} \end{bmatrix} \quad (6)$$

This matrix represents communication between non-QG nodes and QG nodes.

Unlike d , which is an $n \times n$ matrix, d_{NQG} is an $n \times m$ matrix.

There are:

$$n = \text{number of QG nodes}$$

and

$$m = \text{number of non-QG nodes.}$$

Therefore,

$$d_{NQG} \in \mathbb{R}^{n \times m}$$

16.1.8 Meaning of $d_{NQG,ik}$

An element $d_{NQG,ik}$ represents the communication relationship between non-QG node k and QG node i .

A positive value indicates that information flows from non-QG node k to QG node i :

$$d_{NQG,ik} > 0 \Rightarrow NQG_k \rightarrow QG_i$$

For example, if

$$d_{NQG,21} > 0,$$

then the communication relationship is

$$NQG_1 \rightarrow QG_2$$

16.1.9 Zero Value in d_{NQG}

Similarly, if

$$d_{NQG,ik} = 0,$$

then there is no communication channel between non-QG node k and QG node i .

Thus,

$$d_{NQG,ik} = 0 \Rightarrow \text{No communication from } NQG_k \text{ to } QG_i$$

The matrix therefore identifies which non-QG nodes provide information to which QG nodes.

16.1.10 Direction of Communication

The most important distinction between the two communication structures is their direction.

QG-to-QG Communication The QG nodes are capable of **dual communication**.

Conceptually,

$$QG_i \rightleftharpoons QG_l$$

This means that information can be exchanged in both directions when the corresponding communication channels exist.

For example,

$$QG_1 \rightarrow QG_2$$

and

$$QG_2 \rightarrow QG_1$$

can both exist.

Non-QG-to-QG Communication For the non-QG relationship, communication is in one direction:

$$NQG_k \rightarrow QG_i$$

The non-QG node provides information to the QG node.

This corresponds to the communication structure represented by d_{NQG} .

16.1.11 Why Is This Communication Necessary?

The cooperative controller is intended to determine appropriate reactive-power control actions for the different QG nodes.

However, a QG node should not consider only its own voltage condition.

The electrical network is interconnected, so the reactive-power action of one QG node can influence voltages at other nodes.

Therefore, the controller needs information from multiple locations.

The overall information flow can be represented as:

Local Information

↓

Communication Network

↓

Information Exchange



Cooperative Control



Reactive-Power Control

This communication structure allows the cooperative-control algorithm to use information from different parts of the network when determining the control actions.

16.1.12 Overall Meaning of Equations (5) and (6)

The two matrices can be interpreted as follows:

$d = \text{QG-to-QG communication topology}$

and

$d_{NQG} = \text{Non-QG-to-QG communication topology}$

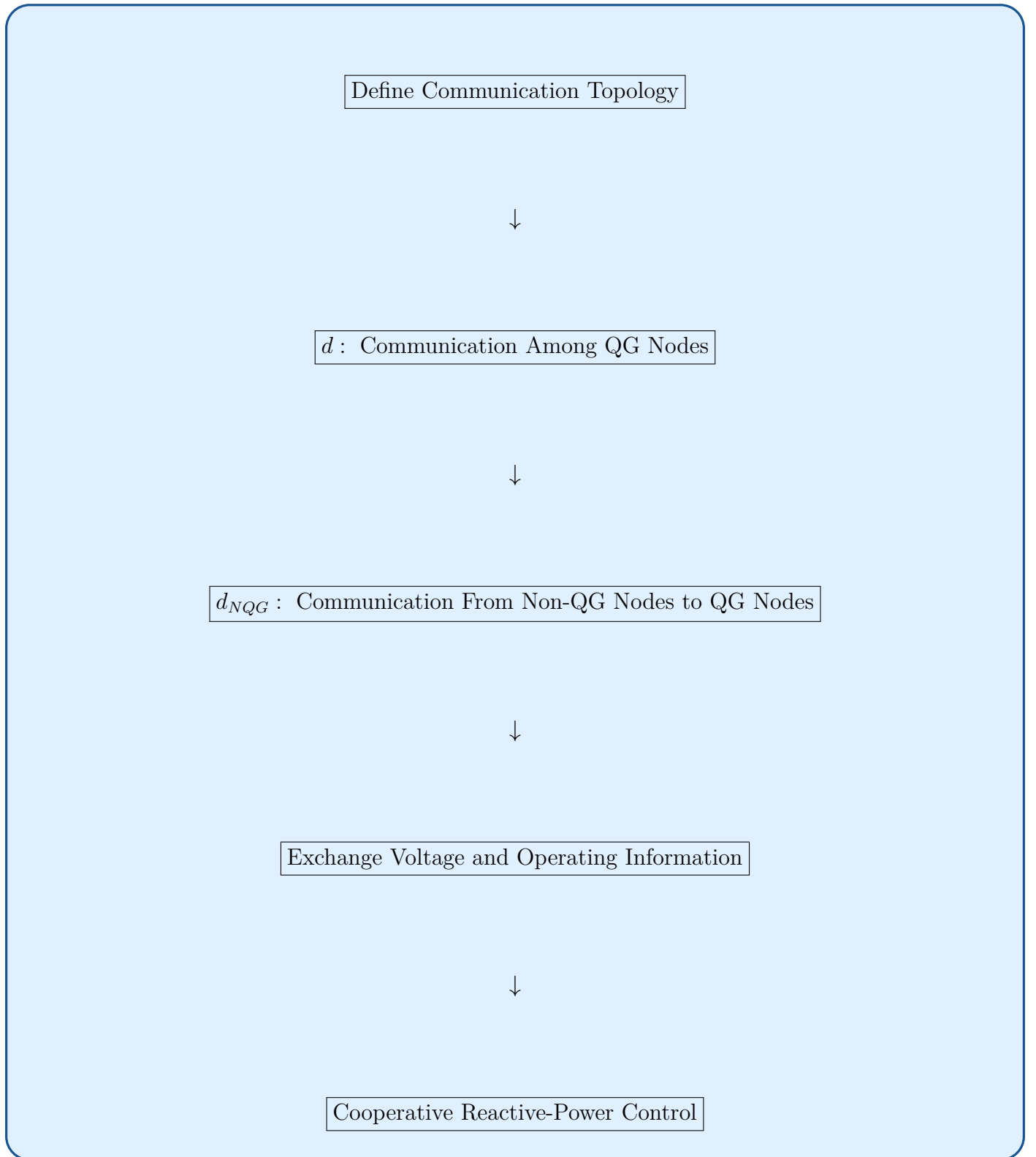
Their roles are different:

Matrix	Dimension	Communication
d	$n \times n$	QG $\rightarrow$ QG
d_{NQG}	$n \times m$	Non-QG $\rightarrow$ QG

A **positive element** means that a weighted communication channel exists, while a **zero element** means that there is no communication channel between the corresponding nodes.

16.1.13 Key Idea of the Communication Methodology

The complete communication process can be summarized as follows:



Methodology Interpretation

The first step of the proposed methodology is to establish the communication topology among the nodes involved in cooperative control. The matrix d represents communication among the QG nodes and is an $n \times n$ matrix, where n is the total number of reactive-power sources participating in the cooperative control. Each element d_{il} represents the communication relationship in which information flows from node l to node i . A positive value indicates the presence of a weighted communication channel, whereas a zero value indicates that no communication channel exists between the corresponding nodes.

The second matrix, d_{NQG} , represents communication between non-QG nodes and QG nodes. Since there are n QG nodes and m non-QG nodes, d_{NQG} is an $n \times m$ matrix. A positive value of $d_{NQG,ik}$ indicates that information flows from non-QG node k to QG node i , while a zero value indicates that no such communication channel exists.

The QG nodes are capable of dual communication, allowing information to be exchanged between connected QG nodes in both directions. In contrast, the non-QG nodes communicate toward the QG nodes in a single direction. This communication structure allows the cooperative controller to consider information from different locations of the distribution network when determining the reactive-power control actions. Thus, equations (5) and (6) establish the communication structure required for the subsequent cooperative-control and reactive-power optimization process.

16.2 Obtaining B_{ii} , B_{ik} , and G_{ik}

This part of the methodology explains how the required electrical-network parameters are obtained from the **bus admittance matrix**, Y_{BUS} . These parameters are needed later by the cooperative-control algorithm to describe the electrical relationship between **QG nodes** and **non-QG nodes**.

16.2.1 Why is Y_{BUS} Required?

The first step is to obtain the bus admittance matrix of the power network:

$$Y_{\text{BUS}}$$

The bus admittance matrix represents the electrical characteristics and interconnections of the distribution network.

In general, it can be written as:

$$Y_{\text{BUS}} = G + jB$$

where G is the conductance matrix and B is the susceptance matrix.

Thus,

$$G = \text{Conductance matrix}$$

and

$$B = \text{Susceptance matrix}$$

Therefore, an individual element of the bus admittance matrix can be expressed as:

$$Y_{ik} = G_{ik} + jB_{ik}$$

where G_{ik} is the real part and B_{ik} is the imaginary part of the corresponding admittance.

16.2.2 Obtaining B_{ii}

The parameter

$$B_{ii}$$

is the **imaginary component of the self-admittance** of QG node i .

The diagonal element of Y_{BUS} corresponding to node i is:

$$Y_{ii} = G_{ii} + jB_{ii}.$$

Therefore:

$$B_{ii} = \text{Im}(Y_{ii})$$

where $\text{Im}(\cdot)$ denotes the imaginary part.

The term **self-admittance** means that the parameter describes the electrical admittance associated with the node itself, including its connections to the rest of the network.

16.2.3 Phase-Level Consideration

If the proposed algorithm is implemented at the **phase level**, the individual phase quantities are considered separately.

However, if the algorithm is **not implemented at the phase level**, for simplification, B_{ii} is calculated as the sum of the imaginary components of the self-admittance of all phases.

Therefore, for a multi-phase node:

$$B_{ii} = \sum_{\phi} \text{Im} (Y_{ii}^{\phi})$$

where ϕ represents the phases associated with the node.

For example, for a three-phase node:

$$B_{ii} = B_{ii}^a + B_{ii}^b + B_{ii}^c$$

when the phase-level information is aggregated into a single value.

Thus, B_{ii} represents the combined imaginary self-admittance associated with QG node i under the simplified non-phase-level implementation.

16.2.4 Obtaining B_{ik}

The next parameter is:

$$B_{ik}$$

This parameter represents the **susceptance between a QG node i and a non-QG node k** .

The corresponding element of the bus admittance matrix is:

$$Y_{ik} = G_{ik} + jB_{ik}.$$

Therefore:

$$B_{ik} = \text{Im}(Y_{ik})$$

This means that B_{ik} is obtained by taking the **imaginary component of the admittance connecting QG node i and non-QG node k** .

The parameter therefore represents the reactive/electrical coupling between these two nodes:

$$\boxed{\text{QG node } i \longleftrightarrow \text{Non-QG node } k}$$

16.2.5 Obtaining G_{ik}

Similarly, the parameter

$$\boxed{G_{ik}}$$

represents the **conductance between QG node i and non-QG node k** .

From:

$$Y_{ik} = G_{ik} + jB_{ik},$$

the conductance is obtained from the real component:

$$\boxed{G_{ik} = \text{Re}(Y_{ik})}$$

where $\text{Re}(\cdot)$ represents the real part.

Therefore:

$$\boxed{G_{ik} = \text{Real component of the admittance between QG node } i \text{ and non-QG node } k}$$

16.2.6 Relationship Between the Three Parameters

The three quantities can be understood directly from the elements of Y_{BUS} .

For the self-admittance of QG node i :

$$\boxed{Y_{ii} \rightarrow B_{ii}}$$

For the admittance between QG node i and non-QG node k :

$$Y_{ik} \rightarrow \begin{cases} G_{ik} = \text{Re}(Y_{ik}) \\ B_{ik} = \text{Im}(Y_{ik}) \end{cases}$$

In compact form:

$$\begin{aligned} B_{ii} &= \text{Im}(Y_{ii}), \\ B_{ik} &= \text{Im}(Y_{ik}), \\ G_{ik} &= \text{Re}(Y_{ik}). \end{aligned}$$

16.2.7 Physical Interpretation

The three parameters have different meanings:

Parameter	Meaning	Obtained from
B_{ii}	Imaginary part of the self-admittance of QG node i	Y_{ii}
B_{ik}	Susceptance between QG node i and non-QG node k	Y_{ik}
G_{ik}	Conductance between QG node i and non-QG node k	Y_{ik}

Thus, the methodology takes the electrical-network information contained in Y_{BUS} and extracts the specific parameters required by the cooperative-control formulation.

16.2.8 Simple Numerical Example

Suppose an element of the bus admittance matrix is:

$$Y_{ik} = 0.05 - j0.20$$

Comparing this with:

$$Y_{ik} = G_{ik} + jB_{ik},$$

we obtain:

$$G_{ik} = 0.05$$

and:

$$B_{ik} = -0.20$$

Similarly, suppose the self-admittance of QG node i is:

$$Y_{ii} = 0.10 + j0.50$$

Then:

$$B_{ii} = 0.50$$

Thus, the required quantities are directly extracted from the appropriate elements of Y_{BUS} .

16.2.9 Overall Process in the Methodology

The procedure described in this subsection can be summarized as:

Power Network



Y_{BUS}



$Y_{ii} \rightarrow B_{ii}$
 $Y_{ik} \rightarrow B_{ik}, G_{ik}$



Parameters required by the cooperative-control algorithm

Key Interpretation

The parameters B_{ii} , B_{ik} , and G_{ik} are obtained from the electrical network model represented by the bus admittance matrix Y_{BUS} . The parameter B_{ii} represents the imaginary component of the self-admittance of QG node i , while B_{ik} and G_{ik} represent the susceptance and conductance, respectively, between QG node i and non-QG node k .

If the algorithm is not implemented at the phase level, B_{ii} is calculated as the sum of the imaginary components of the self-admittance of all phases. These network parameters provide the electrical information required by the subsequent cooperative-control formulation.

16.3 Obtaining Reactive Power and Voltage

This subsection describes the next step in the methodology: obtaining the **reactive power supplied by the reactive-power sources** and the corresponding **voltage measurements at the network nodes**. These quantities are required by the cooperative-control algorithm because the controller uses the measured reactive power and voltage conditions to determine the appropriate control actions.

16.3.1 Reactive Power Injected by a DG or Capacitor Bank

The first quantity that needs to be obtained is the reactive power injected by a **distributed generator (DG)** or a **capacitor bank (CB)**.

This reactive power is represented by

$$Q_{Gi}$$

where Q_{Gi} represents the reactive power injected by the reactive-power source at QG node i , and i represents the index of the QG node.

A QG node is a node containing a controllable reactive-power source, such as a DG inverter or capacitor bank.

The relationship can be represented as:

$$\text{DG/CB at node } i \longrightarrow Q_{Gi}$$

The value of Q_{Gi} indicates the amount of reactive power being supplied by the corresponding reactive-power source to the distribution network.

16.3.2 Voltage at QG and Non-QG Nodes

The second important quantity is the **voltage magnitude** at the network nodes.

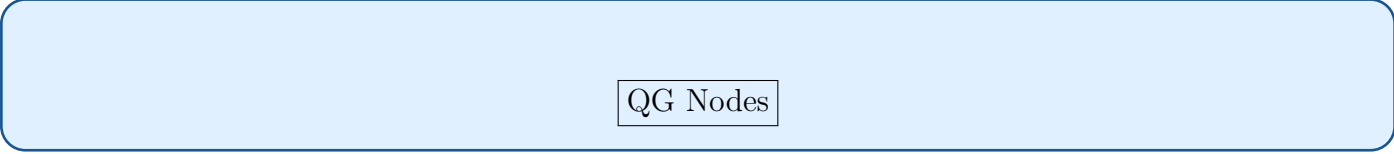
For a QG node i , the voltage is represented by

$$V_i$$

where V_i represents the voltage magnitude at QG node i .

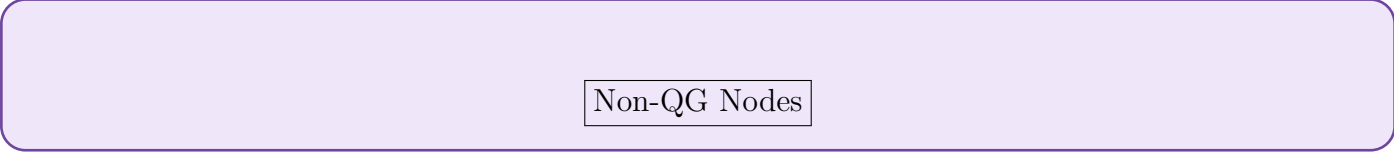
The methodology also considers the voltage at **non-QG nodes**, which are nodes that do not contain a controllable reactive-power source.

Therefore, the controller requires voltage information from both types of nodes:



QG Nodes

and



Non-QG Nodes

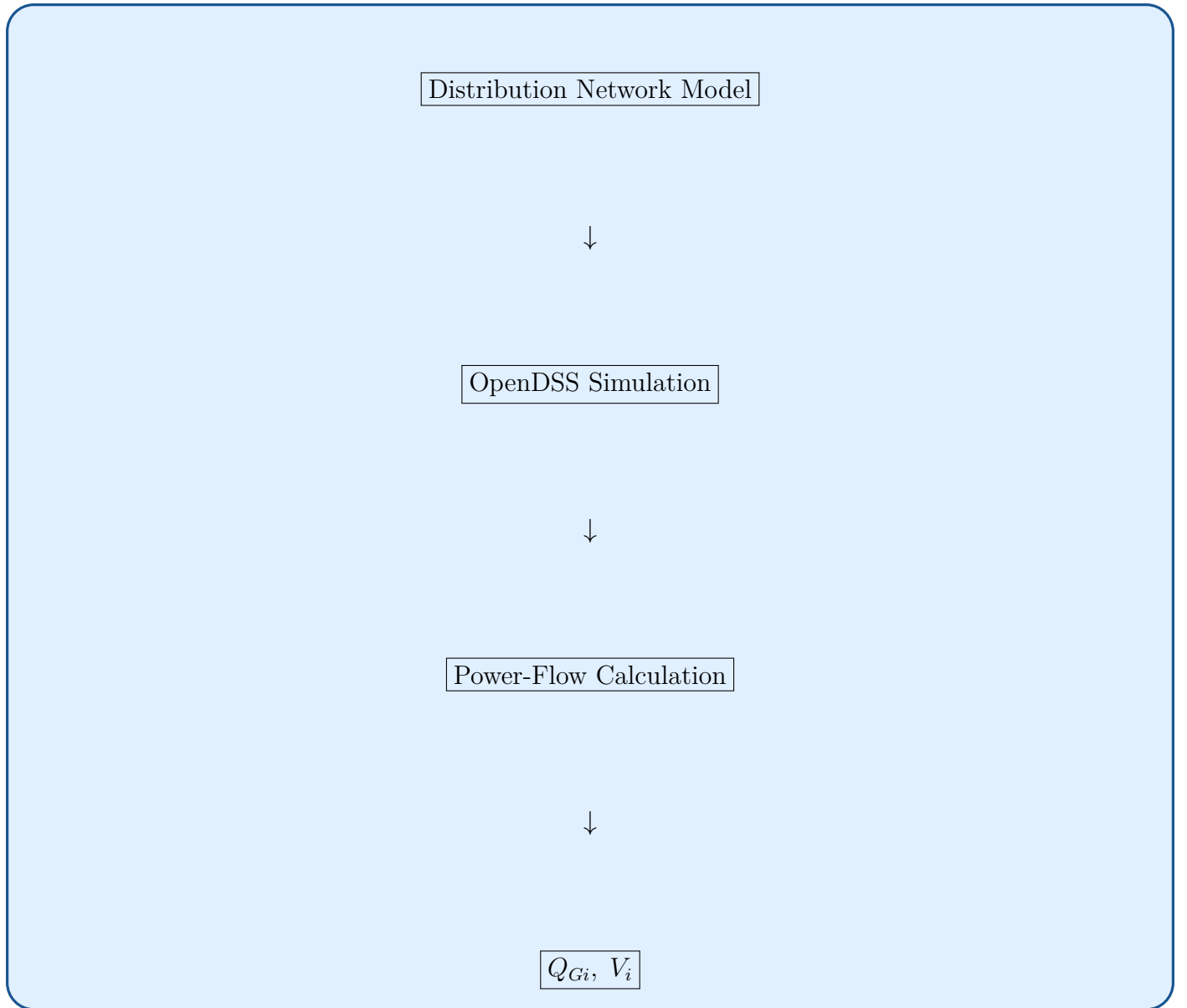
This is important because controlling the voltage only at nodes containing reactive-power sources does not necessarily guarantee that the voltage throughout the entire distribution network remains within the desired range.

16.3.3 How Are These Quantities Obtained?

In practice, the reactive power and voltage values can be obtained from electrical measurements.

However, in the study considered here, these quantities are obtained through an **OpenDSS-based simulation**.

The general process can be represented as:



OpenDSS is therefore used to simulate the distribution system and determine the electrical operating conditions resulting from the specified reactive-power injection.

16.3.4 Reactive Power Calculation in the Non-Phase-Level Implementation

An important consideration in the methodology is whether the cooperative-control algorithm is implemented at the **phase level**.

If the algorithm is not implemented at the phase level, the three-phase quantities are combined into a single value for simplification.

For reactive power, Q_{Gi} is calculated as the **sum of the reactive powers of all phases** associated with the QG node.

For a three-phase node:

$$Q_{Gi} = Q_{Gi}^a + Q_{Gi}^b + Q_{Gi}^c$$

where

Q_{Gi}^a = reactive power in phase a ,

Q_{Gi}^b = reactive power in phase b ,

and

Q_{Gi}^c = reactive power in phase c .

Thus, instead of treating the reactive power of each phase independently, the algorithm uses one aggregated reactive-power value for the QG node.

In general, for N_ϕ phases:

$$Q_{Gi} = \sum_{\phi=1}^{N_\phi} Q_{Gi}^\phi$$

16.3.5 Voltage Calculation in the Non-Phase-Level Implementation

The treatment of voltage is different from the reactive-power calculation.

For a non-phase-level implementation, V_i is calculated as the **average of the voltage magnitudes of all phases**.

For a three-phase QG node:

$$V_i = \frac{|V_i^a| + |V_i^b| + |V_i^c|}{3}$$

where

$|V_i^a|$ = voltage magnitude of phase a ,

$|V_i^b|$ = voltage magnitude of phase b ,

and

$|V_i^c|$ = voltage magnitude of phase c .

More generally, for N_ϕ phases:

$$V_i = \frac{1}{N_\phi} \sum_{\phi=1}^{N_\phi} |V_i^\phi|$$

Therefore, the methodology uses different aggregation operations for reactive power and voltage:

Reactive Power = Sum of Phase Reactive Powers

while

Voltage = Average of Phase Voltage Magnitudes

This distinction is important for the non-phase-level implementation.

16.3.6 Why Are the Quantities Aggregated?

The aggregation simplifies the implementation when the cooperative-control algorithm is not being applied independently to every phase.

Instead of having separate reactive-power variables for each phase:

$$Q_{Gi}^a, \quad Q_{Gi}^b, \quad Q_{Gi}^c,$$

and separate voltage quantities:

$$V_i^a, \quad V_i^b, \quad V_i^c,$$

the controller works with the aggregated quantities:

$$Q_{Gi}$$

and

$$V_i$$

This reduces the number of variables that need to be handled by the cooperative-control algorithm and provides a simplified representation of the three-phase operating condition.

16.3.7 Overall Procedure

The complete procedure described in this subsection can be summarized as follows.

For reactive power:

OpenDSS Distribution-System Model



Reactive Power of Each Phase



$$Q_{Gi} = \sum_{\phi=1}^{N_{\phi}} Q_{Gi}^{\phi}$$

For voltage:

Voltage Magnitude of Each Phase

↓

$$V_i = \frac{1}{N_\phi} \sum_{\phi=1}^{N_\phi} |V_i^\phi|$$

The calculated values of Q_{Gi} and V_i are then supplied to the subsequent stages of the cooperative-control algorithm.

Key Interpretation

The purpose of this step is to obtain the electrical operating conditions required by the cooperative-control algorithm. The reactive power Q_{Gi} represents the reactive power supplied by the DG or capacitor bank associated with QG node i , while V_i represents the corresponding voltage magnitude.

In the study, these quantities are obtained using an OpenDSS-based distribution-system simulation. When the algorithm is not implemented at the phase level, the reactive powers of the individual phases are summed to obtain Q_{Gi} , whereas the voltage magnitudes of the individual phases are averaged to obtain V_i .

Thus, the methodology converts the phase-level electrical operating information obtained from the distribution-system simulation into the aggregated quantities required by the subsequent cooperative-control formulation.

16.4 Gradient at QG and Non-QG Nodes

This subsection explains how the **gradient of the objective function** is calculated with respect to the reactive-power utilization ratio α_{qi} .

The main purpose of calculating these gradients is to determine **how much the objective function changes when the reactive-power utilization of a particular QG node is changed**.

There are two gradients considered:

$$g_{ii}$$

which represents the gradient associated with the **QG node itself**, and

$$g_{ki}$$

which represents the effect of the QG-node control action on a **non-QG node**.

16.4.1 Gradient at a QG Node

The gradient at QG node i is defined as

$$g_{ii} = \frac{\partial f_i}{\partial \alpha_{qi}}$$

where:

- g_{ii} is the gradient at QG node i ,
- f_i is the relevant cost function,
- α_{qi} is the reactive-power utilization ratio of QG node i .

In simple terms,

$$g_{ii} = \text{Change in cost function caused by changing } \alpha_{qi}$$

Therefore, g_{ii} tells the controller how sensitive the objective function is to the reactive-power utilization of QG node i .

16.4.2 Objective Function Components

The gradient is obtained from two components of the objective function:

$$f = f_1 + f_2$$

where f_1 represents the **voltage-deviation cost at QG nodes**, and f_2 represents the **reactive-power utilization cost at QG nodes**.

The first component is:

$$f_1 = \sum_{i=1}^N w_{VQG} \left[\left| \frac{V_i}{V_{bi}} \right| - 1 \right]^2$$

where V_i is the voltage magnitude at QG node i , V_{bi} is the base/reference voltage at QG node i , and w_{VQG} is the weighting factor for QG-node voltage deviation.

The second component is:

$$f_2 = \sum_{i=1}^N w_{\alpha} \alpha_{qi}^2 \overline{Q}_{Gi}$$

where α_{qi} is the reactive-power utilization ratio, $\overline{Q}_{Gi}$ is the maximum available reactive power, and w_{α} is the weighting factor associated with reactive-power utilization.

Thus,

$$\text{QG-node cost} = \text{Voltage deviation cost} + \text{Reactive-power utilization cost}$$

16.4.3 Derivation of g_{ii}

The gradient can be written as:

$$g_{ii} = \frac{\partial f_i}{\partial \alpha_{qi}} = \frac{\partial f_1}{\partial \alpha_{qi}} + \frac{\partial f_2}{\partial \alpha_{qi}}.$$

Since the voltage V_i depends on reactive power Q_{Gi} , and Q_{Gi} depends on α_{qi} , the first term requires the application of the chain rule.

Therefore,

$$g_{ii} = \frac{\partial f_1}{\partial |V_i|} \frac{\partial |V_i|}{\partial Q_{Gi}} \frac{\partial Q_{Gi}}{\partial \alpha_{qi}} + \frac{\partial f_2}{\partial \alpha_{qi}}$$

The dependency chain is:

$$\alpha_{qi} \rightarrow Q_{Gi} \rightarrow |V_i| \rightarrow f_1$$

In other words:

1. Changing α_{qi} changes the reactive power Q_{Gi} .
2. Changing Q_{Gi} changes the voltage magnitude $|V_i|$.
3. Changing $|V_i|$ changes the voltage-deviation cost.

16.4.4 Relationship Between α_{qi} and Q_{Gi}

The reactive power is related to the utilization ratio through:

$$Q_{Gi} = \alpha_{qi} \bar{Q}_{Gi}$$

Therefore,

$$\frac{\partial Q_{Gi}}{\partial \alpha_{qi}} = \bar{Q}_{Gi}$$

This means that the rate at which actual reactive power changes with respect to the utilization ratio is equal to the maximum available reactive-power capability.

Thus, when α_{qi} is increased, the reactive power Q_{Gi} increases according to the available reactive-power capability $\bar{Q}_{Gi}$.

16.4.5 Voltage-Deviation Contribution

The voltage-related component is:

$$f_1 = \sum_{i=1}^N w_{VQG} \left[\left| \frac{V_i}{V_{bi}} \right| - 1 \right]^2.$$

For QG node i , the corresponding voltage-deviation cost depends on:

$$\left| \frac{V_i}{V_{bi}} \right| - 1$$

If the voltage magnitude reaches its base value,

$$|V_i| = V_{bi},$$

then:

$$\left| \frac{V_i}{V_{bi}} \right| - 1 = 0$$

and the voltage-deviation contribution is minimized.

If the voltage differs from its base value, the voltage-deviation cost increases.

16.4.6 Reactive-Power Utilization Contribution

The second contribution comes from:

$$f_2 = \sum_{i=1}^N w_\alpha \alpha_{qi}^2 \bar{Q}_{Gi}.$$

Taking the derivative with respect to α_{qi} gives:

$$\frac{\partial f_2}{\partial \alpha_{qi}} = 2w_\alpha \alpha_{qi} \bar{Q}_{Gi}$$

This term increases as α_{qi} increases.

Therefore, the controller has an incentive to avoid unnecessarily large reactive-power utilization.

The basic relationship is:

$$\alpha_{qi} \uparrow \Rightarrow \text{Reactive-power utilization cost} \uparrow$$

16.4.7 Final Expression for g_{ii}

After applying the chain rule and the electrical-network relationships, the gradient at QG node i is given by:

$$g_{ii} = \frac{2w_{VQG} (|V_i|(|V_i| - V_{bi})Q_{Gi})}{V_{bi}^2 (Q_{Gi} - Q_{Di} - |V_i|^2 B_{ii})} + 2w_\alpha \alpha_{qi} \bar{Q}_{Gi} \quad (7)$$

This expression contains two major contributions.

The first term is:

$$\frac{2w_{VQG}|V_i|(|V_i| - V_{bi})Q_{Gi}}{V_{bi}^2(Q_{Gi} - Q_{Di} - |V_i|^2B_{ii})}$$

This represents the effect of changing α_{qi} on the **voltage-deviation cost at the QG node**.

The second term is:

$$2w_{\alpha}\alpha_{qi}\overline{Q}_{Gi}$$

This represents the increase in the **reactive-power utilization cost** resulting from increasing α_{qi} .

Therefore,

$$g_{ii} = \text{Voltage-related gradient} + \text{Reactive-power-utilization gradient}$$

16.4.8 Gradient at a Non-QG Node

The second gradient is associated with a non-QG node k .

It is defined as:

$$g_{ki} = \frac{\partial f_k}{\partial \alpha_{qi}}$$

This is different from g_{ii} .

The purpose of g_{ki} is to determine how changing the reactive-power utilization at QG node i affects the voltage-deviation cost at non-QG node k .

The chain rule gives:

$$g_{ki} = \frac{\partial f_k}{\partial |V_k|} \frac{\partial |V_k|}{\partial Q_{Gi}} \frac{\partial Q_{Gi}}{\partial \alpha_{qi}}$$

The corresponding relationship is:

$$\alpha_{qi} \rightarrow Q_{Gi} \rightarrow |V_k| \rightarrow f_k$$

This is important because a QG node can affect the voltage at other locations in the distribution network.

16.4.9 Voltage-Deviation Cost at the Non-QG Node

The non-QG-node voltage cost is associated with:

$$w_{V_{NQG}} \left[\left| \frac{V_k}{V_{bk}} \right| - 1 \right]^2$$

where V_k is the voltage magnitude at non-QG node k , V_{bk} is the base voltage of non-QG node k , and $w_{V_{NQG}}$ is the weighting factor for non-QG-node voltage deviation.

The controller therefore tries to keep:

$$|V_k| \approx V_{bk}$$

even though there is no direct reactive-power source located at node k .

16.4.10 Electrical Coupling Between QG and Non-QG Nodes

The voltage at non-QG node k depends on the reactive power injected at QG node i .

This relationship is influenced by the network parameters:

$$G_{ik}$$

and

$$B_{ik}$$

which were obtained previously from Y_{BUS} .

The final gradient is:

$$g_{ki} = \frac{2w_{VNQG}(|V_k| - V_{bk})\overline{Q}_{Gi}}{|V_i|V_{bk}^2(G_{ik}\sin\delta_{ik} - B_{ik}\cos\delta_{ik})} \quad (8)$$

This equation describes how changing the reactive-power utilization ratio at QG node i influences the voltage-deviation cost at non-QG node k .

16.4.11 Meaning of δ_{ik}

The equation contains the term:

$$\delta_{ik}$$

The quantity δ_{ik} represents the difference between the voltage angle at node i and the voltage angle at node k .

Therefore,

$$\delta_{ik} = \delta_i - \delta_k$$

where δ_i is the voltage angle at QG node i , and δ_k is the voltage angle at non-QG node k .

The terms

$$\sin \delta_{ik}$$

and

$$\cos \delta_{ik}$$

therefore describe the influence of the relative voltage-angle difference on the electrical coupling between the two nodes.

16.4.12 Physical Meaning of g_{ki}

The gradient g_{ki} tells the controller how much the non-QG voltage cost changes when the reactive-power utilization of QG node i changes.

Therefore,

$$g_{ki} = \text{Effect of QG node } i \text{ on the voltage cost at non-QG node } k$$

This allows the controller to consider **network-wide voltage effects**, rather than controlling only the voltage at the location where reactive power is directly available.

16.4.13 Difference Between g_{ii} and g_{ki}

The two gradients have different purposes:

Gradient	Represents
g_{ii}	Effect of α_{qi} on the cost at the same QG node i
g_{ki}	Effect of α_{qi} on the cost at a non-QG node k

The first one therefore represents a **local effect**, while the second captures a **network-wide effect**.

Conceptually, for g_{ii} :

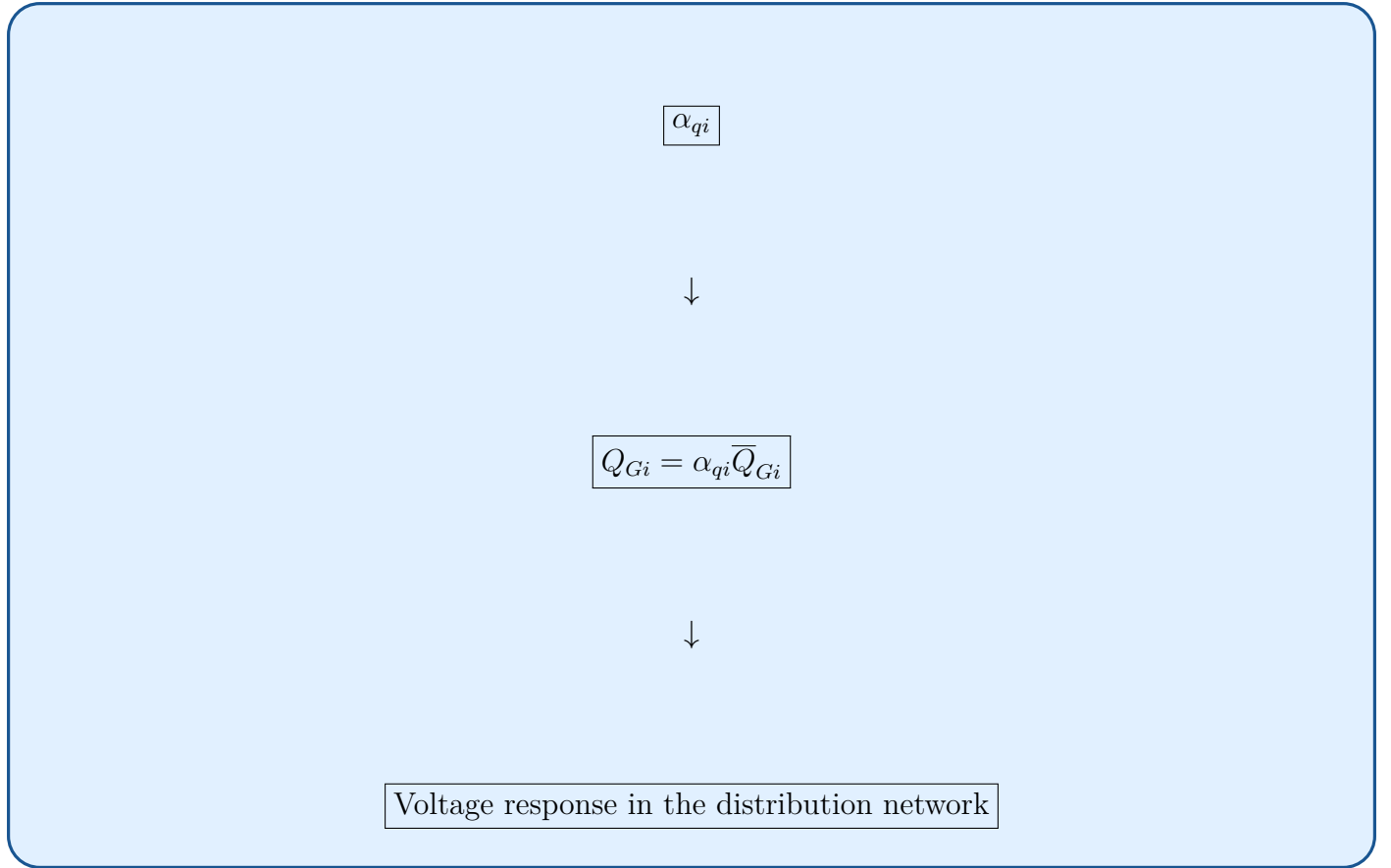
$$\alpha_{qi} \rightarrow Q_{Gi} \rightarrow V_i \rightarrow \text{QG voltage cost}$$

For g_{ki} :

$$\alpha_{qi} \rightarrow Q_{Gi} \rightarrow V_k \rightarrow \text{Non-QG voltage cost}$$

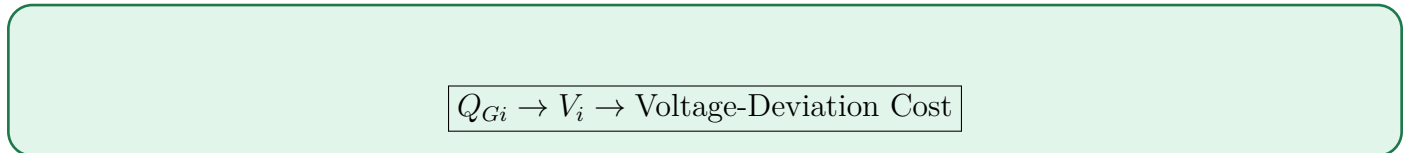
16.4.14 Overall Interpretation

The complete gradient calculation can be understood as:



The change in Q_{Gi} affects the voltages throughout the network.

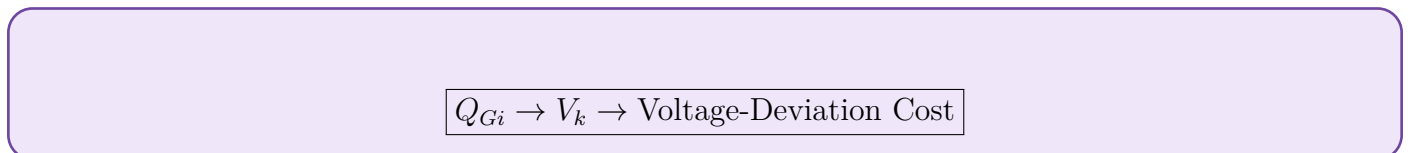
At the QG node:



which contributes to:

$$g_{ii}$$

At a non-QG node:



which contributes to:

$$g_{ki}.$$

Therefore, the gradients provide the cooperative controller with information about **how the objective**

function responds to changes in the reactive-power utilization ratio.

The key idea of this subsection is:

$$\text{Gradient} = \text{Sensitivity of Objective Function to } \alpha_{qi}$$

The methodology therefore considers both the **local voltage effect at QG nodes** and the **network-wide voltage effect at non-QG nodes**.

16.5 Gradient Gain

This subsection explains the purpose of the **gradient gains** used in the cooperative-control algorithm. The gradient gains determine **how strongly the controller responds to the calculated gradients** g_{ii} and g_{ki} .

The two types of gradient gains are:

$$\beta_{QG_i}$$

for a **QG node**, and

$$\beta_{NQGi_k}$$

for a **non-QG node** k with respect to QG node i .

16.5.1 Why is a Gradient Gain Required?

In the previous subsection, the gradients were obtained as

$$g_{ii} = \frac{\partial f_i}{\partial \alpha_{qi}}$$

and

$$g_{ki} = \frac{\partial f_k}{\partial \alpha_{qi}}$$

These gradients indicate the **sensitivity of the objective function** to a change in the reactive-power utilization ratio α_{qi} .

However, the raw gradient does not directly determine how aggressively the controller should change α_{qi} .

A **gradient gain** is therefore introduced to scale the gradient before it is used in the control update. Conceptually,

Gradient $\rightarrow$ Gradient Gain $\rightarrow$ Control Action

16.5.2 Gradient Gain for a QG Node

For a QG node i , the gradient gain is represented by

$$\beta_{QG_i}$$

This gain is associated with the local gradient

$$g_{ii}$$

Therefore, the QG-node control action is influenced by the combination

$$\beta_{QG_i} g_{ii}$$

where β_{QG_i} is the gradient gain associated with QG node i , and g_{ii} is the gradient of the objective function at QG node i .

The gain determines the **strength of the response** to the calculated gradient.

A larger gain produces a stronger response to the gradient, whereas a smaller gain produces a more conservative response.

Conceptually,

$$\beta_{QG_i} \uparrow \Rightarrow \text{Stronger response to } g_{ii}$$

and

$$\beta_{QG_i} \downarrow \Rightarrow \text{More conservative response}$$

16.5.3 Gradient Gain for a Non-QG Node

For a non-QG node k , the gradient gain is represented by

$$\beta_{NQGi k}$$

The subscript ik is important because this gain is associated with the effect of **QG node i on non-QG node k** .

The corresponding gradient is

$$g_{ki}$$

Therefore, the contribution associated with the non-QG node can be scaled by

$$\beta_{NQGi k} g_{ki}$$

where $\beta_{NQGi k}$ is the gradient gain for the communication/interaction between QG node i and non-QG node k , and g_{ki} is the sensitivity of the non-QG-node cost with respect to α_{qi} .

16.5.4 Meaning of the Two Gains

The two gains correspond to two different effects.

QG-node gain The QG-node gradient gain is

$$\beta_{QG_i}$$

and is associated with the local effect

$$\alpha_{qi} \rightarrow Q_{Gi} \rightarrow V_i \rightarrow g_{ii}$$

It determines how strongly QG node i responds to its own gradient.

Non-QG-node gain The non-QG-node gradient gain is

$$\beta_{NQGik}$$

and is associated with the effect of QG node i on non-QG node k :

$$\alpha_{qi} \rightarrow Q_{Gi} \rightarrow V_k \rightarrow g_{ki}$$

It therefore represents the gain associated with the communication/interaction between these two types of nodes.

16.5.5 Why Are Different Gains Used?

The electrical network is not equally sensitive everywhere.

A change in reactive power at one QG node may have:

- a strong effect on a nearby node,
- a weaker effect on a distant node, or
- different effects depending on the network topology and electrical parameters.

Therefore, the controller can use appropriate gradient gains to scale the calculated gradients.

The basic idea is

$$\text{Gradient} \times \text{Gain} = \text{Scaled Gradient}$$

For the QG node,

$$g_{ii} \rightarrow \beta_{QGi} g_{ii}$$

For the non-QG node,

$$g_{ki} \rightarrow \beta_{NQGik} g_{ki}$$

16.5.6 How Are the Gains Obtained?

The gradient gains

$$\beta_{QG_i}$$

and

$$\beta_{NQG_{ik}}$$

can be obtained using either **numerical calculation** or **simulation trials**.

Numerical Calculation The gains can be determined through numerical analysis of the system.

$$\text{Numerical Calculation} \rightarrow \beta_{QG_i}, \beta_{NQG_{ik}}$$

Simulation Trials Alternatively, different gain values can be tested through simulations and their effects on the control performance can be evaluated.

$$\text{Simulation Trials} \rightarrow \text{Evaluate Control Response} \rightarrow \text{Select Gains}$$

Thus, the gains are treated as **selected/tuned parameters** for the gradient-based control process.

16.5.7 Simple Interpretation

Suppose the controller calculates

$$g_{ii} = 0.8.$$

If the selected gradient gain is

$$\beta_{QG_i} = 0.5,$$

then the scaled gradient becomes

$$\beta_{QG_i} g_{ii} = 0.5(0.8) = 0.4$$

The gain therefore controls how strongly the calculated gradient influences the subsequent control action.

Similarly, if

$$g_{ki} = 0.6$$

and

$$\beta_{NQGik} = 0.4,$$

then

$$\beta_{NQGik}g_{ki} = 0.4(0.6) = 0.24$$

This illustrates the role of the gain as a scaling factor.

16.5.8 Connection With the Previous Section

The previous subsection calculated

$$g_{ii}$$

and

$$g_{ki}.$$

This subsection introduces the corresponding gains

$$\beta_{QG_i}$$

and

$$\beta_{NQGik}.$$

Therefore, the methodology can be viewed as:

$$\alpha_{qi}$$



$$Q_{Gi} = \alpha_{qi} \overline{Q}_{Gi}$$



Voltage Response



$$g_{ii}, g_{ki}$$



$$\beta_{QG_i} g_{ii}, \quad \beta_{NQ_{Gik}} g_{ki}$$



Cooperative Control Action

16.5.9 Key Interpretation

The main idea of **gradient gain** is:

Key Interpretation

A gradient gain is a tuning parameter that scales the calculated gradient so that the cooperative controller can determine an appropriate magnitude of control response.

The two gains serve different purposes:

$$\beta_{QG_i} \rightarrow \text{QG-node/local gradient response}$$

and

$$\beta_{NQGi_k} \rightarrow \text{QG-to-non-QG/network-wide gradient response}$$

According to the provided methodology, both β_{QG_i} and β_{NQGi_k} can be obtained through **numerical calculation or simulation trials**.

16.6 Updating Reactive Power Utilization Ratio Based on Consensus Algorithm

This subsection describes the **iterative update of the reactive-power utilization ratio** α_{qi} using a **consensus-based cooperative-control algorithm**.

The main objective is to allow the QG nodes to **coordinate their reactive-power utilization** while considering both the information received from other QG nodes and the influence of non-QG nodes. The utilization ratio is updated repeatedly until the participating QG devices reach an acceptable level of agreement.

16.6.1 Reactive-Power Utilization Ratio

The reactive-power utilization ratio is defined as

$$\alpha_{qi} = \frac{Q_{Gi}}{\bar{Q}_{Gi}}$$

where:

- Q_{Gi} is the actual reactive power provided by QG node i ,
- $\bar{Q}_{Gi}$ is the maximum available reactive-power capability,
- α_{qi} indicates how much of the available reactive-power capability is being utilized.

Therefore,

$$Q_{Gi} = \alpha_{qi} \overline{Q_{Gi}}$$

A higher value of α_{qi} means that a larger portion of the available reactive-power capability is being used.

16.6.2 Iterative Nature of the Algorithm

The paper states that obtaining α_{qi} is an **iterative process**. This means that the controller does not calculate the final value of α_{qi} in one step.

Instead,

$$\alpha_{qi}(n) \rightarrow \alpha_{qi}(n+1) \rightarrow \alpha_{qi}(n+2) \rightarrow \dots$$

At every iteration, the current value is modified according to:

- information received from other QG nodes,
- the local gradient,
- information received from non-QG nodes,
- the corresponding gradient gains.

The iteration continues until the required consensus condition is satisfied.

16.6.3 Update Equation

The update equation used in the cooperative-control algorithm is

$$\alpha_{qi}(n+1) = \left(\sum_{l=1}^{N_{QG}} d_{il} \alpha_{ql}(n) \right) - \beta_{QG_i} g_{ii} - \left(\sum_{k=1}^{N_{NQG}} d_{NQG_{ik}} \beta_{NQG_{ik}} g_{ki} \right) \quad (9)$$

This equation contains three important parts:

QG information exchange – Local gradient correction – Non-QG gradient correction

Each term has a specific role in the cooperative-control process.

16.6.4 First Term: Information From QG Nodes

The first term in (9) is

$$\sum_{l=1}^{N_{QG}} d_{il} \alpha_{ql}(n)$$

This term represents the information received by QG node i from other QG nodes.

Here:

- N_{QG} = total number of QG nodes,
- l = index of the QG node sending information,
- d_{il} = communication coefficient between QG node i and QG node l ,
- $\alpha_{ql}(n)$ = reactive-power utilization ratio of QG node l at iteration n .

Thus, QG node i uses the utilization-ratio information received from the other participating QG nodes.

Conceptually,

$QG_l \rightarrow \text{Information} \rightarrow QG_i$

This is the **consensus component** of the update.

16.6.5 Role of the Communication Matrix

The coefficient

$$d_{il}$$

comes from the communication topology defined earlier.

It determines whether and how strongly QG node i receives information from QG node l .

For example,

$$d_{il} > 0$$

means that a communication relationship exists between the corresponding nodes.

If

$$d_{il} = 0$$

there is no direct communication channel represented between the two nodes.

Therefore, the first term can be interpreted as

Weighted information from communicating QG nodes

16.6.6 Second Term: Local Gradient Correction

The second term is

$$-\beta_{QG_i} g_{ii}$$

This term represents the **local gradient-based correction**.

From the previous subsection,

$$g_{ii} = \frac{\partial f_i}{\partial \alpha_{qi}}$$

represents the sensitivity of the objective function at QG node i with respect to α_{qi} .

The corresponding gradient gain is

$$\beta_{QG_i}$$

Therefore,

$$\beta_{QG_i} g_{ii}$$

is the scaled local gradient.

The negative sign indicates that the gradient term is used in the direction that reduces the objective function.

Conceptually,

Local Gradient $\rightarrow$ Gradient Gain $\rightarrow$ Cost Reduction

16.6.7 Why Is the Local Gradient Important?

The consensus term alone would make the QG nodes move toward agreement, but agreement by itself does not guarantee that the resulting operating point is optimal.

The gradient term provides information about the **objective function**.

Therefore, the controller is not simply trying to make all QG nodes have identical values.

Instead, it attempts to achieve an appropriate coordinated solution while also considering:

- voltage regulation,
- reactive-power utilization,
- network-wide voltage effects.

Thus,

Consensus + Optimization

are combined in the update process.

16.6.8 Third Term: Information From Non-QG Nodes

The third term is

$$- \left(\sum_{k=1}^{N_{NQG}} d_{NQGik} \beta_{NQGik} g_{ki} \right)$$

This term accounts for the influence of **non-QG nodes** on the control update of QG node i .

Here:

- N_{NQG} = total number of non-QG nodes,
- k = index of the non-QG node,

- d_{NQGik} = communication coefficient associated with the QG-node/non-QG-node communication,
- β_{NQGik} = corresponding gradient gain,
- g_{ki} = gradient representing the effect of QG node i on non-QG node k .

Therefore, the non-QG contribution is

$$\sum_{k=1}^{N_{NQG}} d_{NQGik} \beta_{NQGik} g_{ki}$$

16.6.9 Why Are Non-QG Nodes Included?

A non-QG node does not directly control reactive power. However, its voltage can be affected by reactive power injected by a QG node.

The relationship is

$$\alpha_{qi} \rightarrow Q_{Gi} \rightarrow V_k \rightarrow \text{Non-QG Voltage Cost}$$

This effect is represented by

$$g_{ki}$$

Therefore, even though the non-QG node cannot directly change its reactive-power output, its voltage condition contributes to the decision made by the QG node.

This makes the control strategy **network-wide rather than purely local**.

16.6.10 Role of d_{NQGik}

The coefficient

$$d_{NQGik}$$

represents the communication relationship between QG node i and non-QG node k .

When the corresponding communication relationship exists, the non-QG gradient information contributes to the update of α_{qi} .

Thus,

$$d_{NQGik} \rightarrow \text{Communication weighting}$$

while

$$g_{ki} \rightarrow \text{Voltage sensitivity}$$

and

$$\beta_{NQGik} \rightarrow \text{Gradient scaling}$$

Together, these determine the non-QG contribution to the update.

16.6.11 Complete Interpretation of Equation (9)

Equation (9) can therefore be interpreted as

$$\begin{aligned} \text{New } \alpha_{qi} &= \text{Information from QG nodes} \\ &\quad - \text{Local QG gradient correction} \\ &\quad - \text{Non-QG gradient correction} \end{aligned}$$

More specifically,

$$\begin{aligned} \alpha_{qi}(n+1) &= \underbrace{\sum_{l=1}^{N_{QG}} d_{il} \alpha_{ql}(n)}_{\text{QG consensus information}} \\ &\quad - \underbrace{\beta_{QG} g_{ii}}_{\text{local optimization}} \\ &\quad - \underbrace{\sum_{k=1}^{N_{NQG}} d_{NQGik} \beta_{NQGik} g_{ki}}_{\text{non-QG/network-wide effect}} \end{aligned}$$

Therefore, the update is not based on a single piece of information.

It combines

$$\text{Communication} + \text{Optimization} + \text{Network-wide Voltage Effects}$$

16.6.12 Meaning of N_{QG} and N_{NQG}

The paper defines

$$N_{QG}$$

as the **total number of QG nodes**.

Similarly,

$$N_{NQG}$$

is the **total number of non-QG nodes**.

Therefore,

$$l = 1, \dots, N_{QG}$$

represents all QG nodes participating in the QG communication process, while

$$k = 1, \dots, N_{NQG}$$

represents all non-QG nodes considered in the network-wide voltage contribution.

16.6.13 Meaning of $\alpha_{ql}(n)$

The variable

$$\alpha_{ql}(n)$$

represents the reactive-power utilization ratio of QG node l at iteration n .

Therefore, when QG node i calculates its new utilization ratio, it receives information about the current utilization ratios of other QG nodes.

This allows the QG nodes to progressively coordinate their reactive-power utilization.

16.6.14 Consensus Process

The purpose of the consensus mechanism is to allow the participating QG devices to reach an appropriate coordinated solution.

The process can be represented as

$$\alpha_{q1}(0), \alpha_{q2}(0), \dots, \alpha_{qN_{QG}}(0)$$

↓

Information Exchange

↓

Gradient-Based Correction

↓

$$\alpha_{q1}(1), \alpha_{q2}(1), \dots, \alpha_{qN_{QG}}(1)$$

↓

Repeated Iterations

↓

Consensus / Convergence

16.6.15 Fair Utilization of Reactive Power

An important objective stated in the paper is to **fairly utilize the reactive-power capability of each DG and capacitor bank**.

Suppose several QG devices are available:

$$DG_1, \quad DG_2, \quad DG_3, \quad CB.$$

Each device may have a different maximum reactive-power capability:

$$\overline{Q}_{G1}, \quad \overline{Q}_{G2}, \quad \overline{Q}_{G3}, \quad \overline{Q}_{CB}$$

The utilization ratio provides a normalized measure of how much of each device's capability is being used.

Thus, instead of simply comparing the absolute reactive powers, the controller considers

$$\alpha_{qi} = \frac{Q_{Gi}}{\overline{Q}_{Gi}}$$

This allows devices with different reactive-power capacities to be considered on a common utilization basis.

16.6.16 Loss Minimization and Voltage Regulation

The paper states that the cooperative control aims to achieve

$$\text{Loss Minimization} + \text{Voltage Regulation}$$

while fairly utilizing the available reactive-power resources.

The controller therefore does not simply maximize reactive-power injection.

Instead, it searches for a suitable utilization of the available resources that provides effective voltage regulation while avoiding unnecessary reactive-power utilization.

Conceptually,

$$\text{Reactive Power Utilization} \rightarrow \text{Voltage Regulation} \rightarrow \text{Loss Minimization}$$

16.6.17 Stopping Criterion

The iterative process does not continue indefinitely.

The paper states that the iteration is stopped when the **difference between the reactive-power utilization ratios of the QG devices becomes less than a predefined tolerance**.

In general, the convergence condition can be represented as

$$|\alpha_{qi} - \alpha_{qj}| < \varepsilon$$

where:

- α_{qi} and α_{qj} are utilization ratios of participating QG devices,
- ε is the predefined tolerance.

When this condition is satisfied, the QG devices are considered to have reached an acceptable level of consensus.

16.6.18 Overall Methodology Flow

The entire process of this subsection can be summarized as

Initial α_{qi}



Exchange QG Information



Calculate Local Gradient g_{ii}



Calculate Non-QG Gradient g_{ki}



Apply Gradient Gains



Update $\alpha_{qi}(n+1)$

16.6.19 Key Interpretation

The most important point of this subsection is that **the reactive-power utilization ratio is not selected independently for each DG or capacitor bank.**

Instead, each QG node updates its utilization ratio using

Information from other QG nodes

together with

Local gradient information

and

Non-QG voltage-gradient information

Therefore, the overall concept is

Communication $\rightarrow$ Consensus $\rightarrow$ Gradient Correction
 $\rightarrow$ Updated α_{qi} $\rightarrow$ Coordinated Reactive Power

The iteration continues until the QG devices reach the predefined consensus tolerance. This enables the available reactive-power resources to be coordinated for **voltage regulation and loss minimization**, while avoiding unnecessary or unfair utilization of individual DG and capacitor-bank resources.

16.7 Capacitor Switching

This subsection explains how the **capacitor banks (CBs)** are incorporated into the cooperative-control strategy. Unlike inverter-based DGs, whose reactive-power output can generally be varied continuously within their operating limits, capacitor banks operate through **discrete switching of individual capacitor units**.

The main purpose of the switching strategy is to provide the required reactive-power support while **minimizing unnecessary switching operations**. Frequent switching increases mechanical and electrical stress on the switching equipment, which can increase maintenance requirements and reduce the equipment lifetime.

16.7.1 Objective of Capacitor Switching

Capacitor banks are installed in distribution systems to provide reactive-power support. However, utilities generally try to minimize the number of switching operations because excessive switching can increase:

- maintenance requirements,
- equipment wear,
- operating costs, and
- degradation of the switching equipment.

Therefore, the controller must determine not only **how much reactive power is required**, but also **how many capacitor units should actually be switched ON**.

The basic idea can be represented as:

Required Reactive Power → Determine Number of CB Units → Switch Appropriate Units ON

16.7.2 Capacitor Banks in the Studied System

The paper considers capacitor banks at two locations:

Node 675

and

Node 611

At node 675, there are **three single-phase capacitor banks**, with each capacitor unit having a rating of

200 kvar

Therefore, the total available reactive-power capacity at node 675 is

$$Q_{CB,675}^{\max} = 3 \times 200 = 600 \text{ kvar}$$

At node 611, there is a **single capacitor unit** rated at

$$100 \text{ kvar}$$

Therefore,

$$Q_{CB,611}^{\max} = 100 \text{ kvar}$$

16.7.3 Why α_{qi} Must Be Treated Differently for Capacitor Banks

Earlier, the reactive-power utilization ratio was defined as

$$\alpha_{qi} = \frac{Q_{Gi}}{\bar{Q}_{Gi}}$$

For a DG inverter, α_{qi} can generally take a **continuous range of values** because the inverter can continuously adjust its reactive-power output.

For a capacitor bank, however, the available reactive power depends on the number of individual capacitor units that are switched ON.

For example, at node 675, with three 200 kvar units:

$$\begin{aligned} 0 \text{ units ON} &\rightarrow 0 \text{ kvar} \\ 1 \text{ unit ON} &\rightarrow 200 \text{ kvar} \\ 2 \text{ units ON} &\rightarrow 400 \text{ kvar} \\ 3 \text{ units ON} &\rightarrow 600 \text{ kvar} \end{aligned}$$

Thus, the capacitor-bank reactive power is **discrete rather than continuously variable**.

16.7.4 Lower and Upper Limits of α_{qi}

To account for this discrete nature, the paper defines a **lower limit** and an **upper limit** for the capacitor-bank utilization ratio.

The lower limit is calculated using

$$\alpha_{qi, \text{lower limit}} = \frac{\text{No. of units switched on}}{\text{Total number of available units}} \quad (10)$$

The upper limit is calculated using

$$\alpha_{qi, \text{upper limit}} = \frac{\text{No. of units switched on} + 1}{\text{Total number of available units}} \quad (11)$$

These two limits define the interval within which the optimized value of α_{qi} lies.

16.7.5 Meaning of the Lower Limit

Suppose there are N available capacitor units and currently m units are switched ON.

Then,

$$\alpha_{qi, \text{lower limit}} = \frac{m}{N}$$

This represents the utilization ratio corresponding to the **current number of switched-on units**.

For example, if there are three units and one unit is ON,

$$N = 3, \quad m = 1$$

then

$$\alpha_{qi, \text{lower limit}} = \frac{1}{3}$$

Therefore,

$$\alpha_{qi, \text{lower limit}} = 0.3333$$

16.7.6 Meaning of the Upper Limit

The upper limit corresponds to switching **one additional capacitor unit ON**.

It is calculated as

$$\alpha_{qi, \text{upper limit}} = \frac{m + 1}{N}$$

For the same example of three available units with one unit currently ON,

$$N = 3, \quad m = 1$$

we obtain

$$\alpha_{qi, \text{upper limit}} = \frac{1+1}{3} = \frac{2}{3}$$

Therefore,

$$\alpha_{qi, \text{upper limit}} = 0.6667$$

Hence, the possible range is

$$0.3333 \leq \alpha_{qi} \leq 0.6667$$

This interval determines whether the next capacitor unit should be switched ON.

16.7.7 Example for the Capacitor Bank at Node 675

Node 675 has three 200 kvar capacitor units.

Suppose initially one unit is switched ON. Then

$$N = 3, \quad m = 1.$$

The lower limit becomes

$$\alpha_{qi, \text{lower limit}} = \frac{1}{3}$$

and the upper limit becomes

$$\alpha_{qi, \text{upper limit}} = \frac{2}{3}.$$

Therefore,

$$\frac{1}{3} \leq \alpha_{qi} \leq \frac{2}{3}$$

The corresponding capacitor states are

$$\begin{aligned} 1 \text{ unit ON} &\rightarrow \alpha_{qi} = \frac{1}{3}, \\ 2 \text{ units ON} &\rightarrow \alpha_{qi} = \frac{2}{3}. \end{aligned}$$

Thus, the optimized continuous value of α_{qi} can be used to determine whether the capacitor-bank state should remain unchanged or move to the next discrete level.

16.7.8 Three Methods for Updating Capacitor-Bank α_{qi}

The paper describes **three methods** for converting the optimized value of α_{qi} into a practical capacitor-switching command.

Method 1: Assign the Lower Limit In the first method, the value of α_{qi} is assigned the lower limit:

$$\alpha_{qi} = \alpha_{qi, \text{lower limit}}$$

This corresponds to keeping the currently switched-on capacitor units.

Therefore,

$$\text{Optimized } \alpha_{qi} \rightarrow \text{Lower Limit} \rightarrow \text{Existing CB State}$$

This method avoids switching an additional capacitor unit unless it is necessary.

Method 2: Assign the Upper Limit In the second method, the value of α_{qi} is assigned the upper limit:

$$\alpha_{qi} = \alpha_{qi, \text{upper limit}}$$

This corresponds to switching **one additional capacitor unit ON**.

The process is

$$\text{Optimized } \alpha_{qi} \rightarrow \text{Upper Limit} \rightarrow \text{Switch One Additional Unit ON}$$

This allows the capacitor bank to provide a higher level of reactive-power support.

Method 3: Round to the Nearest Integer In the third method, the optimized α_{qi} is rounded to the nearest appropriate discrete value.

Conceptually,

$$\alpha_{qi} \rightarrow \text{Nearest Discrete Level} \rightarrow \text{Number of CB Units}$$

This approach attempts to select the capacitor-bank state that is closest to the optimized continuous utilization ratio.

16.7.9 Why Three Methods Are Required

The optimization algorithm produces an α_{qi} value based on the objective function and network conditions.

However, a capacitor bank cannot necessarily implement an arbitrary continuous value.

For example, with three identical capacitor units,

$$\alpha_{qi} = 0.5$$

would correspond to

$$0.5 \times 3 = 1.5 \text{ units}$$

but it is impossible to switch **1.5 capacitor units** ON.

The actual system can only operate at

$$0, \quad 1, \quad 2, \quad 3$$

units ON.

Therefore, the continuous optimization result must be converted into a **discrete switching command**.

This is the reason for defining the lower limit, upper limit, and nearest discrete value.

16.7.10 Connection With the Consensus Algorithm

The capacitor-switching procedure occurs after the cooperative-control algorithm determines the utilization ratio.

The overall process can therefore be understood as

Consensus Algorithm



Optimized α_{qi}



Determine CB Utilization Range



$\alpha_{qi, \text{lower limit}}$ and $\alpha_{qi, \text{upper limit}}$



Select Switching Method



Number of Capacitor Units ON

16.7.11 Relationship Between Number of Units and α_{qi}

For a capacitor bank containing N identical units, if m units are switched ON,

$$\alpha_{qi} = \frac{m}{N}$$

The corresponding reactive power can be expressed as

$$Q_{CB} = mQ_{\text{unit}}$$

where Q_{unit} is the reactive-power rating of one capacitor unit.

For the node 675 capacitor bank,

$$Q_{\text{unit}} = 200 \text{ kvar}$$

and

$$Q_{CB,675} = 200m \text{ kvar}$$

where

$$m \in \{0, 1, 2, 3\}.$$

Thus,

$$Q_{CB,675} \in \{0, 200, 400, 600\} \text{ kvar}$$

16.7.12 Important Difference Between DG and CB Control

The key distinction is

$$\text{DG inverter} \rightarrow \text{Continuous reactive-power control}$$

whereas

Capacitor bank $\rightarrow$ Discrete reactive-power control

Therefore, the cooperative-control algorithm can directly determine a continuous reactive-power utilization ratio for an inverter, but the capacitor-bank value must ultimately be translated into an integer number of switched-on units.

16.7.13 Overall Interpretation

The main purpose of this subsection is to ensure that the optimized reactive-power utilization obtained from the cooperative controller can be implemented by the **actual discrete capacitor-bank hardware**.

The complete concept can be summarized as

Optimized $\alpha_{qi} \rightarrow$ CB Utilization Range
 $\rightarrow$ Discrete Switching Decision
 $\rightarrow$ Reactive Power $\rightarrow$ Voltage Regulation

At the same time, the switching strategy attempts to avoid unnecessary capacitor switching:

Avoid Unnecessary Switching $\rightarrow$ Less Wear $\rightarrow$ Lower Maintenance $\rightarrow$ Longer Equipment Life

Thus, **capacitor switching provides the discrete implementation mechanism for the reactive-power utilization ratio determined by the cooperative-control algorithm**, while the lower-limit, upper-limit, and rounding methods provide different ways of converting the optimized control variable into a practical switching state.

17 Results

The effectiveness of the proposed **phase-level cooperative control method** is evaluated through simulation studies performed on the modified IEEE 13-bus distribution system. The results are presented for two different control scenarios in order to investigate the effect of including different reactive-power resources in the cooperative-control process.

The simulations are performed using the solar PV-based distributed generators introduced in the modified system, with their active-power output varying according to changes in solar irradiance.

17.1 Simulation Scenarios

Two scenarios are considered for evaluating the proposed cooperative control at the **phase level**.

17.1.1 Scenario 1: DGs and Capacitor Banks Participate

In the first scenario, **both the distributed generators (DGs) and capacitor banks (CBs)** participate in the cooperative-control process.

Therefore, the available reactive-power resources include:

DGs + CBs

The controller can coordinate the reactive-power contribution of both types of resources to improve the voltage profile.

For the capacitor banks, the paper selects the **upper limit of the specified range of α_{qi}** . This establishes the operating condition for the CBs during the simulation.

The basic control structure can therefore be represented as:

Voltage Fluctuation $\rightarrow$ Cooperative Controller
 $\rightarrow \begin{cases} \text{DG Reactive Power} \\ \text{CB Reactive Power} \end{cases}$
 $\rightarrow$ Voltage Regulation

17.1.2 Scenario 2: Only DGs Participate

In the second scenario, only the **distributed generators** participate in the cooperative control.

Therefore:

DGs only

Only the DGs are used as the controllable reactive-power resources.

The capacitor banks are not included as active participants in the cooperative optimization in this scenario.

Comparing the two scenarios allows the effect of capacitor-bank participation to be investigated.

Thus, the comparison is essentially:

Scenario 1: DGs + CBs

versus

Scenario 2: DGs only

This comparison is useful for understanding how the availability of additional reactive-power resources affects voltage regulation.

17.2 Simulation Duration

The simulations are performed over a period of:

15 minutes

or:

900 seconds

Therefore, the active-power variation of the solar PV DGs and the corresponding voltage-control response are evaluated over a **900-second simulation period**.

This duration allows the controller to be tested under changing solar-generation conditions rather than under only a constant operating condition.

17.3 Solar PV Generation Model

The DGs considered in the study are assumed to be **solar photovoltaic (PV) generators**.

Unlike conventional generators whose output may be controlled relatively independently of weather conditions, solar PV generation depends strongly on the available **solar irradiance**.

Consequently, the active power injected by the DGs changes with time:

$$P_{DG} = P_{DG}(t)$$

The paper uses a time-varying active-power profile to represent this behavior.

The active-power profile shown in Figure 3 therefore represents the changing total active power injected by the DGs during the 900-second simulation period.

17.4 Interpretation of Figure 3

Figure 3 presents the **active power injected by the DGs in the modified IEEE 13-bus system**.

The two axes of the graph are:

- **Horizontal axis:** Time in seconds
- **Vertical axis:** Active power in kW

The active-power output varies approximately from **50 kW to 300 kW** over the simulation period.

The important feature of the graph is that the DG output is **not constant**.

Instead, it changes continuously with time, exhibiting both increases and decreases.

This variation represents changes in solar irradiance.

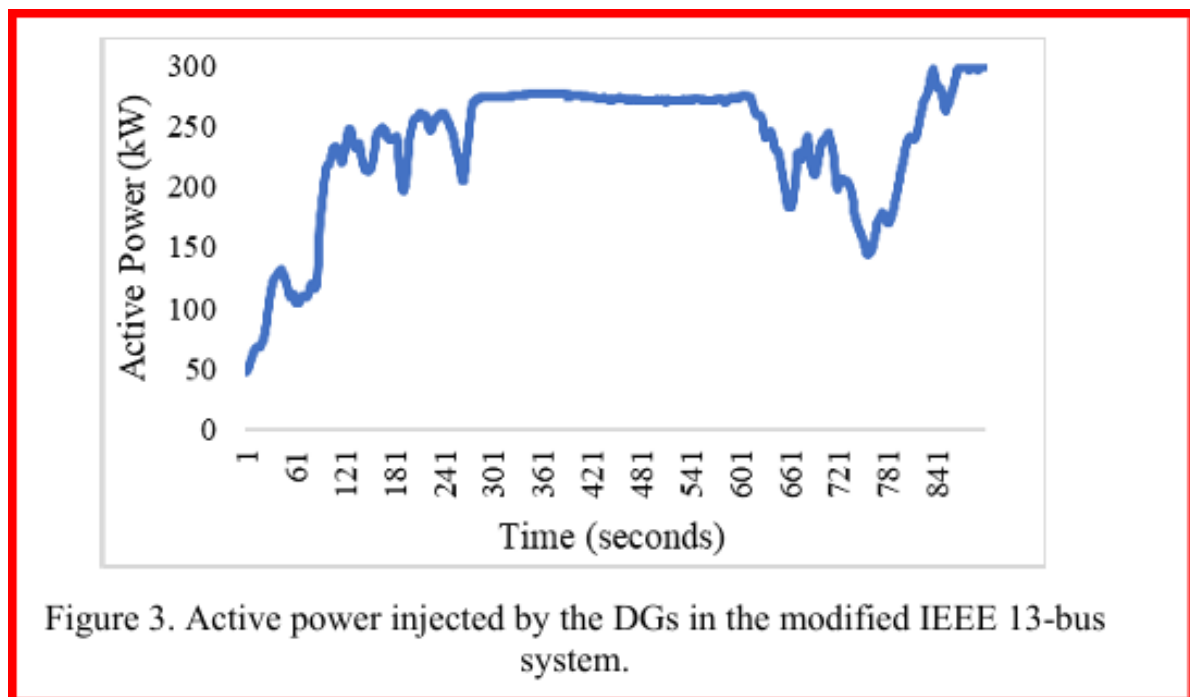


Figure 3: Active power injected by the DGs in the modified IEEE 13-bus system.

17.4.1 Initial Generation

At the beginning of the simulation, the active power is relatively low, approximately around:

50 kW

The power then increases during the early portion of the simulation.

This can be interpreted as an increase in available solar irradiance.

Conceptually:

Increasing solar irradiance → Increasing PV generation

17.4.2 Rapid Variations During the Initial Period

After the initial increase, the graph shows several fluctuations in active power.

The output rises to values above approximately:

200 kW

and then experiences several short-term increases and decreases.

These rapid changes are important for the study because they represent the type of **intermittent renewable-generation behavior** that can produce voltage fluctuations in a distribution system.

Thus, the graph provides the disturbance/input condition under which the proposed cooperative voltage-control method is tested.

17.4.3 Higher and Relatively Stable Generation

In the middle portion of the simulation, the active power reaches approximately:

250–275 kW

and remains relatively close to this level for a significant period.

The curve is not perfectly constant, however, and small variations are still visible.

This represents a period in which the solar irradiance is comparatively stable, resulting in relatively stable PV active-power generation.

17.4.4 Significant Reduction in Active Power

A particularly important feature occurs later in the simulation, where the active power decreases substantially.

The output falls from approximately:

250–275 kW

to approximately:

150 kW

before increasing again.

This significant reduction represents a substantial decrease in solar PV generation.

A possible physical interpretation is a reduction in solar irradiance, such as that produced by changing cloud conditions.

The important relationship is:

Reduction in Solar Irradiance → Reduction in PV Active Power

Such rapid generation changes can produce corresponding voltage variations in the distribution network.

17.4.5 Recovery Toward Maximum Generation

Toward the end of the simulation, the active power increases again and reaches values close to:

300 kW

The graph therefore demonstrates that the solar PV generation can recover after a significant reduction in output.

This produces another change in the power-flow conditions of the distribution network.

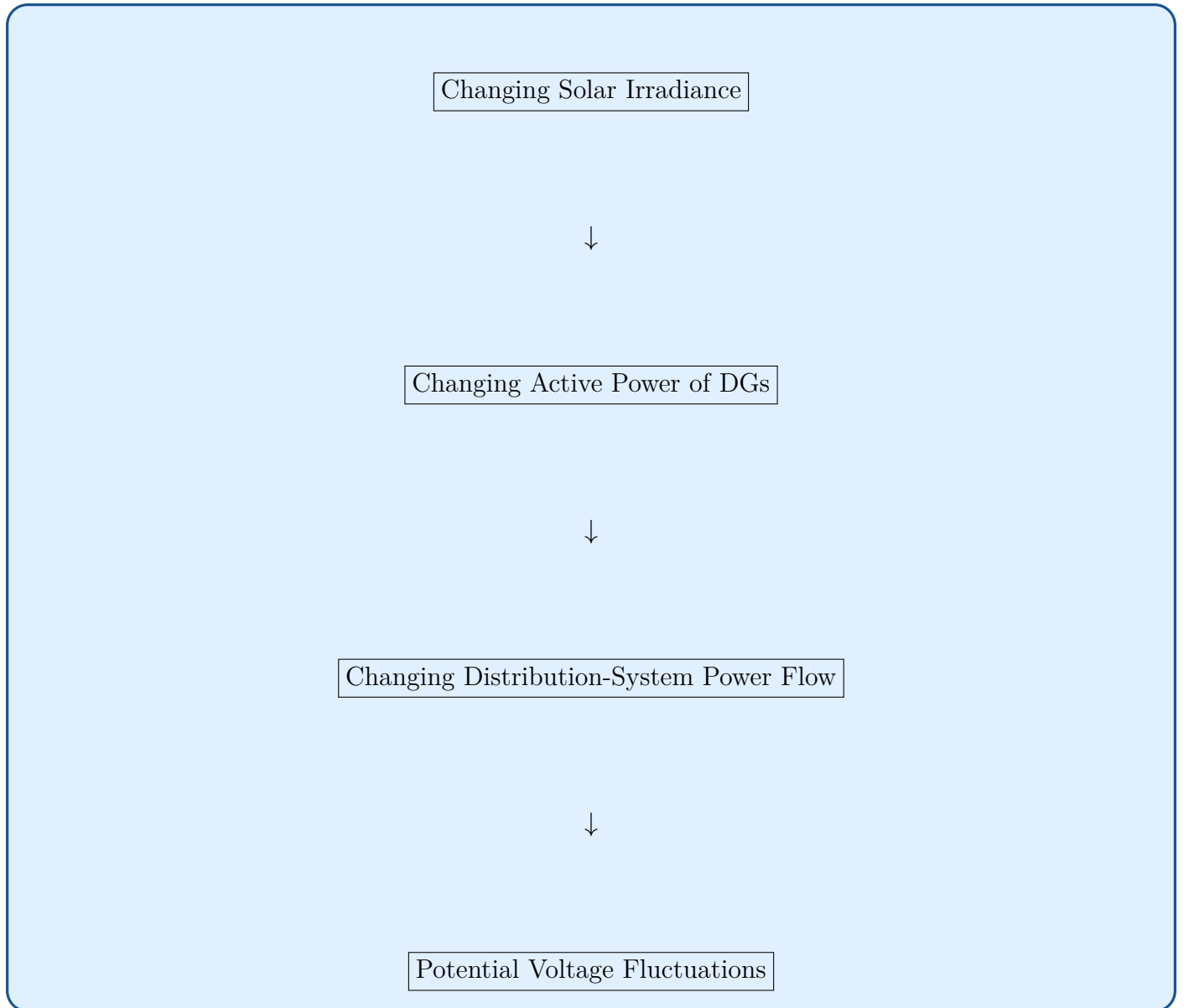
Therefore, throughout the 900-second period, the controller is exposed to:

Increasing generation + decreasing generation
+ rapid generation fluctuations

17.5 Why Figure 3 Is Important for the Results

Figure 3 is important because it establishes the **input disturbance** used to evaluate the voltage-control strategy.

The research problem can be represented as:



The purpose of the cooperative controller is then to use available reactive-power resources to mitigate the resulting voltage variations.

Thus, Figure 3 does **not** directly show the voltage-control performance. Instead, it shows the **time-varying DG active-power input** that creates the challenging operating conditions for the subsequent voltage-control analysis.

17.6 Relationship Between Active and Reactive Power

It is important to distinguish the active power shown in Figure 3 from the reactive power controlled by the proposed method.

The graph shows:

$$P_{DG}$$

which represents the **active power injected by the DGs**.

The cooperative controller, on the other hand, determines the appropriate reactive-power utilization:

$$Q_{Gi}$$

through the utilization ratio:

$$\alpha_{qi} = \frac{Q_{Gi}}{\bar{Q}_{Gi}}$$

Therefore, the overall control process can be viewed as:

$$\text{Solar Irradiance} \rightarrow P_{DG}(t)$$

while simultaneously:

$$P_{DG}(t) \rightarrow \text{Available Reactive-Power Capability}$$

and then:

$$\text{Cooperative Optimization} \rightarrow \alpha_{qi} \rightarrow Q_{Gi}$$

The resulting reactive-power response is used to regulate the voltage.

17.7 Significance of the Two Scenarios

The two simulation scenarios allow the authors to investigate whether adding capacitor banks to the cooperative-control process provides additional benefits.

17.7.1 Scenario 1

$$\text{DGs} + \text{CBs}$$

provides a larger set of reactive-power resources.

17.7.2 Scenario 2



DGs

uses only the reactive-power capability of the DG inverters.

This provides a meaningful comparison because it shows how the voltage-control performance changes when capacitor-bank resources are also available.

The comparison also relates to the practical issue of **capacitor-bank switching**. Although capacitor banks can provide reactive-power support, excessive switching can increase operating and maintenance costs and reduce equipment lifetime. The proposed method therefore considers capacitor-bank operation carefully rather than assuming unlimited switching.

17.8 Overall Interpretation of the Initial Results Section

The results section begins by establishing the simulation conditions before presenting the actual voltage-control results.

The sequence is:

Modified IEEE 13-Bus System



Solar PV DGs



Time-Varying Solar Irradiance



Variable DG Active Power



Voltage Fluctuation



Cooperative Reactive-Power Control

The two scenarios are then used to determine how effectively the proposed method can respond to these changing conditions.

17.8.1 summary

The proposed phase-level cooperative-control method is evaluated through two simulation scenarios on the modified IEEE 13-bus distribution system. In the first scenario, both distributed generators and capacitor banks participate in the cooperative control, while in the second scenario only the distributed generators participate. The upper limit of the specified reactive-power utilization ratio range is selected for the capacitor banks in the first scenario. The simulations are performed over a period of 15 minutes, corresponding to 900 seconds. The DGs are modeled as solar PV generators, and their active-power output varies with changes in solar irradiance. Figure 3 illustrates the resulting time-varying active power injected by the DGs. The active power varies approximately from 50 kW to nearly 300 kW, with several significant increases and decreases during the simulation period. These variations represent the intermittent nature of solar generation and provide the changing operating conditions under which the proposed cooperative voltage-control strategy is tested. The resulting variation in DG active power changes the power-flow conditions of the distribution network and can consequently produce voltage fluctuations. The subsequent results therefore evaluate how the cooperative reactive-power control responds to these variations and improves the voltage profile of the system.

17.9 Scenario 1 – Including DGs and Capacitor Banks in Cooperative Control

In **Scenario 1**, both the **distributed generators (DGs)** and **capacitor banks (CBs)** are included in the cooperative-control process. The purpose of this scenario is to investigate how the proposed controller performs when all available reactive-power resources participate in the optimization.

Three different cases are considered by changing the weighting factors in the objective function. The weighting factors determine the relative importance given to **voltage deviation** and **reactive-power generation/consumption**.

The results are presented in **Table 1** and **Figures 4–6**.

17.9.1 Selection of Weighting Factors

Recall that the objective function contains two major objectives:

1. **Minimization of voltage deviation**
2. **Minimization of reactive-power utilization**

The voltage-related weights for QG and non-QG nodes are selected to be equal:

$$w_{VQG} = w_{VNQG}$$

and their common value is represented by:

$$w_v$$

Therefore, the objective function can be viewed conceptually as:

$$F = w_v(\text{Voltage Deviation}) + w_\alpha(\text{Reactive-Power Utilization})$$

Three cases are investigated.

17.9.2 Case 1: Minimization of Voltage Deviation

For Case 1, the selected weights are:

$$w_v = 1$$

$$w_\alpha = 0$$

Therefore, the objective function gives importance only to **voltage deviation**.

The optimization can be conceptually written as:

$$\min F = \text{Voltage Deviation}$$

while the reactive-power utilization term is not given any weight.

The purpose of this case is to determine how effectively the cooperative controller can improve the voltage profile when **voltage regulation is the only optimization priority**.

The gradient gains obtained for this case are:

$$\beta_{QG_i} = 11$$

$$\beta_{NQG_{ik}} = 11$$

The resulting total cost is:

$$F = 0.0025$$

This is the smallest total cost among the three cases presented in Table 1. However, this value should be interpreted together with the selected objective weights. In Case 1, only the voltage-deviation objective

is active.

17.9.3 Figure 4 – Voltage Regulation for Case 1

Figure 4 presents the voltage responses obtained for:

$$w_v = 1, \quad w_\alpha = 0$$

Four nodes are considered in the figure:

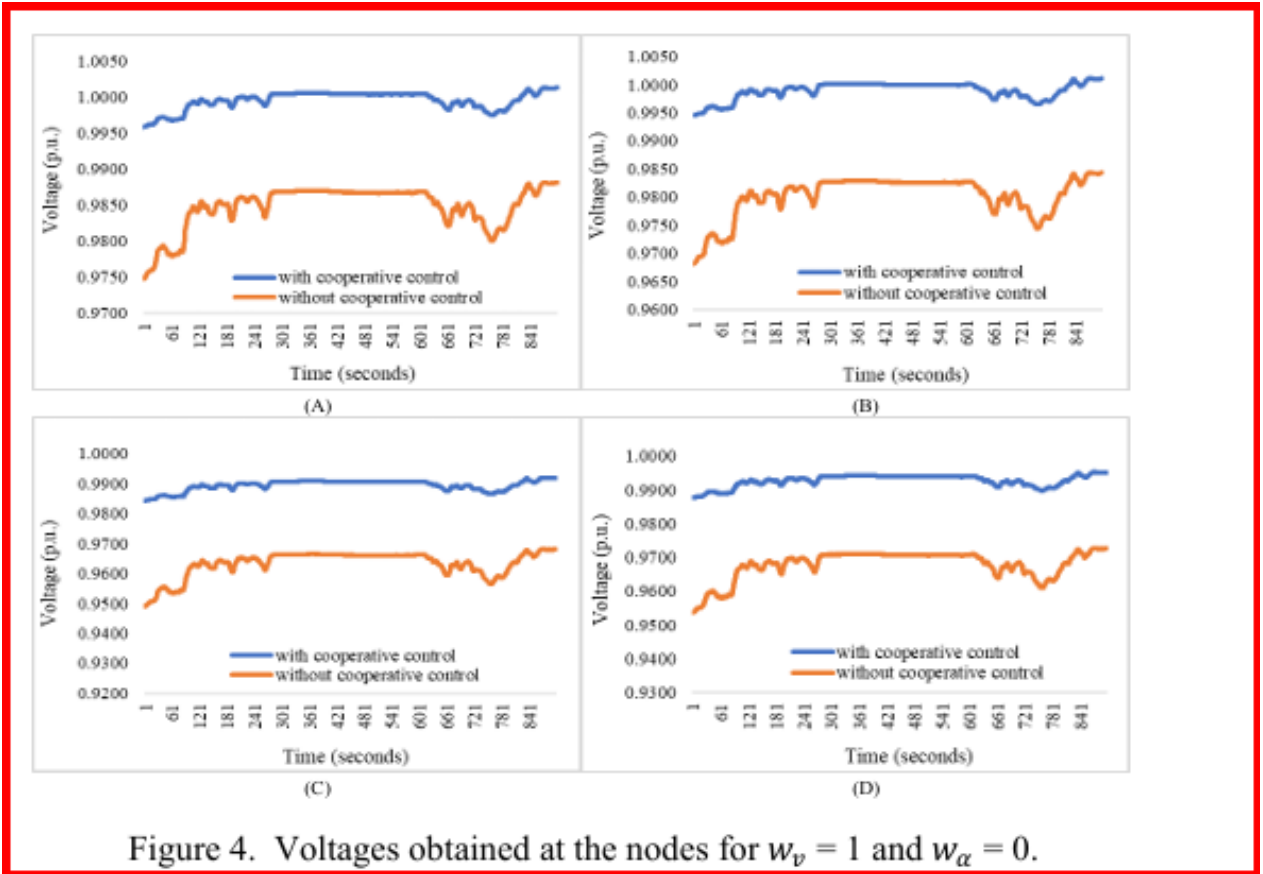


Figure 4: Voltages obtained at the nodes for $w_v = 1$ and $w_\alpha = 0$.

- **Figure 4(A):** QG node 632.1
- **Figure 4(B):** QG node 670.1
- **Figure 4(C):** QG node 675.1
- **Figure 4(D):** non-QG node 692.1

Each graph contains two voltage profiles:

- **With cooperative control**
- **Without cooperative control**

The purpose of this comparison is to directly demonstrate the effect of implementing the proposed cooperative-control strategy.

17.9.4 Figure 4(A) – QG Node 632.1

Figure 4(A) represents the voltage at **QG node 632.1**.

The blue curve represents the voltage obtained **with cooperative control**, while the orange curve represents the voltage **without cooperative control**.

The controlled voltage remains close to:

1.0 p.u.

for a large portion of the simulation.

In contrast, the uncontrolled voltage remains noticeably below 1 p.u. and shows larger variations with time.

Therefore, the comparison demonstrates that cooperative control significantly improves the voltage profile at node 632.1.

17.9.5 Figure 4(B) – QG Node 670.1

Figure 4(B) shows the voltage response at **QG node 670.1**.

Similar to node 632.1, the voltage with cooperative control remains much closer to the desired voltage level.

The uncontrolled case exhibits a lower voltage profile and more noticeable deviations.

Therefore:

Cooperative control → Reduced voltage deviation at 670.1

This confirms that the improvement is not limited to one particular DG location.

17.9.6 Figure 4(C) – QG Node 675.1

Figure 4(C) presents the voltage at **QG node 675.1**.

The controlled voltage is maintained close to the nominal value, whereas the uncontrolled voltage remains at a lower level.

This is particularly relevant because node 675 contains the modified capacitor-bank arrangement.

The result demonstrates that when the reactive-power resources participate cooperatively, the voltage at this QG location can be improved substantially.

17.9.7 Figure 4(D) – Non-QG Node 692.1

Figure 4(D) is particularly important because it represents a **non-QG node**.

Node 692.1 does not directly provide the reactive-power control resource being optimized, yet its voltage also improves when cooperative control is implemented.

The blue curve corresponding to cooperative control remains much closer to the desired voltage level than the uncontrolled case.

This demonstrates an important feature of the proposed method:

Control at QG nodes → Voltage improvement at non-QG nodes

Therefore, the cooperative-control strategy is capable of producing a **network-wide voltage improvement**, rather than improving only the voltages at locations containing DGs or capacitor banks.

17.9.8 Overall Observation From Figure 4

Across all four subfigures, the same general behavior can be observed:

$V_{\text{with cooperative control}} > V_{\text{without cooperative control}}$

for the operating conditions represented in the plots.

More importantly, the controlled voltage profiles remain closer to the desired value of approximately:

1.0 p.u.

The voltage fluctuations caused by the changing solar-PV generation are also mitigated.

Therefore, Case 1 confirms that when voltage deviation is given the highest priority, the proposed cooperative control can effectively improve the voltage profile at both QG and non-QG nodes.

17.9.9 Case 2: Minimization of Reactive-Power Generation/Consumption

In Case 2, the weighting factors are reversed:

$$w_v = 0$$

$$w_\alpha = 1$$

Therefore, the optimization focuses exclusively on **minimizing reactive-power generation or consumption**.

The conceptual objective becomes:

$$\min F = \text{Reactive-Power Utilization}$$

The voltage-deviation term is assigned zero weight.

The gradient gains obtained in this case are:

$$\beta_{QG_i} = 0.0014$$

and

$$\beta_{NQG_{ik}} = 0.0014$$

The resulting total cost is:

$$F = 848.3971$$

The much larger numerical value of F , compared with Case 1, results from the fact that the objective function is now dominated by the reactive-power utilization component rather than the voltage-deviation component.

17.9.10 Figure 5 – Voltage Results for Case 2

Figure 5 presents the voltage profiles for:

$$w_v = 0, \quad w_\alpha = 1$$

Again, the four subfigures correspond to:

- (A): QG node 632.1
- (B): QG node 670.1

- (C): QG node 675.1
- (D): non-QG node 692.1

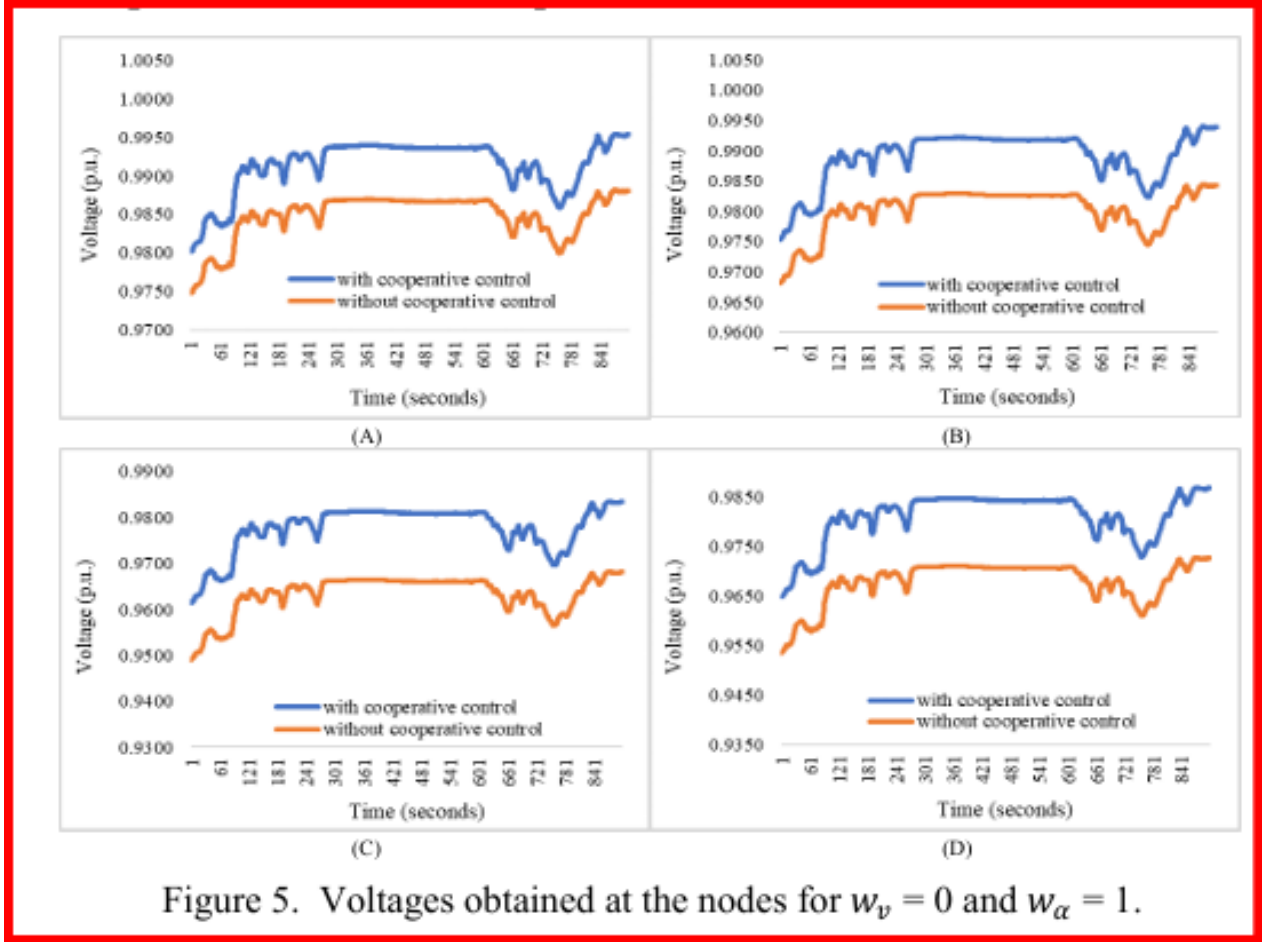


Figure 5: Voltages obtained at the nodes for $w_v = 0$ and $w_\alpha = 1$.

The blue curves represent the results **with cooperative control**, while the orange curves represent the results **without cooperative control**.

Although voltage deviation is not directly included in the objective in this case, the controlled voltage profiles are still improved compared with the uncontrolled condition.

This is an important observation because it demonstrates that reactive-power coordination itself can produce a beneficial effect on the system voltage.

17.9.11 Interpretation of Figure 5

At all four investigated nodes, the cooperative-control curves remain above the corresponding uncontrolled curves and are generally closer to the nominal voltage level.

This demonstrates that:

Reactive-power coordination $\rightarrow$ Voltage improvement

even when the explicit voltage-deviation weight is set to zero.

However, the primary purpose of this case is not to obtain the minimum possible voltage deviation. Instead, the controller is prioritizing the reduction of reactive-power generation or consumption.

Therefore, Case 2 demonstrates the **trade-off between voltage regulation and reactive-power utilization**.

17.9.12 Case 3: Combined Voltage and Reactive-Power Objective

Case 3 considers a combination of the two objectives.

The selected weighting factors are:

$w_v = 1000$

$w_\alpha = 0.005$

Therefore, both voltage deviation and reactive-power utilization contribute to the objective function.

Conceptually:

$$\min F = 1000(\text{Voltage Deviation}) + 0.005(\text{Reactive-Power Utilization})$$

The gradient gains are:

$\beta_{QG_i} = 0.011$

and

$\beta_{NQG_{ik}} = 0.011$

The resulting total cost is:

$F = 6.9429$

This case is particularly important because it represents a **compromise between the two competing objectives**.

17.9.13 Figure 6 – Combined Objective

Figure 6 presents the voltage results for:

$$w_v = 1000, \quad w_\alpha = 0.005$$

The four subfigures again correspond to:

$$632.1, \quad 670.1, \quad 675.1, \quad 692.1$$

with the first three being QG nodes and the last being a non-QG node.

The results show that the cooperative-control voltage profiles remain close to the nominal voltage while the controller simultaneously considers reactive-power utilization.

Thus, Case 3 provides a more balanced control strategy compared with the two extreme cases.

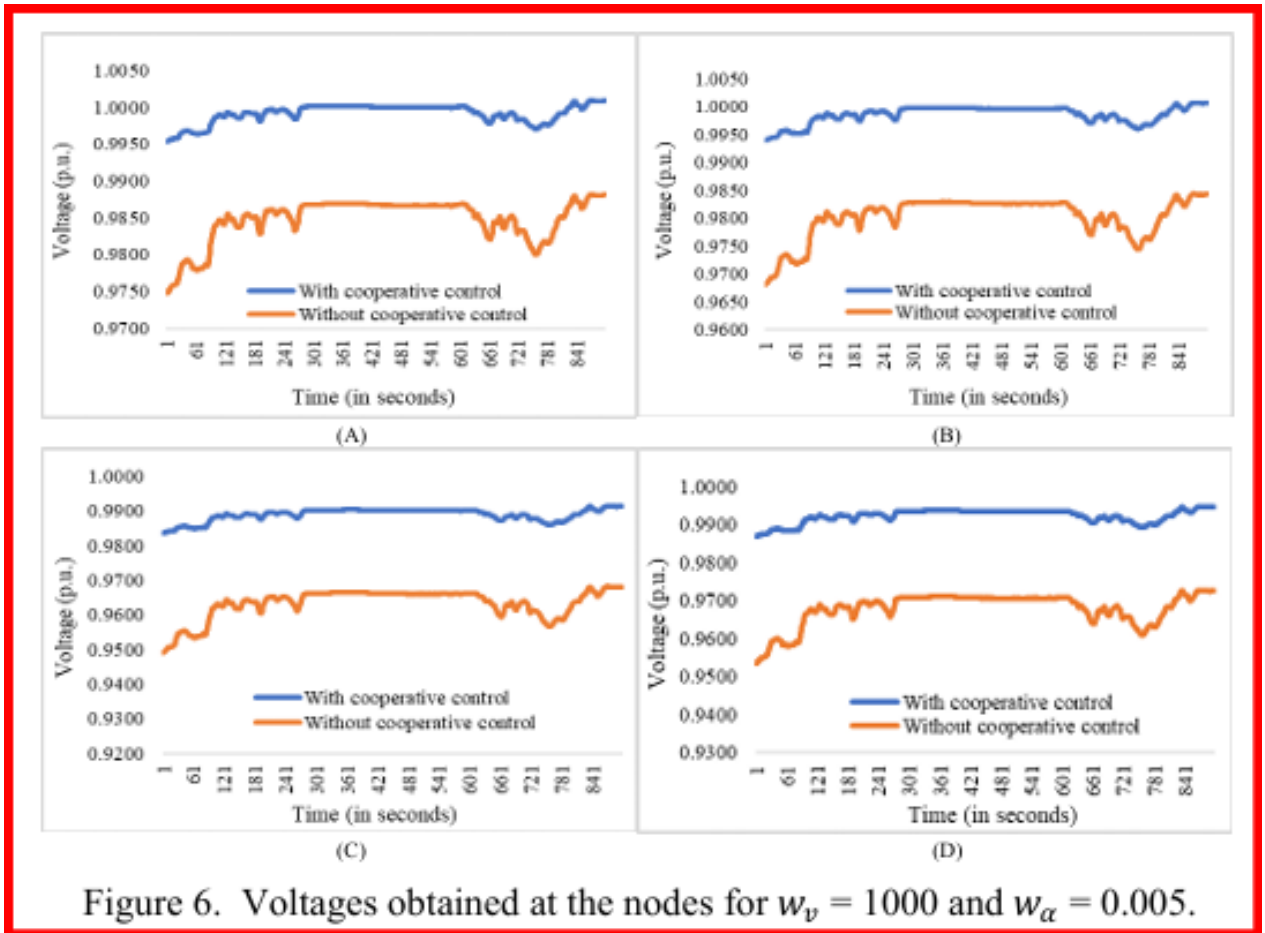


Figure 6. Voltages obtained at the nodes for $w_v = 1000$ and $w_\alpha = 0.005$.

Figure 6: Voltages obtained at the nodes for $w_v = 1000$ and $w_\alpha = 0.005$.

17.9.14 Comparison of the Three Cases

The three cases can be summarized as follows:

Table 1: Comparison of the three weighting-factor cases.

Case	w_v	w_α	Main Objective	β_{QG_i}	$\beta_{NQG_{ik}}$	F
Case 1	1	0	Minimize voltage deviation	11	11	0.0025
Case 2	0	1	Minimize reactive-power generation/consumption	0.0014	0.0014	848.3971
Case 3	1000	0.005	Combined voltage and reactive-power objective	0.011	0.011	6.9429

17.9.15 Significance of the Weight Selection

The results clearly demonstrate that the choice of weighting factors has a significant effect on the optimization behavior.

Case 1

$$w_v = 1, \quad w_\alpha = 0$$

places complete emphasis on voltage regulation.

Therefore, the controller is allowed to use reactive power as necessary to minimize voltage deviation.

Case 2

$$w_v = 0, \quad w_\alpha = 1$$

places complete emphasis on minimizing reactive-power utilization.

Therefore, voltage deviation is not directly penalized by the objective.

Case 3

$$w_v = 1000, \quad w_\alpha = 0.005$$

considers both objectives simultaneously.

Therefore, the controller seeks a balance between:

Maintaining good voltage

and

Avoiding excessive reactive-power utilization

This combined formulation is more representative of the practical control problem because the objective is not simply to obtain the closest possible voltage at any cost.

17.9.16 Role of Gradient Gains

The table also reports the gradient gains:

$$\beta_{QG_i}$$

and

$$\beta_{NQG_{ik}}$$

These gains are used in the iterative cooperative-control process to update the optimization variables.

For the three cases, the gains are:

Case 1

$$\beta_{QG_i} = \beta_{NQG_{ik}} = 11$$

Case 2

$$\beta_{QG_i} = \beta_{NQG_{ik}} = 0.0014$$

Case 3

$$\beta_{QG_i} = \beta_{NQG_{ik}} = 0.011$$

The different gain values are selected according to the corresponding weighting configuration so that the iterative optimization can operate under the respective objective priorities.

17.9.17 Overall Result of Scenario 1

The most important observation from Figures 4–6 is that **cooperative control improves the voltage profile compared with the uncontrolled case.**

This improvement is visible at:

QG nodes

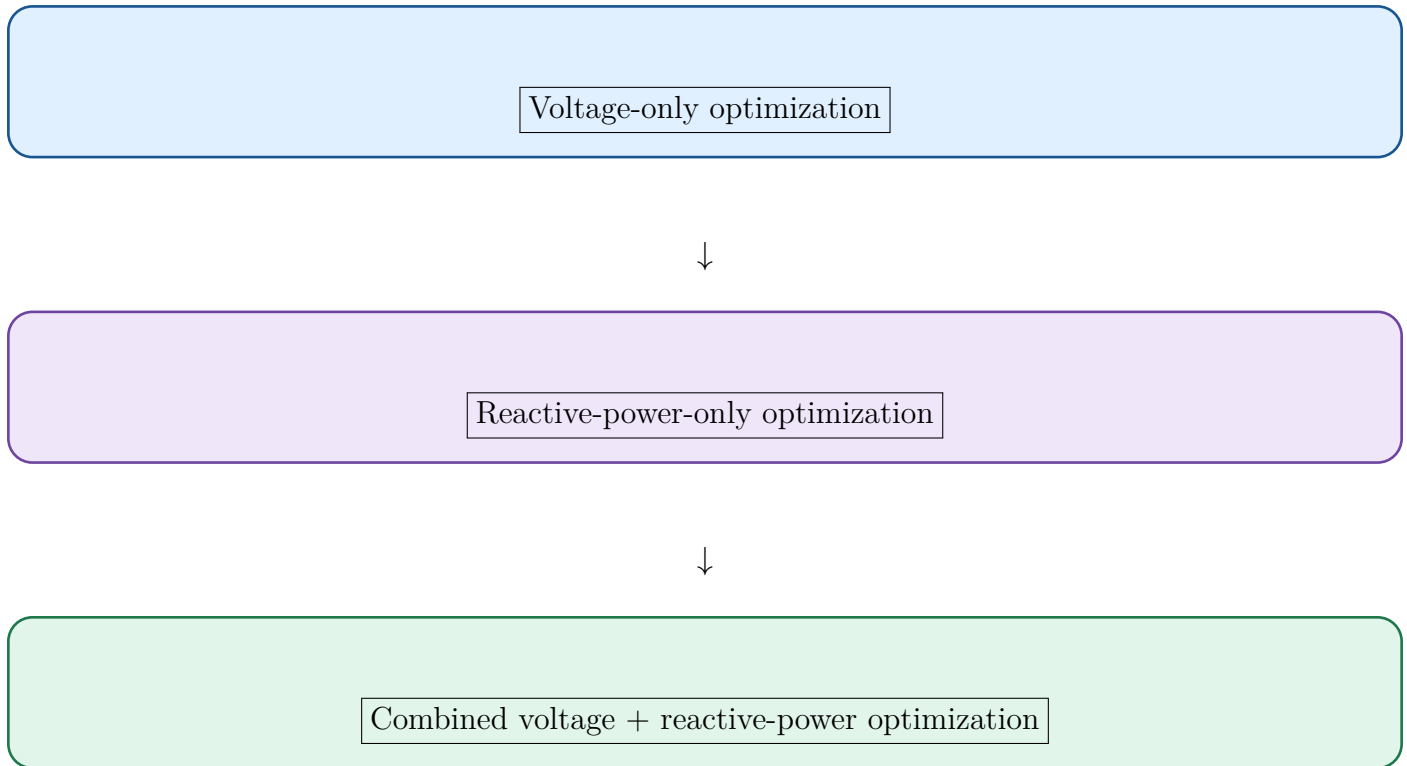
as well as:

Non-QG nodes

Therefore, the proposed method does not simply regulate the voltage at the locations where DGs and capacitor banks are installed. Through the cooperative optimization and communication topology, the effects of the reactive-power control extend to other nodes in the distribution system.

The results also demonstrate that the controller can be configured according to different objectives.

The three cases show the progression:



17.9.18 Main Finding From Scenario 1

The results confirm the effectiveness of the proposed cooperative-control method for mitigating the voltage variations produced by the changing solar-PV generation.

When cooperative control is implemented, the voltage curves are generally maintained closer to the desired voltage level than in the case without cooperative control.

At the same time, the weighting factors provide flexibility in deciding whether the controller should prioritize:

- voltage regulation,
- reduction of reactive-power utilization, or
- a combination of both.

This is particularly important because excessive reactive-power operation can increase inverter utilization and contribute to aging, while insufficient reactive-power support can result in larger voltage deviations.

Therefore, the combined objective in **Case 3** provides a mechanism for considering both requirements simultaneously.

17.9.19 summary

Scenario 1 evaluates the proposed phase-level cooperative control with both DGs and capacitor banks participating as reactive-power resources. Three cases are investigated by varying the weighting factors of voltage deviation and reactive-power utilization.

In **Case 1**, $w_v = 1$ and $w_\alpha = 0$, so the controller exclusively minimizes voltage deviation, resulting in a total cost of 0.0025.

In **Case 2**, $w_v = 0$ and $w_\alpha = 1$, and the optimization focuses exclusively on minimizing reactive-power generation or consumption, resulting in a total cost of 848.3971.

In **Case 3**, $w_v = 1000$ and $w_\alpha = 0.005$, allowing both voltage deviation and reactive-power utilization to be considered, with a total cost of 6.9429.

Figures 4–6 show the voltage profiles at QG nodes 632.1, 670.1, and 675.1 and non-QG node 692.1. In all three cases, the voltage profiles obtained with cooperative control are improved compared with the corresponding uncontrolled profiles.

The results demonstrate that the proposed cooperative-control strategy can effectively improve voltage conditions at both QG and non-QG nodes, while the weighting factors provide flexibility to balance voltage regulation against reactive-power utilization.

17.10 Scenario 2 – Including Only DGs in the Cooperative Control

In **Scenario 2**, the cooperative-control strategy is modified so that only the **distributed generators (DGs)** participate in the cooperative-control process. The capacitor banks (CBs) are excluded from the cooperative optimization.

The purpose of this scenario is to evaluate whether the DGs alone can provide sufficient reactive-power support for voltage regulation and to compare the result with **Scenario 1**, where both DGs and CBs participated.

17.10.1 Configuration of Scenario 2

The main difference between Scenario 1 and Scenario 2 is the set of reactive-power resources participating in the controller.

Scenario 1

DGs + CBs → Cooperative Control

DGs only → Cooperative Control

Scenario 2

Therefore, in Scenario 2, the reactive-power contribution of the capacitor banks is not optimized by the cooperative controller.

The capacitor-bank utilization ratio is fixed at

$$\alpha_{qi} = 1$$

for all capacitor banks.

This means that the capacitor banks are maintained at their specified maximum utilization condition rather than being treated as optimization variables in the cooperative-control process.

17.10.2 Weighting Factors and Gradient Gains

Three different cases are considered in Scenario 2. The weighting factors determine which component of the objective function receives greater importance.

Table 2: Weighting factors, gradient gains, and total cost for Scenario 2.

Case	w_v	w_α	β_{QG_i}	$\beta_{NQG_{ik}}$	F
Case 1	1	0	0.1	0.1	0.0037
Case 2	0	1	0.0013	0.0013	0.0026
Case 3	1000	0.005	0.002	0.0002	4.5565

These three cases correspond to three different optimization priorities.

17.10.3 Case 1 – Voltage Regulation Priority

For Case 1, the weighting factors are

$$w_v = 1, \quad w_\alpha = 0$$

Therefore, the optimization gives priority to **voltage regulation**.

Conceptually, the objective becomes

$$\min F = \text{Voltage Deviation}$$

The reactive-power utilization term does not contribute directly to the weighted objective because

$$w_\alpha = 0.$$

The gradient gains used in this case are

$$\boxed{\beta_{QG_i} = 0.1} \quad \boxed{\beta_{NQG_{ik}} = 0.1}$$

The resulting total cost is

$$\boxed{F = 0.0037}$$

Interpretation The controller is therefore allowed to use the reactive-power capability of the DGs as required to achieve better voltage regulation.

The basic control philosophy is

Voltage deviation $\rightarrow$ DG reactive-power adjustment $\rightarrow$ Improved voltage profile

17.10.4 Figure 7 – Voltage Profile for Case 1

Figure 7 presents the voltage profiles obtained for

$$\boxed{w_v = 1, \quad w_\alpha = 0}$$

The four subfigures represent the same four nodes considered previously:

- (A) QG node 632.1
- (B) QG node 670.1
- (C) QG node 675.1
- (D) non-QG node 692.1

Each graph compares the voltage obtained:

- with cooperative control, and
- without cooperative control.

The blue curve represents the voltage with cooperative control, whereas the orange curve represents the voltage without cooperative control.

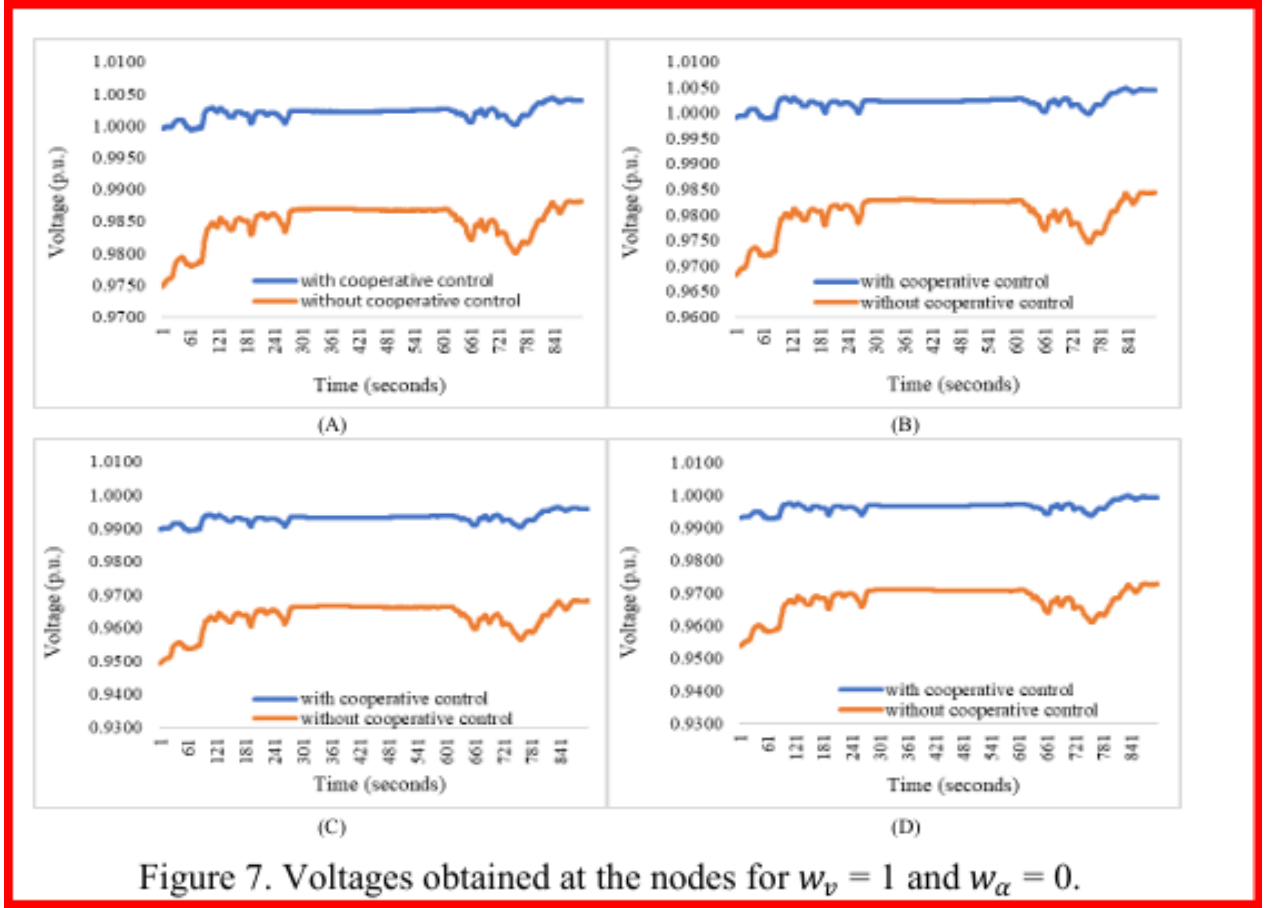


Figure 7: Voltages obtained at the nodes for $w_v = 1$ and $w_\alpha = 0$.

17.10.5 QG Node 632.1

In Figure 7(A), the controlled voltage is maintained close to the nominal value of approximately

$$1.0 \text{ p.u.}$$

The uncontrolled voltage remains considerably lower.

Therefore, the DG reactive-power control provides voltage support at this QG node.

17.10.6 QG Node 670.1

Figure 7(B) shows a similar behavior.

With cooperative control, the voltage remains close to the desired voltage level, whereas the uncontrolled voltage is lower.

This indicates that the reactive-power response of the DGs is capable of improving the voltage at the QG node.

17.10.7 QG Node 675.1

Figure 7(C) also demonstrates an improvement in voltage when cooperative control is applied.

The controlled voltage is maintained close to the nominal value, while the uncontrolled case shows a larger voltage deviation.

This is particularly useful for evaluating the effect of removing the capacitor banks from the cooperative-control optimization.

17.10.8 Non-QG Node 692.1

Figure 7(D) is particularly important because node 692.1 is a **non-QG node**.

Although the node itself does not contain a controllable reactive-power source participating in the optimization, its voltage is nevertheless affected by the reactive-power changes produced by the DGs elsewhere in the network.

The improvement demonstrates that

DG reactive-power control → Network power-flow change → Non-QG voltage improvement

Thus, the cooperative controller provides a network-wide effect rather than only regulating the voltage directly at DG locations.

17.10.9 Case 2 – Reactive-Power Utilization Priority

For Case 2,

$$w_v = 0, \quad w_\alpha = 1$$

Therefore, the optimization focuses on **reactive-power generation/consumption** rather than directly minimizing voltage deviation.

Conceptually,

$$\min F = \text{Reactive-Power Utilization}$$

The gradient gains are

$$\beta_{QG_i} = 0.0013 \quad \beta_{NQG_{ik}} = 0.0013$$

The resulting total cost is

$$F = 0.0026$$

Interestingly,

$$0.0026 < 0.0037.$$

However, the numerical values of the total cost should be interpreted according to the corresponding objective weighting because the objective functions are different between the two cases.

17.10.10 Figure 8 – Voltage Profile for Case 2

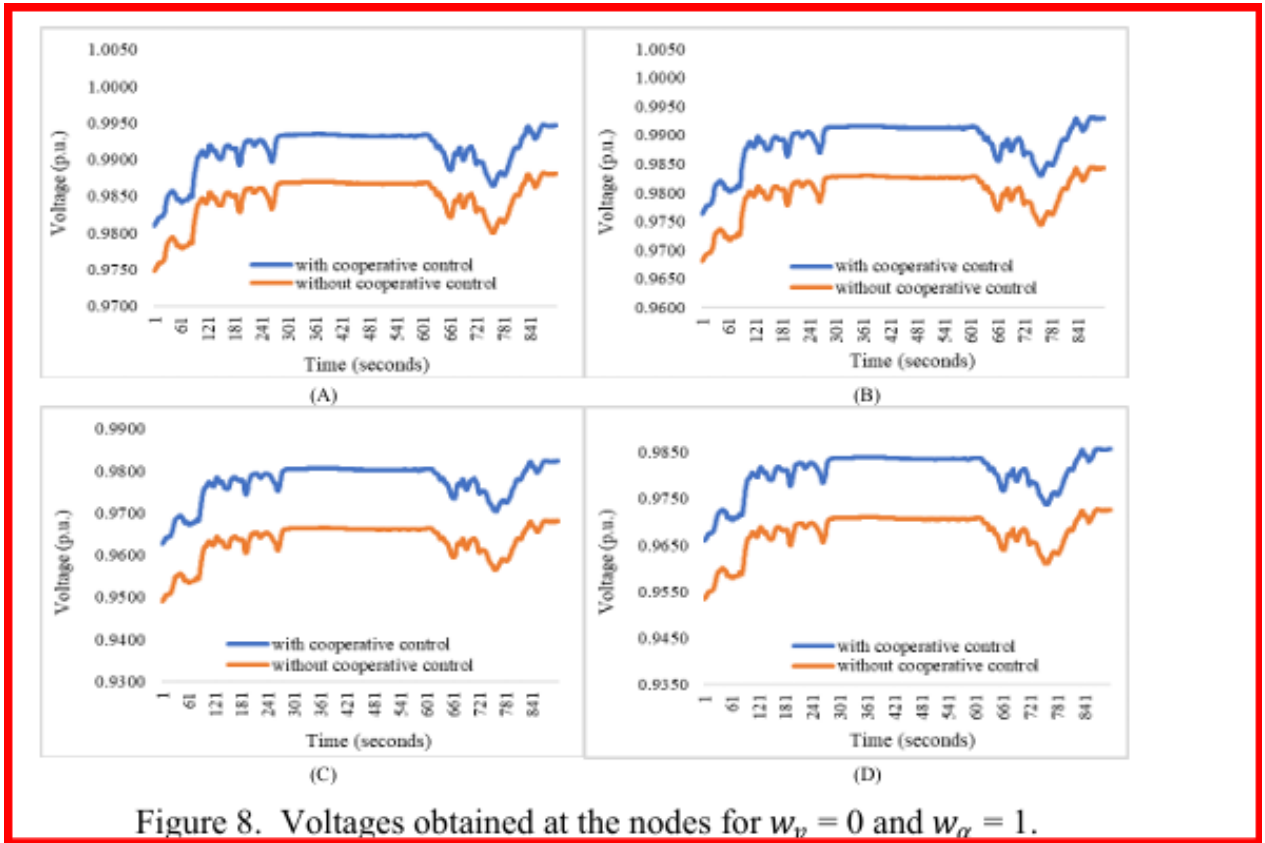


Figure 8: Voltages obtained at the nodes for $w_v = 0$ and $w_\alpha = 1$.

Figure 8 presents the voltage profiles for

$$w_v = 0, \quad w_\alpha = 1$$

The four nodes considered are

$$632.1, \quad 670.1, \quad 675.1, \quad 692.1$$

The blue curves show the results with cooperative control, while the orange curves represent the corresponding uncontrolled condition.

Even though voltage deviation is not explicitly prioritized in the objective for this case, the voltage profiles with cooperative control still show improvement.

This indicates that

Reactive-power coordination → Improved network voltage

can occur even when the controller's primary objective is reactive-power utilization.

17.10.11 Case 3 – Combined Voltage and Reactive-Power Objective

Case 3 considers both voltage regulation and reactive-power utilization.

The selected weighting factors are

$$w_v = 1000, \quad w_\alpha = 0.005$$

Therefore, the objective function can be conceptually expressed as

$$\min F = 1000(\text{Voltage Deviation}) + 0.005(\text{Reactive-Power Utilization})$$

This case represents a **compromise between the two objectives**.

The gradient gains are

$$\beta_{QG_i} = 0.002 \quad \beta_{NQG_{ik}} = 0.0002$$

The resulting total cost is

$$F = 4.5565$$

17.10.12 Figure 9 – Voltage Profile for Case 3

Figure 9 presents the voltage results for

$$w_v = 1000, \quad w_\alpha = 0.005$$

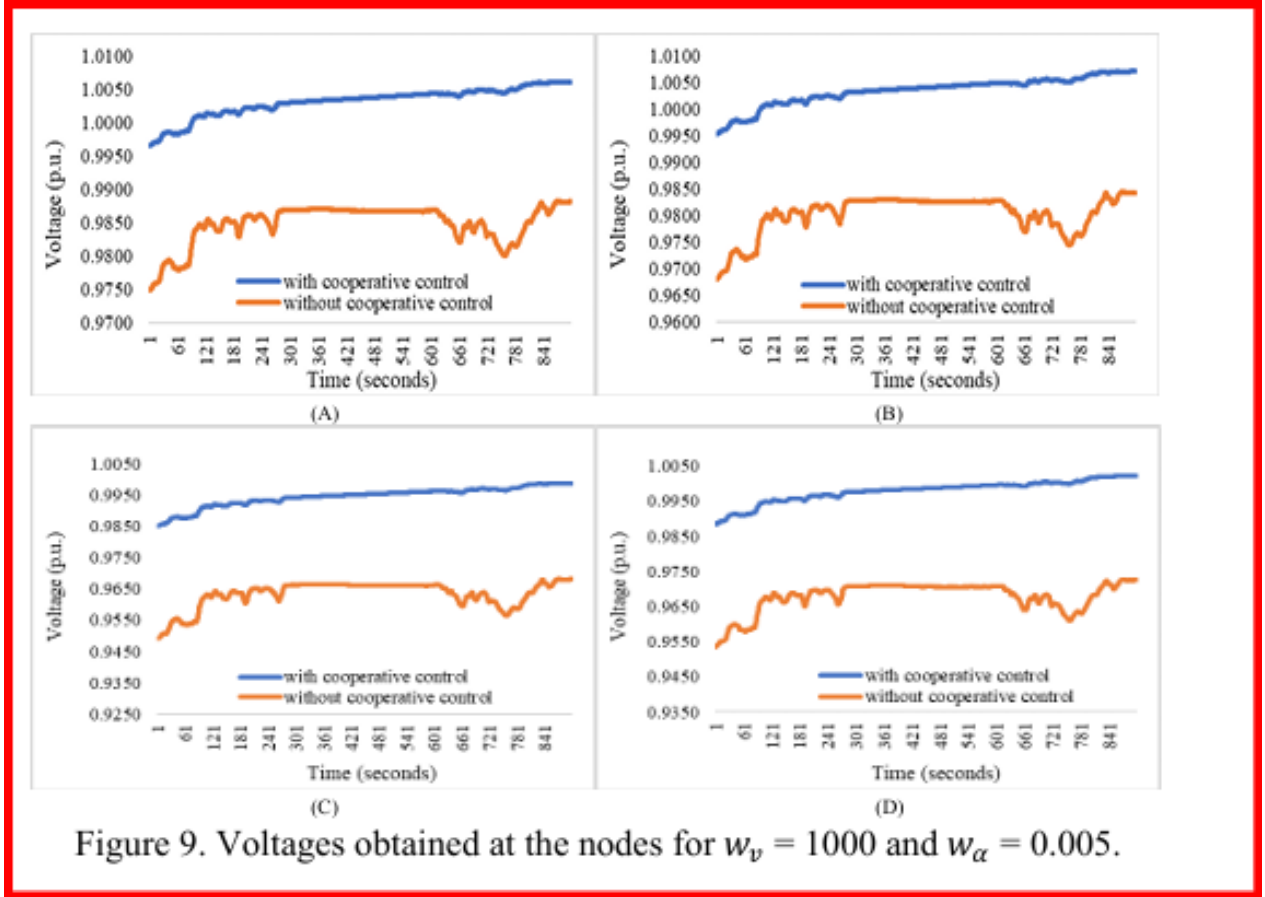


Figure 9: Voltages obtained at the nodes for $w_v = 1000$ and $w_\alpha = 0.005$.

The four subfigures correspond to

$$632.1, \quad 670.1, \quad 675.1, \quad 692.1$$

The cooperative-control voltage profiles remain substantially closer to the desired voltage level than the uncontrolled profiles.

The important difference is that the controller is now considering both

$$\text{Voltage deviation}$$

and

$$\text{Reactive-power utilization}$$

rather than focusing exclusively on one of them.

17.10.13 Comparison of the Three Cases

The three cases can be summarized as follows.

Case 1

$$w_v = 1, \quad w_\alpha = 0$$

Main priority:

Voltage regulation

Total cost:

$$F = 0.0037$$

Case 2

$$w_v = 0, \quad w_\alpha = 1$$

Main priority:

Minimization of reactive-power utilization

Total cost:

$$F = 0.0026$$

Case 3

$$w_v = 1000, \quad w_\alpha = 0.005$$

Main priority:

Combined voltage regulation and reactive-power utilization

Total cost:

$$F = 4.5565$$

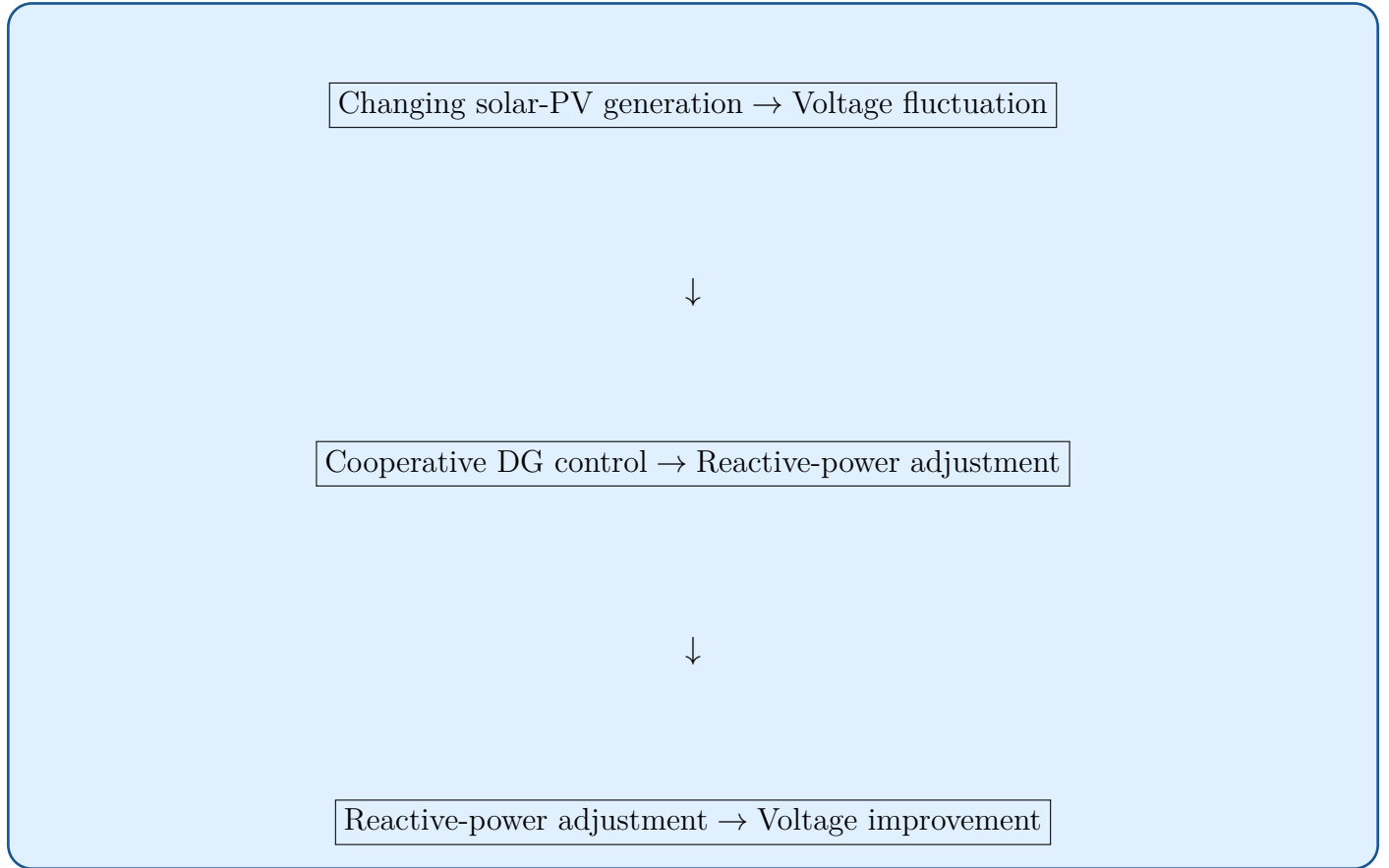
17.10.14 Most Important Observation From Scenario 2

The most important conclusion from the three figures is that the **voltage improves in Scenario 2 as well**.

In all three cases, the voltage obtained with cooperative control is generally closer to the desired voltage than the corresponding voltage without cooperative control.

This means that the DGs themselves possess sufficient reactive-power control capability to provide significant voltage support.

The fundamental mechanism is



17.10.15 Why Is Scenario 2 Different From Scenario 1?

The major difference is the participation of capacitor banks.

Scenario 1

DGs + CBs

participate in cooperative control.

Scenario 2

DGs only

participate.

In Scenario 2, the capacitor-bank utilization ratios are fixed at

$$\alpha_{qi} = 1$$

for all CBs.

Thus, the controller does not optimize the CB utilization ratio.

The DGs are therefore responsible for the cooperative reactive-power adjustments.

17.10.16 Why Is the Total Cost Lower for Case 2 in Scenario 2?

The paper specifically notes that the total cost obtained for

$$w_v = 0, \quad w_\alpha = 1$$

in Scenario 2 is significantly lower than the corresponding result in Scenario 1.

The reason is the **non-participation of the capacitor banks**.

In Scenario 1, the objective includes the reactive-power contributions associated with both DGs and CBs.

In Scenario 2, only the DGs participate in the cooperative-control optimization.

Therefore, the optimization problem has fewer participating reactive-power resources.

The comparison can be represented as

Scenario 1: DGs + CBs → More reactive-power resources included in optimization

Scenario 2: DGs only → CBs excluded from cooperative optimization

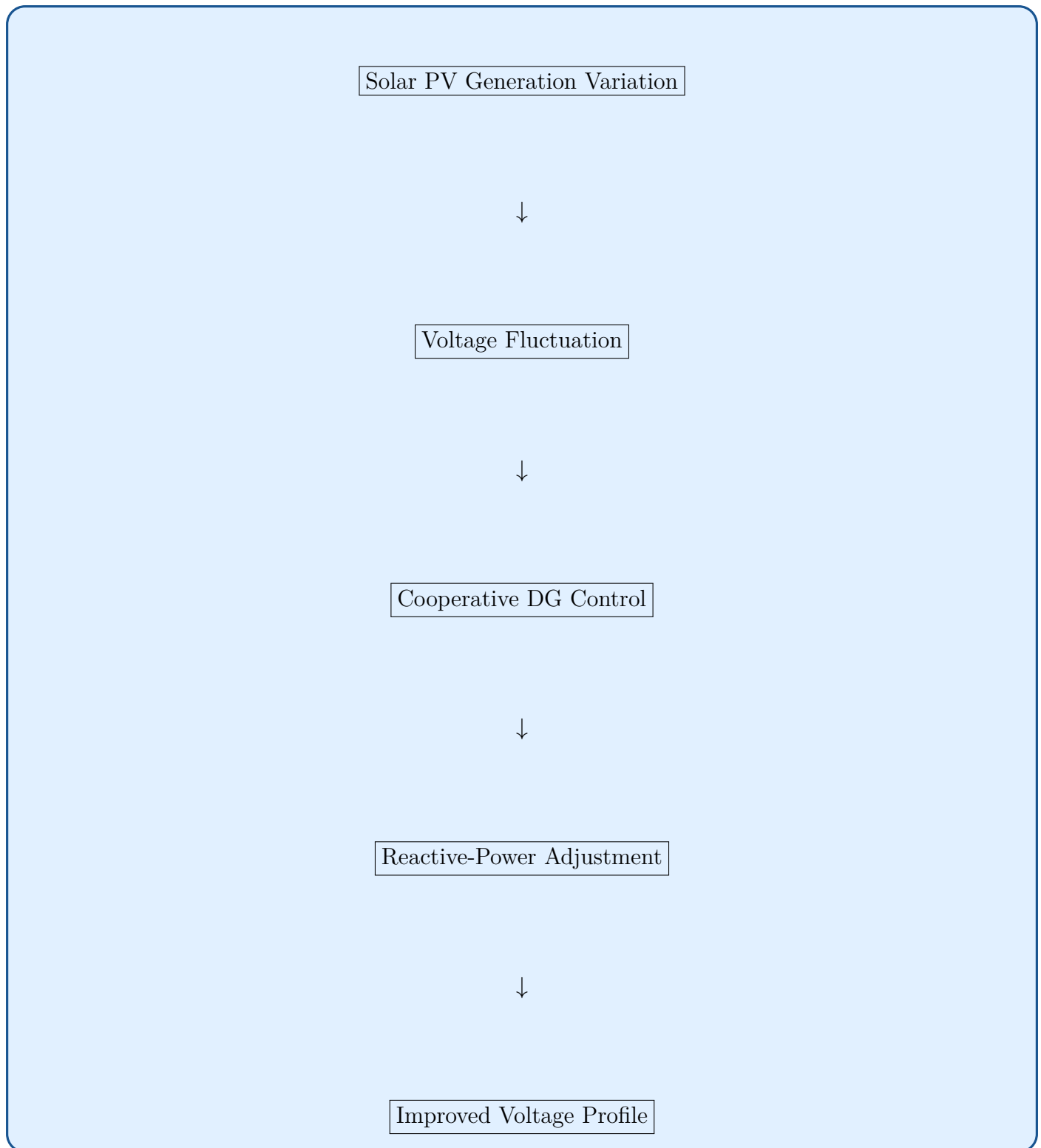
This difference affects the calculated objective value.

17.10.17 Overall Interpretation of Scenario 2

Scenario 2 demonstrates that **capacitor banks are not essential for the cooperative-control algorithm to improve the voltage profile**.

Even when only DGs participate, the controller can coordinate the reactive-power capability of the DG inverters and improve the voltage at both QG and non-QG nodes.

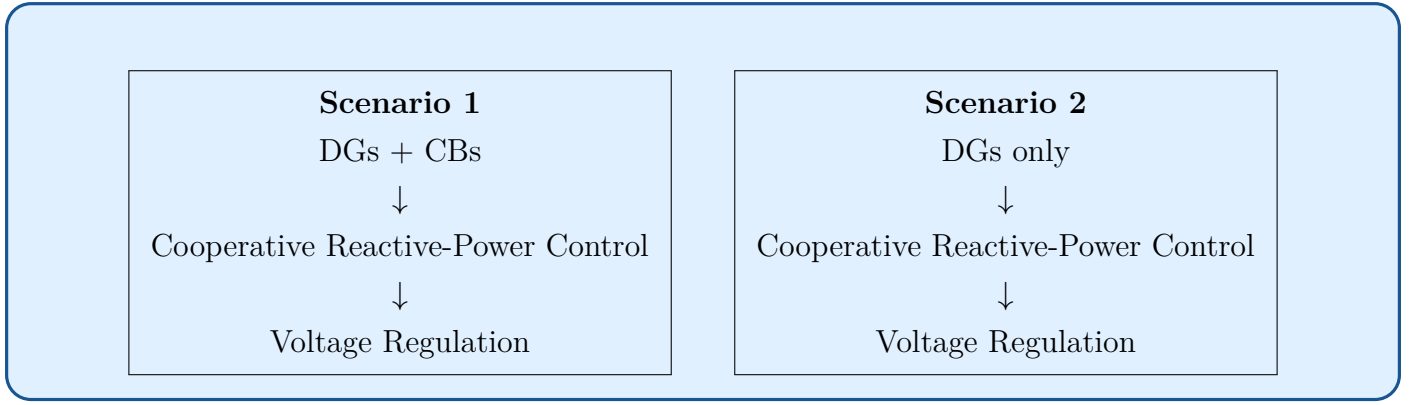
The overall process can be summarized as



Therefore, Scenario 2 provides an important validation of the proposed method: the DG inverters alone can participate cooperatively to mitigate voltage fluctuations, even when the capacitor banks are not part of the cooperative-control optimization.

17.10.18 Final Comparison

The two scenarios can be compared as follows:



The results shown in Figures 7–9 demonstrate that both scenarios can improve the voltage profile, while Scenario 2 specifically demonstrates the effectiveness of using **DG reactive-power capability alone** for cooperative voltage regulation.

17.11 Results Discussion

The results obtained from the different cases in Scenario 1 provide important insights into the behavior of the proposed cooperative-control strategy. In particular, the results demonstrate the **trade-off between voltage regulation and reactive-power utilization**. The discussion is based primarily on the objective function given in Equation (2) and the voltage profiles presented in Figures 4–6.

17.11.1 Effect of the Reactive-Power Weight

The objective function for a QG node is:

$$f_i = w_{VQG} \left[\left| \frac{V_i}{V_{bi}} \right| - 1 \right]^2 + w_\alpha \alpha_{qi}^2 \overline{Q}_{Gi}$$

The two weighting factors determine the relative importance of the two objectives:

$$w_v \rightarrow \text{Voltage deviation}$$

and

$$w_\alpha \rightarrow \text{Reactive-power utilization}$$

A particularly important case is:

$$w_\alpha = 1, \quad w_v = 0$$

which corresponds to **Case 2**.

In this case, the objective function does not assign any weight to voltage deviation. Instead, the optimization focuses entirely on minimizing the reactive-power utilization term:

$$\alpha_{qi}^2 \overline{Q}_{Gi}$$

Because the optimization attempts to minimize this term, the optimal reactive-power utilization ratio tends toward:

$$\alpha_{qi} = 0$$

This means that the controller attempts to minimize reactive-power generation or consumption by the DGs.

17.11.2 Consequence of $\alpha_{qi} = 0$

Although minimizing reactive-power utilization may reduce the stress associated with reactive-power operation, it creates an important consequence for voltage regulation.

If

$$\alpha_{qi} = 0$$

then from

$$Q_{Gi} = \alpha_{qi} \overline{Q}_{Gi}$$

we obtain:

$$Q_{Gi} = 0$$

Therefore, there is effectively no reactive-power injection from the DG for mitigating the voltage fluctuations.

As a result:

No reactive-power support $\rightarrow$ Voltage fluctuations remain

The paper therefore points out that voltage fluctuations would still exist when

$$w_\alpha = 1, \quad w_v = 0$$

because there is no reactive-power injection available through the DGs to mitigate those fluctuations.

17.11.3 Interpretation of Figure 5

This behavior can be observed in Figure 5, which corresponds to:

$$w_v = 0, \quad w_\alpha = 1$$

Even though cooperative control is implemented, noticeable voltage fluctuations remain in the voltage profiles.

This is an important result because it demonstrates that **cooperative control by itself does not guarantee minimum voltage deviation**. The result depends strongly on what the controller is instructed to optimize.

In Case 2, the controller has been instructed to prioritize:

Minimum reactive-power utilization

rather than:

Minimum voltage deviation

Therefore, the optimization naturally avoids significant reactive-power utilization, even though reactive power would be useful for improving the voltage.

This demonstrates the importance of appropriately selecting the objective-function weights.

17.11.4 Trade-Off Between Voltage Regulation and Reactive-Power Utilization

The results demonstrate a fundamental trade-off.

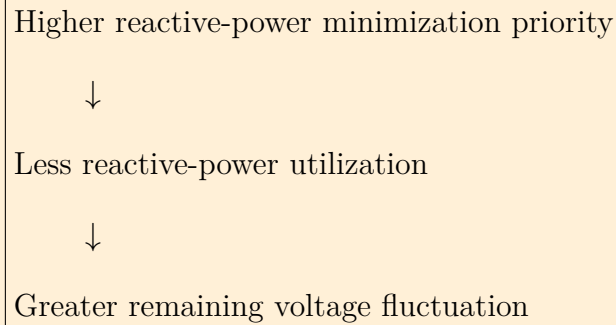
If the controller gives greater importance to voltage regulation, it can use more reactive power to compensate for voltage fluctuations.

Thus:

Higher voltage-regulation priority → More reactive-power support → Better voltage profile

However, greater reactive-power utilization is undesirable because the paper considers its potential impact on inverter aging.

On the other hand:



Therefore, neither objective should necessarily be considered independently in a practical implementation.

This motivates the use of a **combined objective function**, as investigated in Case 3.

17.11.5 Why Is the Cooperative-Control Voltage Higher Than the Base Case?

The paper observes that the voltage obtained using cooperative control is generally **higher than the base case**.

Here, the base case refers to the condition **without cooperative control**.

This behavior occurs because the cooperative-control strategy coordinates the reactive-power resources to improve the voltage profile.

However, the paper also explains that the voltage obtained under cooperative control is influenced by the **capacitor banks**.

The capacitor banks are subject to restrictions on their switching operations. Because their switching cannot occur arbitrarily at every instant, the controller may use capacitor banks in a way that results in a higher voltage level than the uncontrolled/base case.

Therefore, the observed voltage increase is not simply due to unrestricted DG reactive-power injection. It also reflects the coordinated operation and switching constraints of the capacitor-bank units.

17.11.6 Role of Capacitor-Bank Switching Restrictions

The capacitor banks are discrete reactive-power resources. Their operation involves switching individual capacitor units.

If switching were allowed continuously and without restrictions, the controller could theoretically respond to every small voltage fluctuation by changing the capacitor-bank state.

However, this is not desirable because frequent switching can increase:

- Equipment wear,
- Operation and maintenance requirements, and
- Long-term equipment degradation.

Therefore, the proposed method considers restrictions on capacitor-bank switching.

The paper explains that the voltage obtained with cooperative control can be higher than the base case because of the **use of capacitor banks under the imposed switching restrictions**.

This illustrates an important practical consideration:

Voltage Regulation $\neq$ Only DG Reactive-Power Control

Instead, the voltage response results from the coordinated operation of:

DGs + Capacitor Banks + Switching Constraints

17.11.7 Strong Correlation Assumption Between Solar Generators

The study assumes that the solar generators have **strong correlations** in their generation profiles.

In other words, the solar PV generators are assumed to experience similar variations in solar irradiance.

Conceptually:

Solar irradiance at $DG_1 \approx$ Solar irradiance at $DG_2 \approx$ Solar irradiance at DG_3

Therefore, their active-power outputs tend to vary in a correlated manner.

This assumption simplifies the simulation scenario used in the present study.

However, actual solar PV installations may experience different irradiance conditions depending on their geographical locations, cloud movement, orientation, and other environmental factors.

For this reason, the authors identify the examination of different solar-irradiance profiles as an area for future investigation.

17.11.8 Future Investigation With Different Solar-Irradiance Profiles

The paper proposes that future work should examine the effectiveness of the presented cooperative-control method under **different solar-irradiance profiles**.

This would provide a more comprehensive evaluation of the controller.

For example, future simulations could consider situations where:

$$P_{DG1}(t) \neq P_{DG2}(t) \neq P_{DG3}(t)$$

because the individual PV generators experience different irradiance conditions.

Such a situation would produce a more diverse generation pattern and could create more complicated voltage variations.

The objective would be to determine whether the proposed cooperative-control strategy remains effective when the DGs do not behave in a strongly correlated manner.

17.11.9 Need for Improved Consensus-Based Cooperative Control

Another future research direction identified by the authors is the development of appropriate methods to extend the **consensus algorithm** used in the cooperative control.

The current approach allows QG nodes to cooperate in determining the reactive-power utilization.

However, a more complicated situation can arise when different QG nodes require **opposite control actions**.

For example:

QG_1 : Increase reactive-power generation

while simultaneously:

QG_2 : Decrease reactive-power generation

Both actions may be necessary to maintain an appropriate voltage profile across the distribution network.

This means that the optimal solution may require some QG nodes to increase their reactive-power contribution while other QG nodes decrease theirs.

17.11.10 Why Different QG Nodes May Need Different Reactive-Power Actions

The reason for this behavior is that the QG nodes are located at different electrical positions in the distribution network.

Reactive power injected at one node does not affect every point in the network equally.

For example, suppose

$$QG_1$$

is located close to a low-voltage region, while

$$QG_2$$

is located in an area where voltage is already relatively high.

In such a situation, increasing reactive power at both locations may not be appropriate.

The controller may instead determine:

$$QG_1 : Q \uparrow$$

and

$$QG_2 : Q \downarrow$$

to obtain a better overall voltage profile.

Therefore, future improvements to the consensus algorithm should be able to accommodate these **opposite reactive-power control requirements**.

17.11.11 Future Dual-Layer Voltage–Var Control

The paper also identifies a future research direction involving a **dual-layer voltage-var control scheme**.

The proposed concept divides the control process into two stages:

Layer 1: Higher-Level Optimization

and:

Layer 2: Local Control

The first layer would determine optimal settings for the available voltage-control resources.

The second layer would implement those settings locally.

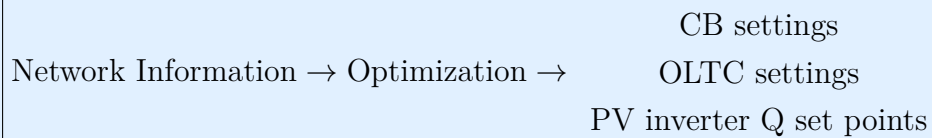
17.11.12 First Layer – Optimal Settings

The first layer would focus on determining the optimal operating settings for devices such as:

- Capacitor banks (CBs)
- On-load tap changers (OLTCs)
- Reactive-power set points for PV inverters

The output of this optimization stage would therefore include suitable operating values for the different voltage-control devices.

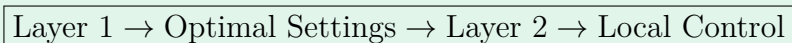
Conceptually:



17.11.13 Second Layer – Local Control

The second layer would be responsible for implementing the optimal settings obtained from the first layer.

Thus:



The paper refers to this as a **dual-layer control structure**, with the local-control stage operating as the second layer.

This approach could provide a way of combining system-level optimization with faster local responses.

17.11.14 Overall Significance of the Results Discussion

The results discussion provides several important conclusions about the proposed cooperative-control method.

First The selection of the objective-function weights is critical.

$$w_v, w_\alpha$$

determine the balance between voltage regulation and reactive-power utilization.

Second If reactive-power utilization is minimized exclusively:

$$w_v = 0, \quad w_\alpha = 1$$

the optimal reactive-power utilization tends toward zero, and voltage fluctuations remain.

Third The participation of capacitor banks affects the resulting voltage profile because their switching is subject to restrictions.

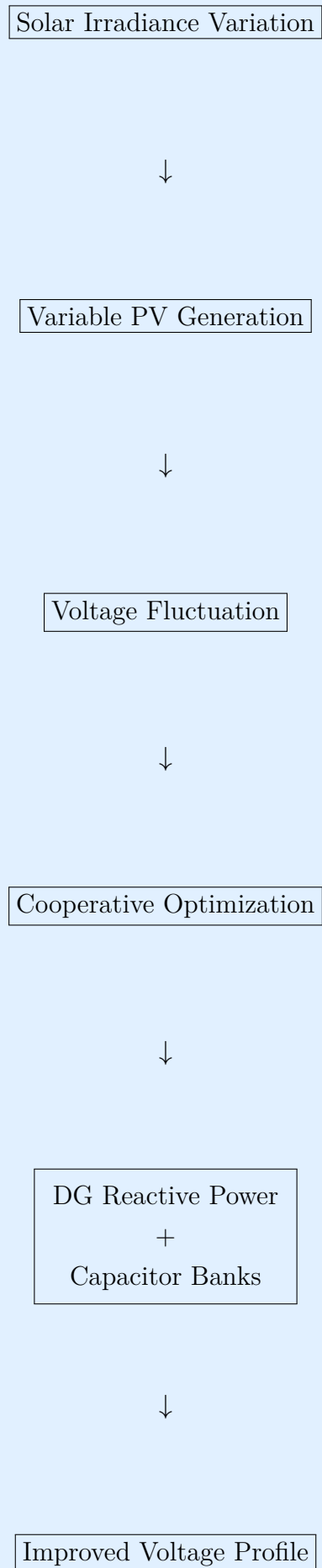
Fourth The proposed approach is based on the assumption that solar generators have strongly correlated generation profiles, and the authors identify different irradiance profiles as an important area for future research.

Fifth Future improvements should allow different QG nodes to make different, and potentially opposite, reactive-power adjustments.

Sixth The authors propose extending the approach toward a dual-layer voltage-var control architecture in which optimal settings for CBs, OLTCs, and PV-inverter reactive-power set points are determined at a higher level and then transferred to local controllers.

17.11.15 Overall Control Concept

The complete interpretation of the results discussion can be summarized as:



while simultaneously considering:

Reactive-Power Utilization

and:

Capacitor Switching Constraints

The results therefore demonstrate that the proposed method provides a flexible framework for balancing **voltage performance, reactive-power utilization, and practical equipment constraints**.

18 Conclusions

The study presents a **cooperative-control-based method for mitigating fast voltage variations in distribution systems with solar photovoltaic (PV) distributed generators**. The main source of the voltage variation considered in the study is the change in solar irradiance, which causes the active-power output of the PV-based DGs to vary with time.

The overall concept of the proposed method is:

Solar Irradiance Variation → PV Power Variation → Voltage Variation → Cooperative Reactive-Power C

18.1 Mitigation of Fast Voltage Variations

One of the primary objectives of the research is to address **fast voltage variations** caused by changes in solar irradiance.

Because solar PV generation is dependent on irradiance, a rapid change in irradiance produces a corresponding change in DG active-power output. This variation can influence the power flow in the distribution network and consequently cause voltage fluctuations.

The proposed cooperative-control method addresses this problem by coordinating the available reactive-power resources.

Thus, instead of attempting to control the voltage only through conventional devices such as OLTCs and capacitor banks, the method makes use of the reactive-power capability of the DGs.

18.2 Minimization of Voltage Deviation

The proposed control method attempts to maintain the voltage close to its desired or base value.

The voltage-deviation component of the objective function is based on:

$$\left[\left| \frac{V}{V_b} \right| - 1 \right]^2$$

Therefore, the controller attempts to minimize the deviation between the actual voltage magnitude and its corresponding base voltage.

The simulation results demonstrate that the voltage profiles at the investigated QG and non-QG nodes are improved when cooperative control is implemented.

Hence,

Cooperative Control → Reduced Voltage Deviation

18.3 Consideration of Reactive-Power Utilization and Aging

An important feature of the proposed method is that it does not consider voltage regulation alone.

The objective function also considers the **reactive-power utilization ratio**:

$$\alpha_{qi}$$

through the term:

$$w_\alpha \alpha_{qi}^2 \overline{Q}_{Gi}$$

This is important because the paper considers the **aging effect of reactive current on power-electronic components**.

Therefore, the controller attempts to achieve a suitable voltage profile without unnecessarily increasing reactive-power utilization.

The basic trade-off is:

Better Voltage Regulation ↔ Reactive-Power Utilization

Consequently, the objective-function weighting factors can be used to determine the relative importance of these two requirements.

18.4 Phase-Level Implementation

Another major contribution is that the cooperative-control method is implemented at the **phase level**.

This makes the proposed method applicable to **unbalanced three-phase distribution systems**.

Instead of treating all three phases as an identical balanced system, the controller can consider the voltage condition of individual phases.

Conceptually,

$$V_A, \quad V_B, \quad V_C$$

can be considered individually.

This is particularly important for distribution systems because the voltage conditions of the three phases may not always be identical.

Therefore, phase-level cooperative control provides greater flexibility for maintaining the desired voltage profile and addressing voltage differences between phases.

18.5 Updating the Reactive-Power Utilization Ratio

The proposed cooperative-control algorithm updates the reactive-power utilization ratio for the available reactive-power sources.

The relationship is:

$$\alpha_{qi} = \frac{Q_{Gi}}{\bar{Q}_{Gi}}$$

where Q_{Gi} is the reactive power generated or supplied by the source and $\bar{Q}_{Gi}$ is its maximum available reactive-power capability.

The controller determines appropriate values of α_{qi} so that the required reactive power can be injected or consumed to achieve the desired voltage profile.

Thus,

$$\alpha_{qi} \rightarrow Q_{Gi} \rightarrow \text{Voltage Regulation}$$

This provides the mechanism through which the DGs and capacitor banks participate in the cooperative voltage-control process.

18.6 Improvement at QG Nodes

The results show that the voltage at the **QG nodes** is improved through cooperative control.

QG nodes are the nodes containing DGs or capacitor banks and therefore have direct access to reactive-power resources.

The cooperative controller can modify the reactive-power utilization of these resources according to the system voltage conditions.

Therefore,

QG Node → Reactive-Power Control → Voltage Improvement

The conclusion of the paper confirms that the voltage at the QG nodes participating directly in the cooperative control was improved.

18.7 Improvement at Selected Non-QG Nodes

An important result is that improvement is not limited to the QG nodes.

The paper also observes improvement at the **selected non-QG nodes** participating in the communication and cooperative-control process.

Although these nodes do not directly control DG or capacitor-bank reactive power, their information is communicated to the QG nodes and incorporated into the optimization.

Therefore,

Non-QG Information → QG Cooperative Optimization
→ Reactive-Power Adjustment → Non-QG Voltage Improvement

This demonstrates the importance of the communication topology developed in the study.

18.8 Improvement at Other Non-QG Nodes

The paper further observes that the voltage at **other non-QG nodes** also improves, although the improvement could potentially be greater if those nodes were able to communicate directly with the QG nodes.

This is an important observation regarding the effect of the communication topology.

The proposed method does not necessarily provide the same level of control influence at every node because the amount of information available to the cooperative controller depends on the communication

connections.

Therefore,

Better Communication → More Network Information → Potentially Better Optimization

This establishes a direct relationship between the communication architecture and the effectiveness of the cooperative voltage-control method.

18.9 Importance of Communication Topology

The communication topology is therefore not simply an additional component of the proposed method; it can directly influence its performance.

The QG nodes exchange information through the defined communication network, while non-QG nodes can provide information to QG nodes when communication links exist.

The conclusion suggests that future research should investigate the effects of:

Communication Topology

and:

Communication Speed

on the overall effectiveness of the proposed control method.

This is particularly relevant because the proposed control is iterative. If information exchange is delayed or the communication structure is not sufficiently connected, the optimization may not respond as effectively to rapidly changing system conditions.

18.10 Overall Contribution of the Research

The major contributions demonstrated by the study can be summarized as follows:

1. Fast Voltage Variation Mitigation The method addresses voltage fluctuations resulting from changes in solar irradiance.

2. Cooperative Reactive-Power Control Multiple DGs and capacitor banks can participate in determining the appropriate reactive-power response.

3. Phase-Level Operation The method is implemented at the phase level, making it applicable to unbalanced three-phase systems.

4. Voltage Deviation Minimization The objective function attempts to maintain the voltage close to the desired/base voltage.

5. Reactive-Power Utilization Consideration The method considers reactive-power utilization and its potential effect on the aging of power-electronic components.

6. Communication-Based Cooperation Information from QG and selected non-QG nodes is incorporated into the cooperative optimization.

7. Improvement Beyond QG Nodes The results show that voltage improvement can extend to selected non-QG nodes and, to some extent, other non-QG nodes.

18.11 Future Research Directions

The conclusion identifies several areas for future research.

18.11.1 Different Communication Topologies

Future work will investigate how different communication-network structures affect the performance of the cooperative controller.

This can be expressed as:

Communication Topology → Information Availability → Control Performance

Different communication structures may result in different control performance.

18.11.2 Communication Speed

The authors also propose studying the effect of **communication speed** on the effectiveness of the method.

This is particularly important because the voltage fluctuations being addressed can occur rapidly.

If communication is too slow:

Slow Information Exchange → Delayed Control Response

whereas faster communication can potentially allow the controller to respond more effectively to changing operating conditions.

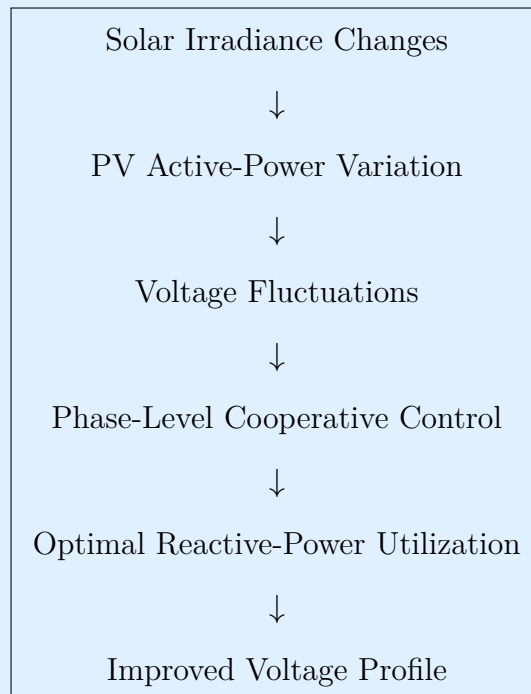
The paper therefore identifies communication topology and communication speed as important factors for future investigation.

18.12 Overall Conclusion

The proposed cooperative-control method provides a systematic approach for regulating voltage in distribution systems containing intermittent solar PV generation. By implementing the control at the **phase level**, the method can be applied to unbalanced three-phase systems. The controller updates the reactive-power utilization ratio of the available reactive-power sources and uses cooperative optimization to determine appropriate reactive-power injection or consumption.

The results demonstrate that the proposed method improves the voltage profile at the QG nodes and selected non-QG nodes participating in the cooperative control. Other non-QG nodes also show improvement, although the paper indicates that their performance could potentially be enhanced if they were directly connected to the QG communication network.

The overall control philosophy can therefore be summarized as:



At the same time, the method considers the utilization of reactive power and the potential aging effects associated with reactive current in power-electronic components.

The study presents a cooperative-control-based method for mitigating fast voltage variations caused by changes in solar irradiance in distribution systems with inverter-based distributed generators. The proposed method minimizes voltage deviation while simultaneously considering the reactive-power utilization and its associated impact on the aging of power-electronic components. The control is implemented at the phase level, making the method applicable to unbalanced three-phase distribution systems. By updating the reactive-power utilization ratio of the available DGs and capacitor banks, the cooperative controller determines appropriate reactive-power injection or consumption to achieve the desired voltage profile. The simulation results demonstrate that the voltage at QG nodes and selected non-QG nodes participating in the cooperative control is improved. Voltage improvement is also observed at other non-QG nodes, although the results indicate that further improvement may be possible if these nodes communicate directly with the QG nodes. Therefore, the communication structure plays an important role in the effectiveness of the proposed method. Future research will investigate the influence of communication topology and communication speed on the performance of the cooperative-control strategy. Overall, the proposed method provides an effective framework for coordinated phase-level voltage regulation in distribution systems with intermittent solar PV generation while considering both voltage performance and reactive-power utilization.